

Exploring the Social World - Quantitative Block: Statistics

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Welcome

This is the website for “Exploring the Social World - Quantitative Block: Statistics” (module **ENVS225**) at the University of Liverpool. This block of the module is designed and delivered by Dr. Gabriele Filomena and Dr. Zi Ye from the Geographic Data Science Lab at the University of Liverpool. The module seeks to provide hands-on experience and training in introductory statistics for human geographers.

The website is **free to use** and is licensed under the [Attribution-NonCommercial-NoDerivatives 4.0 International](#). A compilation of this web course is hosted as a GitHub repository that you can access:

- As an [html website](#).
- As a [GitHub repository](#).

Contact

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Overview

Aim and Learning Objectives

This sub-module aims to provide training and skills on a set of basic quantitative research methods for data collection, analysis, and interpretation. You will learn how to define coherent, relevant research questions, utilise various research quantitative methods, and identify appropriate methodologies to tackle your research questions. **This block serves as the foundation for the dissertation and fieldwork modules.**

Background

Data and research are key pillars of the global economy and society today. We need rigorous approaches to collecting and analysing both the statistics that can tell us ‘how much’ and if there are observable relationships between phenomena; and the information gives us a nuanced understanding of cultural contexts and human dynamics. Quantitative skills enable us to explore and measure socio-economic activities and processes at large scales, while qualitative skills enable understanding of social, cultural, and political contexts and diverse lived experiences. Rather than being in opposition, qualitative and quantitative research can complement one another in the investigation of today’s pressing research questions.

To these ends, this block will help you develop your quantitative (statistical) skills, as critical tools. This course will help you understand what quantitative statistical researchers use and develop a set of research techniques that can be used in your field classes and dissertations.

Learning objectives:

- Understand how to explore a dataset, containing a number of observations described by a set of variables.
- Demonstrate an understanding in the application and interpretation of commonly used quantitative research methods.
- Demonstrate an understanding of how to work with quantitative data to address real-world research questions.

Module Structure

Staff: Dr Zi Ye and Dr Gabriele Filomena

Where and When

Week 1-6 Monday @ Central Teaching Hub PCTC

- Lecture (9 – 10 am).
- Practical PC session (10 am – 12 pm).

Lectures will introduce and explain the fundamentals of quantitative methods, with the opportunity to apply the method introduced in the practical session straight after.

The computer practical sessions, will give you the chance to use and apply quantitative methods to real-world data. These are primarily self-directed sessions, but with support on hand if you get stuck. Support and training in R will be provided through these sessions. Weekly sessions will be driven by empirical research questions.

Week	Topic	Format	Staff
1	Introduction & Review	Lecture and Computer Lab Practical	GF
2	Simple & Multiple Linear Regression	Lecture and Computer Lab Practical	GF
3	Multiple Linear Regression with Categorical Variables	Lecture and Computer Lab Practical	ZY
4	Logistic Regression	Lecture and Computer Lab Practical	ZY
5	Data Visualisation	Lecture and Computer Lab Practical	GF
6	Summary and Assessment Support	Lecture and Computer Lab Practical	ZY

Software and Data

We work in **RStudio in your web browser** (GitHub Codespaces): R, RStudio, the packages and the course data are ready to use, with nothing to install. **Follow the [Setup](#) guide before the first practical.**

RStudio is also installed on University Machines, and you can install [R](#) and [RStudio Desktop](#) on your own laptop if you prefer (see the end of the [Setup](#) guide).

This software is already installed on University Machines. But you will need it to run the analysis on your personal devices.

Data

Example datasets could be accessed through Canvas or (some) on [GitHub](#) Repository of the module. These include:

- 2021 UK Census Data.
- 2021 Annual Population Survey (APS) - only on Canvas.
- 2016 Family Resource Survey (FRS) - only on Canvas.
- 2011 Sample of Anonymised Records (SAR).

Note: The Annual Population Survey requires the completion of a quiz prior to its usage, as it is licensed.

Setup: RStudio in your Browser

In this module you use **RStudio in your web browser**. You don't need to install anything. R, RStudio and all the course files are ready for you.

This works on any computer: university PCs, Windows laptops, Macs and Chromebooks.

It runs on **GitHub Codespaces**. A *codespace* is your own computer on the internet. Only you can use it, and it keeps your files between sessions.

1. First time only

Step 1. Make a GitHub account.

Go to github.com/signup and create a free account. You can use your university email. Then sign in.

Step 2. Create your codespace.

Open this link, then click the green **Create codespace** button:

<https://codespaces.new/GDSL-UL/stats?quickstart=1>

Wait while your codespace is built. **The first time this takes several minutes**, because R, RStudio and the course packages are installed for you. Don't close the tab. Next time it opens much faster. A blue and grey window opens in your browser. This is called VS Code. **You don't need to use it**, but keep it open in the background.

Step 3. Open RStudio.

When everything is ready, you have two ways to open RStudio:

- A small message appears in the bottom-right corner. Click **Open in Browser**.
- Or look at the panel at the bottom of the window. Find the box that says **RStudio is ready**. Hold Ctrl (Cmd on a Mac) and click the link in the box.

RStudio opens in a new tab.



Tip

The new tab is stuck on a GitHub page that says “authenticating”? This is normal. Try these in order:

1. Press F5 to refresh the page.
2. Wait a minute: RStudio may still be starting. Then refresh again.
3. Use a normal browser window, not a private or incognito one. Chrome or Edge work best. Some browsers (Safari, Brave, or Firefox/Edge with strict tracking protection) block the cookie GitHub needs to sign you in: allow cookies for `github.dev` or switch browser.
4. In the VS Code window, click the **Ports** tab in the bottom panel, right-click the **RStudio** row (port 8787) and choose **Open in Browser**.

Step 4. Bookmark the RStudio tab.

This is your personal RStudio address. Use it to come back quickly next time.

2. Every time you work

1. Go to github.com/codespaces and click your codespace. You can also click the button in Step 2 again: it takes you back to the codespace you already have.
2. Open RStudio with your bookmark, or as in Step 3.
3. Work in the RStudio tab.
4. When you finish, go back to github.com/codespaces. Click the ... next to your codespace and choose **Stop codespace**. This saves your free hours. Your files are not deleted.

3. Your files

In RStudio, the **Files** panel (bottom right) shows the course folder:

```
stats/
  data/      the datasets
  img/       pictures used in the labs
  labs/      one lab file for each week
  myLabs/    your own copies of the labs: you work here
```

Everything is already in the right place. **You don’t need to download, move or set up anything.**

The **labs** folder updates automatically each time you open your codespace, so you always have the latest version of the labs. **Don't work in the files in labs:** if you change one, the whole **labs** folder can stop updating. Work on your own copy in **myLabs** instead (see below). Nothing in **myLabs** is ever changed or replaced.

4. Your first lab (Week 1)

1. In the **Files** panel, click **labs**, then click **00.introR.qmd**. The lab opens in the top-left panel. It shows you around RStudio and how to run code.
2. **Make your own copy.** Click **File -> Save As...**, go to the **myLabs** folder, and click **Save**. The copy is now open in RStudio: work in this one.
3. Then open **01.DataExploration.qmd** from **labs** and copy it to **myLabs** in the same way. This is the main practical for Week 1.
4. Go through each lab from top to bottom and run each code chunk. Write your notes and answers in the white space between the chunks.
5. Save often: press **Ctrl+S** (**Cmd+S** on a Mac).

Each week, open that week's lab from **labs**, save a copy in **myLabs**, and work on the copy. Next time, open your copy straight from **myLabs**.

Your changes are only saved in your own codespace. They don't change the website, and other students can't see them.

Note

How the labs find the data

The lab code loads data like this:

```
attacks <- read.csv("../data/attacksUK.csv")
```

`../` means "go up one folder". Your copy of the lab is in **myLabs**, so R goes up to **stats** and then into **data**. You don't need to change these lines.

5. Adding data from Canvas

Some datasets are only on Canvas, for example the Family Resource Survey (FRS). To add one:

1. Download the file from Canvas to your computer.
2. In RStudio's **Files** panel, click **data**.

3. If the lab asks for a new folder (for example **FRS**), click **New Folder**, type the name, and click **OK**. Then click the new folder to open it.
4. Click **Upload**, then **Choose File**. Pick the file you downloaded and click **OK**.

Now the lab code will find the file.

6. Saving a copy on your computer

To download a file from RStudio (for example to keep a copy, or to submit your work):

1. In the **Files** panel, tick the box next to the file.
2. Click **More**, then **Export...**, then **Download**.

For the assessment, download the report template from Canvas and upload it to your **myLabs** folder (the same way as in section 5). Write your report there. You submit an **.html** file, which you make with the **Render** button at the top of your file. See [How to submit](#).

Warning

Keep copies of your work. If you don't open your codespace for 30 days, GitHub deletes it, and your files are deleted too. Download your files from **myLabs** from time to time, and always before holidays and before the assessment deadline.

7. If something goes wrong

- **RStudio is stuck on “authenticating”.** Press F5 to refresh the page. If that doesn't help, see the tips in Step 3 of section 1.
- **RStudio says it lost the connection.** Your codespace has probably stopped. Open it again from github.com/codespaces, then refresh the RStudio tab.
- **R says it can't find a file.** Check that the file is really in the folder named in the code. You can see this in the **Files** panel.
- **GitHub says you have no free hours left.** The free plan gives you about 60 hours a month. Always stop your codespace when you finish (see section 2). If you run out, use RStudio on a university PC (see below).

Another option: RStudio on a university PC or your own laptop

RStudio is already installed on university PCs. To use it:

1. Go to the [course page on GitHub](#). Click the green **Code** button, then **Download ZIP**.
2. Unzip the file. On a university PC, put the folder in your M: drive, so you can open it from any PC on campus.
3. In RStudio, click **File -> Open File...** and open the lab file from the `labs` folder. Then work as described above.

To use your own laptop, first install [R](#) and then [RStudio Desktop](#).

Assessment

Deadline: Thursday 12th November 2026 - 2pm. **Word count:** 2000 words - including tables, excluding references. **Format:** a single `.html` file rendered from your `.qmd` (see [How to submit](#)).

The assignment **Data Exploration and Analysis** consists of writing a research report using one of the regression techniques learned during the module. The basic idea is to put in practice the methods learned during the quantitative block of the module. You are required to apply a linear or logistic regression model to the data provided for the module. The report needs to include the following sections (in brackets, % of the whole length):

- Introduction (5%).
- Literature Review (20%).
- Methods and data (30%).
- Results and discussion (40%).
- Conclusion (5%).
- Reference List.

Required Report Structure

1. Introduction

- Context: Why is the topic relevant or worth being investigated?
- Brief discussion of existing literature.
- Knowledge gap and Aim.
- 1 Research question.

2. Literature review

- More detailed Literature review, i.e. what do we already know about this subject
- Rationale for including certain predictor variables in the model.
- What knowledge gap remains that this article will address? (includes “not studied before in this area”). *Note: there is no expectation on totally original research. The focus is on a clean, sensible, data analysis situated in existing ideas.*

3. Methodology:

- A brief introduction to the dataset being analysed (who collected it? When? How many responses? etc.)
- A description of the variables chosen to be analysed.
- A description of any transformation made to the original data, i.e. turning a continuous variable of income into intervals, or reducing the number of age groups from 11 to 3.
- A description and justification of the statistical techniques used in the subsequent analysis (i.e. the multivariate regression model: Multiple Linear or Logistic Regression).

4. Results and Discussion

- Descriptive statistics and summary of the variables employed, supported by tables or/and charts.
- Correct interpretation of correlation coefficients, statistical significance, and model fit.
- Usage and results of an appropriate multivariate regression model.
- Interpretation of the results, including links and contrasts to existing literature.
- Selective illustrations (graphs and tables) to make your findings as clear as possible.

5. Conclusion

- Summary of main findings.
- Limitations of study (self-critique).
- Highlight any implications derived from the study.

Follow this structure and include **ALL** these points, do not make your life harder.

See [example assignments](#) from previous years on Canvas (you need to be logged in with your UoL account).

How to get there?

The first stage is to identify **ONE** relevant research question to be addressed. Based on the chosen question, you will need to identify a dependent (or outcome) variable which you want to explain, and at least two relevant independent variables that you can use to explain the chosen dependent variable. The selection of variables should be informed by the literature and empirical evidence.

To detail in the Methods Section: Once the variables have been chosen, you will need to describe the data and **appropriate** type of regression to be used for the analysis. You need to explain any transformation done to the original data source, such as reclassifying variables, or changing variables from continuous to nominal scales. You also need to briefly describe the data use: source of data, year of data collection, indicate the number of records used, state if

you are using individual records or geographical units, explain if you are selecting a sample, and any relevant details. You also need to identify type of regression to be used and why.

To detail in the Results and Discussion Section: Firstly, you need to provide two types of analyses. First, you need to provide a descriptive analysis of the data. Here you could use tables and/or plots reporting relevant descriptive statistics, such as the mean, median and standard deviation; variable distributions using histograms; and relationships between variables using correlation matrices or scatter plots. Secondly, you need to present an estimated regression model or models and the interpretation of the estimated coefficients. You need a careful and critical analysis of the regression estimates. You should think that you intend to use your regression models to advice your boss who is expecting to make some decisions based on the information you will provide. As part of this process, you need to discuss the model assessment results for the overall model and regression coefficients. Remember to substantiate your arguments using relevant literature and evidence, and present results clearly in tables and graphs.

How is it graded?

Grade	Score Range	UG	Descriptor	Assignment Expectations
Fail	0-34%	Fail	Inadequate	Literature Review: Lacks relevance and fails to justify variable choice. Evidence is irrelevant or missing, providing no support to the research question. Methods: Data is not described, and the regression model is entirely missing. No appropriate statistical method is applied. Results and Discussion: No descriptive statistics, graphs, or tables are provided. Model results and interpretation are absent. Structure and References: Report is disorganized with significant referencing and citation errors throughout.

Grade	Score Range	UG	Descriptor	Assignment Expectations
Narrow Fail	35-39%	Fail	Highly Deficient	Literature Review: Review is present but lacks coherence and fails to justify variable choice. Evidence is poorly aligned with the research question and mostly irrelevant. Methods: Minimal data description; the regression model is missing but some statistical methods are mentioned. Results and Discussion: Few or no descriptive statistics or visuals are present. Statistical methods are unclear or incorrectly applied. Results are vague and lack meaningful interpretation. Structure and References: Report structure is poor, with referencing errors in multiple sections.

Grade	Score Range	UG	Descriptor	Assignment Expectations
Third / Fail	40-49%	Third (UG)	Deficient	Literature Review: Relevant literature is partially addressed but lacks depth, with limited justification for variable choice. Evidence is minimally aligned with the research question. Methods: A very basic data description is provided, but the selected regression model is deeply inadequate or incorrect (e.g., multiple linear regression for a categorical outcome; logistic regression for a continuous outcome). Results and Discussion: Descriptive statistics or visuals may be present but insufficient. Model results are presented with little to no interpretation. Structure and References: Report structure is present but lacks clarity, with inconsistencies

Grade	Score Range	UG	Descriptor	Assignment Expectations
2.2 / Pass	50-59%	2.2 (UG)	Adequate	Literature Review: Addresses relevant literature but with limited justification of variable choices. Evidence generally supports the research question but lacks detail. Methods: Data description is present but brief; a regression model is included but applied illogically or incorrectly (e.g., multiple linear regression for a categorical outcome; logistic regression for a continuous outcome) and with little explanation. Results and Discussion: Basic descriptive statistics, graphs, or tables are presented; the regression model is applied with some inaccuracies and/or interpretation is minimal. Structure and References: Report is mostly organized

Grade	Score Range	UG	Descriptor	Assignment Expectations
2.1 / Merit	60-69%	2.1 (UG)	Good	Literature Review: Relevant literature is discussed, with some justification for variable choice. Evidence supports the research question well. Methods: Data is described with some detail, though potential data transformations are under-explored. The regression model is appropriate for the selected variable types. Results and Discussion: Descriptive statistics and visuals are provided. Model results are discussed, though interpretation lacks depth. Findings are compared to existing literature. Structure and References: Report is logically structured and clear, with mostly correct citations.

Grade	Score Range	UG	Descriptor	Assignment Expectations
First / Distinction	70-79%	First (UG)	Very Good	Literature Review: Strong grasp of relevant literature, with well-justified variable selection. Evidence aligns well with the research question. Methods: Data is comprehensively described with consideration of relevant transformations. The regression model is appropriate and well-justified. Results and Discussion: Descriptive statistics and clear visuals support findings. Model results are accurately interpreted with strong connections to existing literature. Structure and References: Report has a coherent, professional structure with only minor referencing errors.

Grade	Score Range	UG	Descriptor	Assignment Expectations
High First / High Distinction	80-100%	High First (UG)	Excellent to Outstanding	Literature Review: Critical and thorough literature review with strong, well-justified variable selection. Evidence fully supports the research question with insightful connections. Methods: Detailed data description and transformation steps are clearly articulated. Regression model is expertly applied and justified. Results and Discussion: Comprehensive descriptive statistics, graphs, and tables are provided. Model results are innovatively interpreted with strong links to existing research. Structure and References: Report is professionally structured, with flawless citations and a high standard of organization.

In summary:

1. **Introduction:** Should establish the topic's relevance, present a concise literature overview, identify a knowledge gap, and outline research questions.
2. **Literature Review:** Requires an in-depth review of relevant studies, justification for chosen independent variables, and identification of a potential knowledge gap or unexplored area aligned with the chosen research question.
3. **Methods and Data:** Should describe the dataset, variable transformations, and justify the regression technique. Key transformations, such as reclassifying variables, should be explained with clarity and relevance.
4. **Results and Discussion:** Involves presenting descriptive statistics, followed by a clear regression analysis. Discussion should interpret results, compare findings with existing literature, and include meaningful tables and graphs.
5. **Conclusion:** Summarize findings, discuss limitations, and suggest future directions.
6. **Referencing:** Requires correct and consistent citations and a well-structured reference list.

Employing a novel dataset, i.e. not employed during the practical sessions, for the assignment will be awarded with a higher grade. For example, see other quantitative dataset from [Secondary datasets for Human Geography](#).

Assessment: How to submit

Normally you don't need to do so, but if something does not work, you may have to install [Quarto](#). Recent versions of RStudio do not require you to do so.

.html file

Submit a single, self-contained .html file. PDF submissions are not accepted.

The .html file is a rendered version of a Quarto Markdown file (`qmd` file). This will allow you to write a research paper that also includes your working code, without the need of including the data (the code in a `.qmd` file is executed before the document is converted to HTML).

To do so, at the top of your document, include something like this in a YAML cell (see [Template](#), in RStudio, “visual view”, **Insert**, **YAML block**)

```
---
title: "ENVS225 Assignment"
author: "Anonymous" # do not change
format:
  html:
    self-contained: true
---
```

Once your document is ready with your text and code, render the Quarto document: Click the **Render** button in the RStudio toolbar (blue arrow). This, if everything works (code, existing data, etc), will create an `.html` file in the same location as your `.qmd` file

Assessment Template

Suggested Report Structure — This template follows the requested sections. Each section contains guidance text and an empty R code cell for your analysis. Download it from [here](#), then upload it to the myLabs folder in RStudio (see [Setup](#), section 5) and work on it there.

Start your .qmd file with this YAML header:

```
---
title: "ENVS225 Assignment"
author: "Anonymous"
date: "2026-09-28"
format:
  html:
    self-contained: true
    toc: true
    toc-depth: 2
    code-fold: show
execute:
  warning: false
  message: false
---
```

Setup

Load commonly used libraries for statistical analysis and data manipulation.

```
# Core tidyverse
library(tidyverse) # includes ggplot2, dplyr, tidyr, readr, purrr, tibble, stringr, forcats
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
```

```

v ggplot2 4.0.0      v tibble 3.2.1
v lubridate 1.9.3    v tidyr 1.3.1
v purrr 1.0.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
# Tables & reporting
library(kableExtra)

```

Warning: package 'kableExtra' was built under R version 4.4.2

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```

# Stats helpers
library(broom)

## Already installed in your codespace. On your own PC, install them first.

```

Load your data. If R cannot find the file, see [Setup](#).

```
# df <- read.csv("../data/your_file.csv") # uncomment and change to your file
```

! Important

Where figures and tables go. Put each figure or table in the section that uses it, straight after the paragraph where you first discuss it. **Do not collect them at the end of the report or in an appendix.** Each section below says which figures and tables it needs, if any. Most reports need about **3 tables and 2–3 charts in total**. More charts do not earn more marks: choose each one because it shows something your text discusses (see [Lab: Data Visualisation](#), Parts 3 and 4). Remember that tables count towards the word count.

Once all the chunks execute correctly (no errors when running all the code, Ctrl + Alt + R) you can render the qmd file. Follow the instructions [here](#) to render the document to an HTML file. Remove this message and any irrelevant text from the final submission.

Introduction

Context. Explain why the topic matters (theory, policy, practice), briefly drawing on existing literature.

Aim. State clearly what you want to investigate (e.g., association between X and Y).

Gap & Research Question (RQ). Identify what is missing in the current knowledge and pose 1 RQ (avoid yes/no questions; ask *how, to what extent, which factors*).

Structure. Briefly preview how the rest of the report is organised.

Figures and tables: none. This section is text only.

Literature Review

What we know. Summarise key findings from prior studies relevant to your RQ.

Predictor rationale. Justify the inclusion of variables (theory-driven, prior empirical evidence).

Remaining gap. Specify precisely the gap this report addresses (e.g., population, geography, time period, variable, method).

Note: The goal is a clean, sensible analysis grounded in existing ideas—not necessarily novel theory.

Figures and tables: none. This section is text only.

Methods and Data

Figures and tables in this section: one table (the variables table below). No charts.

Data

Describe the dataset: who collected it, when, whether records are individuals (e.g. FRS, SAR) or geographical units (e.g. Census districts), how many records you use, and any filtering or sampling you applied.

Table 3: Variables used in the analysis. Source: ...

```
# a small table describing your variables (e.g. build a data.frame and show it with kbl());
```

Variables

Name your dependent variable and your independent variables, with their type (numerical or categorical) and units. For categorical variables, state the reference category. Present them in **one short table** (Variable, Description, Type, Units or categories): it describes the variables, it does not report results.

Transformations

Describe any recoding or aggregation (e.g., bin income into bands; reduce age groups from 11 to 3) and justify it. No table or chart needed: if you recode a variable, show its new categories in the variables table above.

Analysis

State which model you use and why, based on your dependent variable: **multiple linear regression** for a numerical dependent variable, **logistic regression** for a binary one.

Important: do not report results here. Descriptive statistics, model output and charts of your data go in Results and Discussion.

Results and Discussion

Figures and tables in this section: one summary table and 2–3 charts (descriptive statistics), then **one model table** (regression model). This is where almost all of your figures and tables belong.

Table 4: Descriptive statistics of the variables in the model. Source: ...

```
# one summary table of all model variables; refer to it as @tbl-descriptives
```

Descriptive statistics

Summarise all the variables in your model in **one table**: mean, standard deviation, median and range for numerical variables; counts and percentages for categorical variables (e.g. `vtable::st()` or `kbl()`, see the Week 5 lab, [Data Visualisation](#)).

Then add **2–3 charts**, each chosen for the variables it shows. Pick from:

- **Distribution of the dependent variable:** histogram (numerical) or bar chart (categorical or binary).
- **Dependent vs a key independent variable:** scatterplot (both numerical), boxplot (categorical vs numerical), or stacked percentage bar chart (both categorical).
- **Associations between independent variables:** correlation matrix or correlogram for numerical variables; Chi-squared test and Cramér's V for categorical variables. This also flags variables that are too closely related to include together.

Comment on what each table and chart shows. Do not overload the report with graphs, and do not show the same result in both a table and a chart. Put each chart in its own code chunk, straight after the paragraph that discusses it.

```
# one chart per chunk; refer to this chart in your text as @fig-dependent
```

Regression model

Present the model in **one table**:

- **Multiple linear regression:** coefficients, standard errors (or 95% confidence intervals), p-values, adjusted R^2 , number of observations.
- **Logistic regression:** odds ratios with 95% confidence intervals, p-values, number of observations.

For each independent variable, report the direction and size of the association in the units of the dependent variable (e.g. percentage points, or odds for logistic regression), whether it is statistically significant, and that it holds with the other variables held constant. Interpret categorical variables against their reference category. Then report the model fit and, in a sentence or two, what the diagnostic checks showed (see the Week 2 lab, [Multiple Linear Regression](#)): describe them in words, you do not need to include the diagnostic plots. The regression model should be the core of this section!

Table 5: Regression model of ... Source: ...

```
# run the model and show a table summarising it (e.g. tidy() + kbl()); refer to it as @tbl-model
```

Discussion

Answer your research question. Compare and contrast your findings with the literature: where do they agree, where do they differ, and why might that be? Remember that regression shows associations, not causes. No new figures or tables here: refer back to the ones above (e.g. “as @tbl-model shows...”).

Captions and labels (all figures and tables): give each one a number and a descriptive title (the `fig-cap` / `tbl-cap` options above do this), label axes with units, state the data source, and refer to each one in the text.

Conclusion

Summary. Recap the main findings vis-à-vis the RQ (do not introduce new results).

Limitations. Offer a brief, honest self-critique (data, measurement, design, external validity).

Implications. Indicate what the findings suggest for practice, policy, or future research.

Figures and tables: none. This section is text only.

References

List every source you cite, in one consistent referencing style (e.g. Harvard). References are not included in the word count.

Samples, Uncertainty and P-values

This page picks up where the Week 1 practical left off. You do not need it to complete any lab, and nothing here is examined on its own. It is here because the ideas below decide whether the numbers in your report actually mean what you say they mean, and because three of them are very commonly got wrong in assignments.

Read it at some point before you write up your results. It should take about twenty minutes.

What you need to have done first: the section *From Sample to Population* in the [Week 1 practical](#), where we established that a sample mean wobbles, and that the size of the wobble is called the standard error.

Setting Up

Everything below uses the same FRS data as the Week 1 practical, so we load it the same way. Check the path — you may need to remove the ...

```
library(dplyr)
library(ggplot2)

frs_data <- read.csv("../data/FRS/FRS16-17_labels.csv")
frs_independent <- frs_data %>% filter(dependent != "Dependent")
```

A note before we start: none of the code below will give you exactly the numbers printed on this page, because every chunk draws a fresh random sample. That is the entire point of the page. Your figures should be *close* to these, not identical, and if you run a chunk twice you will get two different answers.

Why Sample Size Matters

In the practical we found that samples of 500 people missed the true mean income by about £1,200 on average, and that this figure — the standard error — can be calculated as the standard deviation divided by the square root of the sample size.

That square root has a practical consequence. Let's repeat the exercise with smaller and larger samples:

```
means_100 <- replicate(1000, mean(sample(frs_independent$income_gross, size = 100)))
means_2000 <- replicate(1000, mean(sample(frs_independent$income_gross, size = 2000)))

sd(means_100) # standard error with n = 100
```

```
[1] 2605.884
```

```
sd(means_2000) # standard error with n = 2000
```

```
[1] 583.0039
```

Roughly £2,800 for samples of 100, against £600 for samples of 2000. Bigger samples give more precise answers — but look at the *rate*. To halve your uncertainty you must **quadruple** your sample.

This is why national surveys are expensive. It is also why you should be sceptical of confident claims made about small subgroups: a survey of 40,000 people may contain only 90 people in the particular category you want to talk about, and the uncertainty for that group is set by the 90, not the 40,000.

Worth doing: find a subgroup in the FRS that interests you — one region, say, or one age group. How many people are in it? Using the formula above, how precisely can you estimate its mean income? If the answer is “not very”, that is a limitation worth stating in your report.

Confidence Intervals

Reporting “£22,544” on its own pretends to a precision we do not have. A **confidence interval** reports the estimate together with its uncertainty, and R will build one for us:

```
one_sample <- sample(frs_independent$income_gross, size = 500)

t.test(one_sample)
```

One Sample t-test

```
data: one_sample
t = 9.7075, df = 499, p-value < 2.2e-16
```

```

alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 18636.87 28095.11
sample estimates:
mean of x
 23365.99

```

Ignore most of that output and look at the part labelled **95 percent confidence interval**.

The interval is, roughly, “our estimate, give or take about two standard errors”. Its meaning is best stated as a statement about the *method*: if we repeated this whole exercise many times, about 95 out of every 100 intervals built this way would contain the true population mean.

A confidence interval is almost always more informative than a single number, and reporting one in your assignment is a mark of careful work.

Seeing What “95%” Means

That definition is a mouthful, and it is far easier to believe once you have watched it happen. So let’s build not one confidence interval, but a hundred.

The idea is the same as before: take a sample of 500 people, build a confidence interval from it, write it down, and repeat another 99 times. Because we are pretending the FRS is the whole population, we know the true answer, so afterwards we can simply count how many of our intervals caught it.

```

# Take 100 samples of 500 people, and keep the confidence interval from each one
intervals <- replicate(100, t.test(sample(frs_independent$income_gross, 500))$conf.int)

```

`t.test(...)$conf.int` runs the test but keeps *only* the confidence interval, throwing the rest of the output away. The `$` picks one piece out of a result, just as `Grades$Name` picked one column out of a table in the practical.

The result, `intervals`, has two rows and 100 columns: one column per sample, with its lower limit on the first row and its upper limit on the second. We can pull those two rows out to build an ordinary table:

```

many_intervals <- data.frame(
  sample_number = 1:100,
  lower = intervals[1, ], # all of row 1: the lower limits
  upper = intervals[2, ]  # all of row 2: the upper limits
)

```

```
head(many_intervals)
```

	sample_number	lower	upper
1	1	20037.25	25035.52
2	2	20185.51	30836.89
3	3	20251.08	24060.20
4	4	19431.08	23462.05
5	5	19994.24	23661.84
6	6	19734.94	24325.70

Now work out the true answer, and check each interval against it:

```
true_mean <- mean(frs_independent$income_gross)

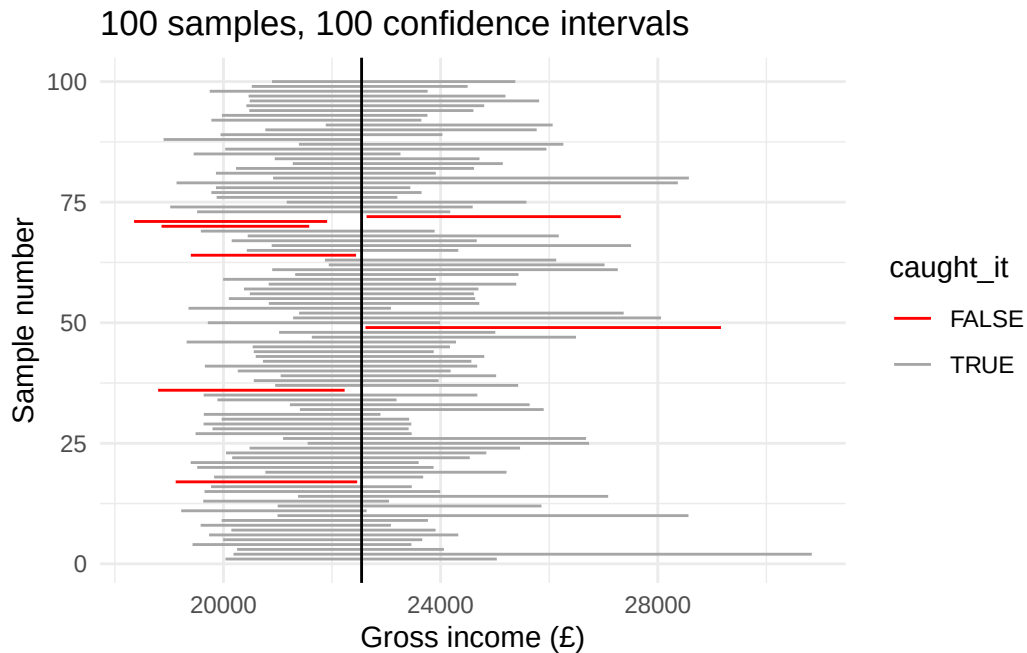
# TRUE if the interval caught the true mean, FALSE if it missed
many_intervals$caught_it <- many_intervals$lower < true_mean & many_intervals$upper > true_m

table(many_intervals$caught_it)
```

FALSE	TRUE
7	93

A handful missed — usually somewhere between about three and ten of the hundred. Let's draw them. Each horizontal line is one sample's confidence interval, and the vertical line is the true mean income of the population:

```
ggplot(many_intervals, aes(y = sample_number, colour = caught_it)) +
  geom_linerange(aes(xmin = lower, xmax = upper)) +
  geom_vline(xintercept = true_mean) +
  scale_colour_manual(values = c("red", "grey65")) +
  labs(
    title = "100 samples, 100 confidence intervals",
    x = "Gross income (£)",
    y = "Sample number"
  ) +
  theme_minimal()
```



This picture *is* the definition of a confidence interval:

- Every interval is a different width and sits in a different place, because every sample was different.
- Most straddle the true mean. A few — the red ones — do not.
- Nothing is wrong with the red intervals. They were built in exactly the same way as the others. They were simply unlucky samples.

And here is the part that matters for your report. **In real research you only ever get one of these lines.** You never find out whether yours is grey or red, because you never learn the true answer. What “95% confident” buys you is not certainty about your own interval; it is the knowledge that the *procedure* you used lands on the truth about 19 times out of 20.

Worth doing: change 500 to 2000 in the code above and run the whole thing again.

You should find the intervals roughly halve in width, because we quadrupled the sample. But the number of red ones does **not** shrink in the same way — it stays somewhere around five, bouncing about from run to run because that count is itself a random quantity.

This is the point. Bigger samples buy you *precision*: a narrower interval, a sharper answer. They do not buy you *reliability*: the 95% was fixed by us when we asked for a 95% interval, and it stays at 95% whatever the sample size.

One honest footnote. If you count carefully you will often find rather more than five red intervals. That is not a mistake either. Incomes are so badly skewed — remember the handful of people earning over a million — that confidence intervals built this way come out slightly

too narrow, and catch the truth about 93% of the time rather than 95%. This is a small example of a large lesson: statistical tools come with assumptions attached, and skewed data strains them.

Significance and Sample Size

In the practical we found a gender income gap of about £9,300 across 33,836 people, with a p-value too small for R to print. Now watch what happens when we ask the identical question of only 60 people:

```
small_sample <- frs_independent %>% slice_sample(n = 60)
t.test(income_gross ~ sex, data = small_sample)
```

Welch Two Sample t-test

```
data: income_gross by sex
t = -4.2399, df = 49.93, p-value = 9.644e-05
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -18295.247 -6532.988
sample estimates:
mean in group Female    mean in group Male
      14397.88          26812.00
```

Worth doing: run those two lines four or five times over, and watch the p-value each time.

You will see it jump around wildly. Sometimes the gap comes out statistically significant; most of the time it does not — in fact, with 60 people this test only reaches $p < 0.05$ about a third of the time. The confidence interval for the difference will usually include zero, which is the same message in a different form.

Nothing about the UK changed between the big test and the small one. There really is a gender income gap. Only our ability to *detect* it changed.

This is the lesson to take away: “not statistically significant” means “*this sample cannot settle the question*”. It does **not** mean “there is no difference”.

In your report, this is a matter of wording, and markers notice. “No significant association was found” is defensible. “There is no association” is not, and it is a much stronger claim than your data can support.

What a P-value Is Not

Three misreadings are extremely common in student reports, and each one costs marks:

- A p-value is **not** the probability that the null hypothesis is true, and it is **not** the probability that your finding is correct. It is calculated *assuming* the null hypothesis is true, so it cannot tell you how likely that assumption is. Three asterisks do not mean “99.9% certain there is a relationship”.
- A small p-value does **not** mean the effect is large or important. With a big enough sample, a trivially small effect will come out “highly significant”. Significance tells you an effect is *detectable*; only the coefficient itself tells you whether it *matters*. Always report and interpret both.
- A p-value says nothing at all about **causation**. It cannot distinguish a real causal relationship from a confounded one. That judgement comes from your literature review and your research design, never from the output.

A Note on Census Data

Everything on this page assumes your data are a *random sample* of some larger population. That is true of the FRS, the APS and the SAR.

It is not true of the Census. When you run a regression on the 2021 Census district data in Week 2, you are working with every district in England and Wales, not a sample of them. Strictly speaking there is no sampling uncertainty to quantify, because there was no sampling.

Statisticians handle this by treating the observed pattern as one realisation of a social process that could have turned out differently — the districts we see are one roll of the dice, and the p-value asks how easily that roll could have come up by chance. That is a reasonable way to reason, but it rests on a weaker footing than a genuine random sample, and it is a judgement rather than a fact about the data.

You do not need to resolve this. But if you use Census data in your assignment, a sentence acknowledging it in your limitations section is exactly the kind of critical awareness the higher marking bands reward.

1 Lab: Introduction to R

The following material has been readapted from:

- <https://dereksonderegger.github.io/570L/1-introduction.html> by Derek L. Sonderegger;

Open this lab in RStudio: in the Files pane, click labs and then 00.introR.qmd. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk. (First time? Follow the [Setup](#) guide first.)

This short lab shows you how to find your way around RStudio and how to run R code. It takes about 20 minutes. Then move on to [Lab: Exploring a Dataset](#), the main practical for Week 1.

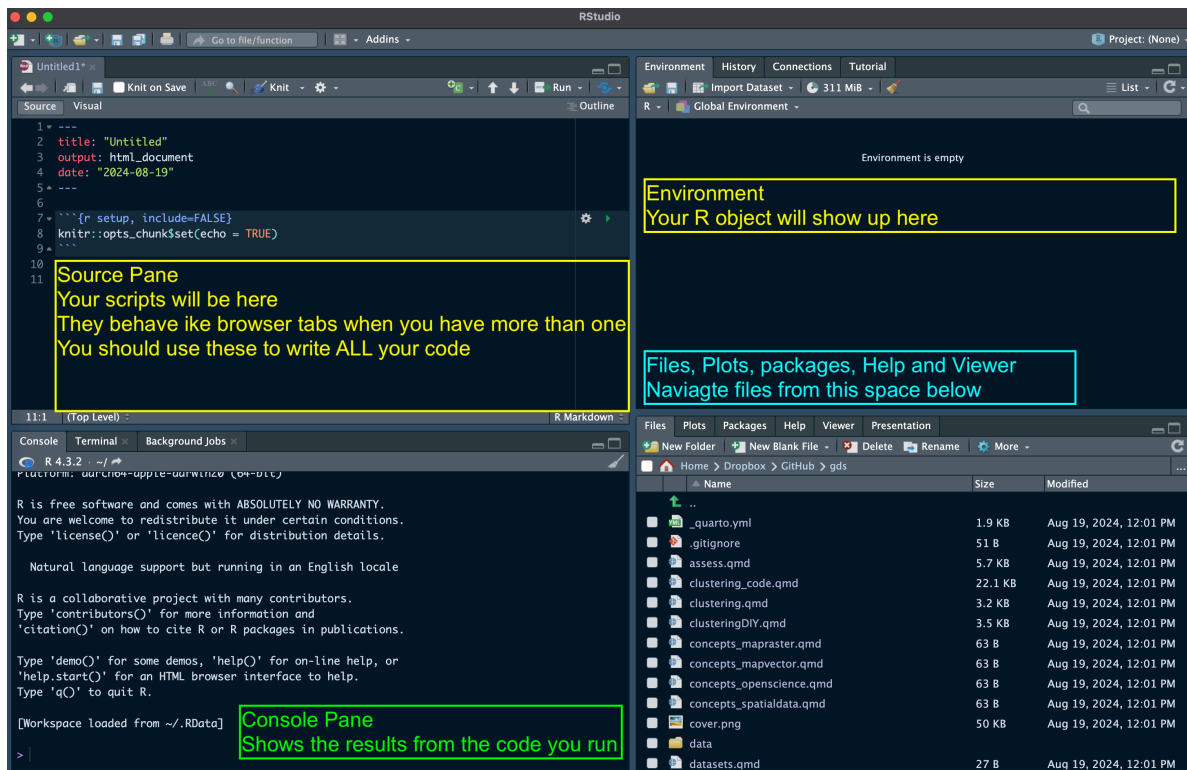
1.1 R and RStudio

R is a free program for statistics and data analysis. You use it by writing code, not by clicking menus. This takes a little getting used to, but it has a big advantage: your code is an exact record of every step of your analysis, so you (or anyone else) can run it again and get the same result.

RStudio is the program you use to write and run R code. In this module it runs in your browser, and R and all the packages you need are already installed.

1.1.1 The four panels

RStudio looks something like this:



- **Top left, Source:** the file you are working on, for example this lab.
- **Bottom left, Console:** where R runs code and prints the results. You can also type code here directly and press Enter. Ignore the **Terminal** tab next to it: you don't need it.
- **Top right, Environment:** the data and variables you have created so far.
- **Bottom right, Files / Plots / Packages / Help:** your course folder, your charts, and help pages.

You can change the colours and font size in Tools -> Global Options -> Appearance.

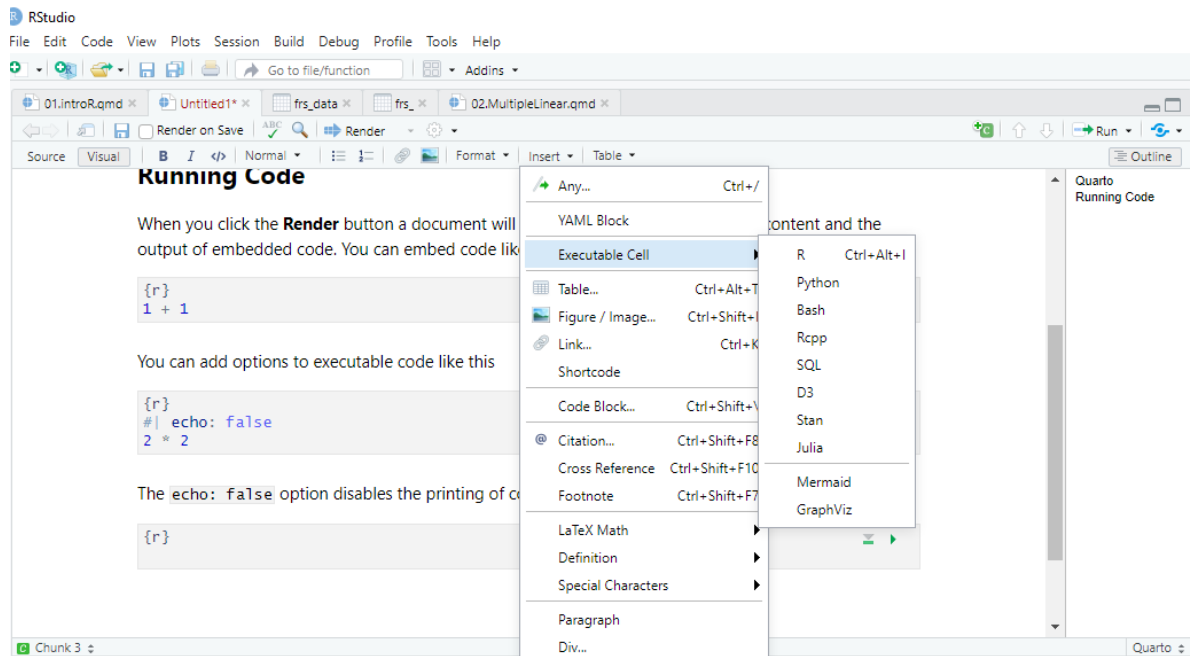
1.2 Quarto documents and code chunks

The labs are **Quarto documents** (.qmd files). A Quarto document mixes normal text, like this paragraph, with **code chunks**: the grey boxes that contain R code. You will write your assessment report in the same kind of file.

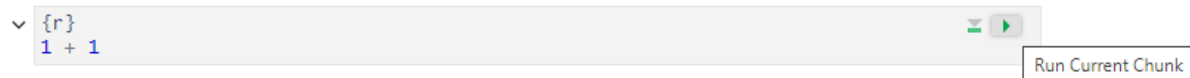
A code chunk starts with ````{r}` and ends with `````:



The buttons **Source** and **Visual** at the top left of the file switch between two views of the same file. In the **Visual** view you can also insert a chunk from **Insert -> Executable Cell -> R**:



To run a chunk, click the green arrow in its top-right corner. The result appears under the chunk.



Task: Run the chunk below.

```
print("hello world")
```

```
[1] "hello world"
```

Some useful shortcuts (use Cmd instead of Ctrl on a Mac):

Shortcut	What it does
Ctrl+Enter	Run the line where your cursor is
Ctrl+Shift+Enter	Run the whole chunk
Ctrl+Alt+I	Insert a new chunk (or use Code -> Insert Chunk)
Ctrl+S	Save the file

To add your own notes, just type in the white space between chunks. To try out your own code, insert a new chunk.

1.3 R as a calculator

Task: Run each chunk in this section and check that you understand the result.

```
# Some simple addition. Text after a # is a comment: R ignores it.
2+3
```

```
[1] 5
```

```
6*8
```

```
[1] 48
```

```
4^3
```

```
[1] 64
```

```
exp(1) # exp() is the exponential function
```

```
[1] 2.718282
```

R has most constants and mathematical functions you could want. For example, **abs()** gives the absolute value of a number, and **round()** rounds it to the nearest whole number.

```
pi # the constant 3.14159265...
```

```
[1] 3.141593
```

```
abs(-1.77)
```

```
[1] 1.77
```

```
round(1.77)
```

```
[1] 2
```

1.3.1 Functions and arguments

`abs()` and `round()` are **functions**. The values you give a function inside the brackets are its **arguments**, separated by commas. Some arguments are required and some are optional.

The `log()` function calculates a logarithm. It takes the number `x` and the `base`. If you name the arguments, the order doesn't matter:

```
log(x=5, base=10)
```

```
[1] 0.69897
```

```
log(base=10, x=5)
```

```
[1] 0.69897
```

If you don't name them, R assumes that the first value is `x` and the second is `base`:

```
log(5, 10)
```

```
[1] 0.69897
```

```
log(10, 5)
```

```
[1] 1.430677
```

1.3.2 Getting help

To see what a function does and which arguments it takes, type `?` followed by its name in the console, for example `?log`. The help page opens in the **Help** tab, bottom right.

1.4 Variables

To use a result later, save it in a **variable**. R uses the arrow `<-` for this (`=` also works, but pick one and stick with it).

```
var <- 2*7.5      # create two variables
another_var <- 5  # notice they appear in the Environment panel
var
```

```
[1] 15
```

```
var * another_var
```

```
[1] 75
```

Variable names cannot start with a number or contain spaces, and they are case sensitive: `var` and `Var` are two different variables.

1.5 Packages

Packages add extra functions to R. For example, `dplyr` has functions for working with data and `ggplot2` makes charts. **All the packages you need in this module are already installed in your codespace.** You just need to load a package with `library()` before you use it, once per session:

```
library(dplyr)  # load the dplyr package; we will use it in the next lab
```

Note

If you use RStudio on a university PC or your own laptop instead of the codespace, a package may be missing. You will see an error like **there is no package called 'dplyr'**. Install the package once with **Tools -> Install Packages...**, then run `library()` again.

1.6 Next

That's all you need to start. Save this file (Ctrl+S), then open `01.DataExploration.qmd` in the `labs` folder and save your copy in `myLabs`: [Lab: Exploring a Dataset](#).

2 Lab: Exploring a Dataset

The following material has been readapted from:

- <https://pietrostefani.github.io/gds/envIRONR.html> by Elisabetta Pietrostefani and Carmen Cabrera-Arnau
- https://raw.githubusercontent.com/dereksonderegger/570L/master/07_DataImport.Rmd

The lecture's slides can be found [here](#).

Before completing the practical, please take this [quiz](#).

Open this lab in RStudio: in the Files pane, click labs and then `01.DataExploration.qmd`. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk and adding your notes and answers in the file. (First time? See the [Setup](#) guide, then do [Lab: Introduction to R](#), which shows how to run code chunks.)

Follow the practical below. You can describe what you are doing in normal text. See [here](#) for how to format normal text in Markdown documents.

2.1 Practice: Dataset and Dataframes

Within this module we will be working with data stored in so-called datasets. A dataset is a structured collection of data points that represent various measurements or observations, often organized in a tabular format with rows and columns. A dataset might contain information about different locations, such as neighborhoods or cities, with each row representing a place and each column detailing characteristics like population density, average income, or number of green parks. For example, a dataset could be compiled to study patterns in urban mobility, where the data includes the number of daily commuters, the distance they travel, and the mode of transport they use. Datasets provide the essential building blocks for statistical analysis; they enable exploring relationships, identifying patterns, and drawing conclusions about certain phenomena.

Examples of everyday datasets:

- **Premier League Standings:** Each row represents a team, with columns for points, games played, wins, draws, and losses.

- **Movie Dataset:** Each row represents a movie, with columns showing its title, genre, release year, director, and rating.
- **Weather Dataset:** Each row shows a day's weather in a city, with columns for temperature, humidity, wind speed, and precipitation.

Usually, data is organized in

- **Columns** of data representing some trait or variable that we might be interested in. In general, we might wish to investigate the relationship between variables.
- **Rows** represent a single object on which the column traits are measured.

For example, in a grade book for recording students scores throughout the semester, there is one row for every student and columns for each assignment. A greenhouse experiment dataset will have a row for every plant and columns for treatment type and biomass.

2.1.1 Datasets in R

In R, we want a way of storing data where it feels just as if we had an Excel Spreadsheet where each row represents an observation and each column represents some information about that observation. We will call this object a `data.frame`, an R representation of a data set. The easiest way to understand data frames is to create one.

Task: Run the chunk below. It creates a `data.frame` that represents an instructor's grade book, where each row is a student, and each column represents some sort of assessment.

```
library(dplyr)

Grades <- data.frame(
  Name = c('Bob', 'Jeff', 'Mary', 'Valerie'),
  Exam.1 = c(90, 75, 92, 85),
  Exam.2 = c(87, 71, 95, 81)
)
# Show the data.frame
# View(Grades) # show the data in an Excel-like tab. Doesn't work when knitting
Grades        # show the output in the console. This works when knitting
```

	Name	Exam.1	Exam.2
1	Bob	90	87
2	Jeff	75	71
3	Mary	92	95
4	Valerie	85	81

R allows two different ways to access elements of the `data.frame`. First is a matrix-like notation for accessing particular values.

Format	Result
<code>[a,b]</code>	Element in row <code>a</code> and column <code>b</code>
<code>[a,]</code>	All of row <code>a</code>
<code>[,b]</code>	All of column <code>b</code>

Because the columns have meaning and we have given them column names, it is desirable to want to access an element by the name of the column as opposed to the column number.

Task: Run:

```
Grades[, 2]      # print out all of column 2
```

```
[1] 90 75 92 85
```

```
Grades$Name      # The $-sign means to reference a column by its label
```

```
[1] "Bob"      "Jeff"     "Mary"     "Valerie"
```

2.1.2 Importing Data in R

Usually we won't type the data in by hand, but rather load the data from some package. Reading data from external sources is a necessary skill.

Comma Separated Values Data

To consider how data might be stored, we first consider the simplest file format: the comma separated values file (`.csv`). In this file type, each of the "cells" of data are separated by a comma. For example, the data file storing scores for three students might be as follows:

```
Able, Dave, 98, 92, 94
Bowles, Jason, 85, 89, 91
Carr, Jasmine, 81, 96, 97
```


Typically when you open up such a file on a computer with MS Excel installed, Excel will open up the file assuming it is a spreadsheet and put each element in its own cell. However, you can also open the file using a more primitive program (say Notepad in Windows, TextEdit on a Mac) you'll see the raw form of the data.

Having just the raw data without any sort of column header is problematic (which of the three exams was the final??). Ideally we would have column headers that store the name of the column.

```
LastName, FirstName, Exam1, Exam2, FinalExam
Able, Dave, 98, 92, 94
Bowles, Jason, 85, 89, 91
Carr, Jasmine, 81, 96, 97
```

Reading (.csv) files

To make R read in the data arranged in this format, we need to tell R three things:

1. Where does the data live? Often this will be the name of a file on your computer, but the file could just as easily live on the internet (provided your computer has internet access).
2. Is the first row data or is it the column names?
3. What character separates the data? Some programs store data using tabs to distinguish between elements, some others use white space. R's mechanism for reading in data is flexible enough to allow you to specify what the separator is.

The primary function that we'll use to read data from a file and into R is the function `read.csv()`. This function has many optional arguments but the most commonly used ones are outlined in the table below.

Argument	Default	Description
<code>file</code>	Required	A character string denoting the file location.
<code>header</code>	TRUE	Specifies whether the first line contains column headers.
<code>sep</code>	","	Specifies the character that separates columns. For <code>read.csv()</code> , this is usually a comma.

Argument	Default	Description
<code>skip</code>	0	The number of lines to skip before reading data; useful for files with descriptive text before the actual data.
<code>na.strings</code>	"NA"	Values that represent missing data; multiple values can be specified, e.g., <code>c("NA", "-9999")</code> .
<code>quote</code>	"	Specifies the character used to quote character strings, typically " or '.
<code>stringsAsFactors</code>	FALSE	Controls whether character strings are converted to factors; FALSE means they remain as character data.
<code>row.names</code>	NULL	Allows specifying a column as row names, or assigning NULL to use default indexing for rows.
<code>colClasses</code>	NULL	Specifies the data type for each column to speed up reading for large files, e.g., <code>c("character", "numeric")</code> .
<code>encoding</code>	"unknown"	Sets the text encoding of the file, which can be useful for files with special or international characters.

Most of the time you just need to specify the file.

Task: Let's read in a dataset of terrorist attacks that have taken place in the UK:

```
attacks <- read.csv(file = '../data/attacksUK.csv') # where the data lives
View(attacks)
```

2.2 Practice: Descriptive Statistics

2.2.1 Summarizing Data

It is very important to be able to take a data set and produce summary statistics such as the mean and standard deviation of a column. For this sort of manipulation, we use the package `dplyr`. This package allows chaining together many common actions to form a particular task.

The fundamental operations to perform on a data set are:

- **Subsetting** - Returns a dataframe with only particular columns or rows
 - `select` - Selecting a subset of columns by name or column number.
 - `filter` - Selecting a subset of rows from a data frame based on logical expressions.
 - `slice` - Selecting a subset of rows by row number.
- `arrange` - Re-ordering the rows of a data frame.
- `mutate` - Add a new column that is some function of other columns.
- `summarise` - calculate some summary statistic of a column of data. This collapses a set of rows into a single row.

Each of these operations is a function in the package `dplyr`. These functions all have a similar calling syntax:

- The first argument is a data set.
- Subsequent arguments describe what to do with the input data frame and you can refer to the columns without using the `df$column` notation.

All of these functions will return a data set.

Let's consider the `summarise` function to calculate the mean score for `Exam.1`. Notice that this takes a data frame of four rows, and summarizes it down to just one row that represents the summarized data for all four students.

```
library(dplyr) # load the library
Grades %>%
  summarize( Exam.1.mean = mean( Exam.1 ) )
```

```
Exam.1.mean
1      85.5
```

Similarly you could calculate the **standard deviation** for the exam as well.

```
Grades %>%
  summarize( Exam.1.mean = mean( Exam.1 ),
             Exam.1.sd   = sd(   Exam.1   ) )
```

```
Exam.1.mean Exam.1.sd
1          85.5    7.593857
```

Task: Insert a new chunk below and calculate the mean and standard deviation of Exam.2. Type the code yourself instead of copying it: it is the best way to learn it.

The %>% operator works by translating the command `a %>% f(b)` to the expression `f(a,b)`. This operator works on any function `f`. This is useful when we want to start with `x`, and first apply a function `f()`, then `g()`, and then `h()`; the usual R command would be `h(g(f(x)))` which is hard to read. Using the pipe command %>%, this sequence of operations becomes `x %>% f() %>% g() %>% h()`.

Below, the code takes the Grades dataframe and calculates a column for the average exam score, and then sorts the data according to that average score

```
Grades %>% mutate( Avg.Score = (Exam.1 + Exam.2) / 2 ) %>% arrange( Avg.Score )
```

	Name	Exam.1	Exam.2	Avg.Score
1	Jeff	75	71	73.0
2	Valerie	85	81	83.0
3	Bob	90	87	88.5
4	Mary	92	95	93.5

You don't have to memorise this.

Let's go back to the terrorist attacks dataframe. There are attacks perpetrated by several different groups. Each record is a single attack and contains information about who perpetrated the attack, what year, how many were killed and how many were wounded. You can get a glimpse of the dataframe with the function `head`

```
head(attacks, n = 10)
```

	nrKilled	nrWound	year	country	group
1	0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades
2	0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades
3	0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades

4	0	0	2005	United Kingdom	Abu Hafs al-Masri Brigades
5	0	1	1982	United Kingdom	Abu Nidal Organization (ANO)
6	0	0	2014	United Kingdom	Anarchists
7	0	0	2014	United Kingdom	Anarchists
8	0	0	2014	United Kingdom	Anarchists
9	0	0	2014	United Kingdom	Anarchists
10	0	0	2014	United Kingdom	Anarchists
				attack	target
1				Bombing/Explosion	Transportation
2				Bombing/Explosion	Transportation
3				Bombing/Explosion	Transportation
4				Bombing/Explosion	Transportation
5				Assassination	Government (Diplomatic)
6				Facility/Infrastructure Attack	Business
7				Facility/Infrastructure Attack	Business
8				Facility/Infrastructure Attack	Business
9				Facility/Infrastructure Attack	Private Citizens & Property
10				Facility/Infrastructure Attack	Police
				weapon	
1				Explosives/Bombs/Dynamite	
2				Explosives/Bombs/Dynamite	
3				Explosives/Bombs/Dynamite	
4				Explosives/Bombs/Dynamite	
5				Firearms	
6				Incendiary	
7				Incendiary	
8				Incendiary	
9				Incendiary	
10				Incendiary	

We might want to compare different actors and see the mean and standard deviation of the number of people wounded, by each group's attack, across time. To do this, we are still going to use `summarise()`, but we will precede it with `group_by(group)` to tell the subsequent `dplyr` functions to perform the actions separately for each group.

```
attacks %>%
  group_by( group) %>%
  summarise( Mean = mean(nrWound),
             Std.Dev = sd(nrWound))
```

```
# A tibble: 38 x 3
  group
```

```
Mean Std.Dev
```

	<chr>	<dbl>	<dbl>
1	Abu Hafs al-Masri Brigades	0	0
2	Abu Nidal Organization (ANO)	1	NA
3	Anarchists	0	0
4	Animal Liberation Front (ALF)	0.0833	0.282
5	Animal Rights Activists	0.167	0.389
6	Armenian Secret Army for the Liberation of Armenia	0.2	0.447
7	Black September	0.667	0.577
8	Continuity Irish Republican Army (CIRA)	0.7	3.12
9	Dissident Republicans	0.0758	0.267
10	Informal Anarchist Federation	0	0

i 28 more rows

Task: Run the chunk above. Then insert a new chunk and try out another categorical variable instead of `group` (e.g. `year`) and `nrKilled` instead of `nrWound`.

Let's now move to another dataset to address a research question.

For illustration purposes, we will use the **Family Resources Survey (FRS)**. The FRS is an annual survey conducted by the UK government that collects detailed information about the income, living conditions, and resources of private households across the United Kingdom. Managed by the Department for Work and Pensions (DWP), the FRS provides data that is essential for understanding the economic and social conditions of households and informing public policy.

Consider questions such as:

- How many respondents (persons) are there in the 2016-17 FRS?
- How many variables (population attributes) are there?
- What types of variables are present in the FRS?
- What is the most detailed geography available in the FRS?

Task: To answer these questions, download `FRS16-17_labels.csv` from the [FRS page on Canvas](#) and upload it to a new folder `data/FRS` in RStudio (see [Setup](#), section 5). Then load and inspect the dataset.

```
# the FRS dataset should be already loaded, otherwise
frs_data <- read.csv("../data/FRS/FRS16-17_labels.csv")

# Display basic structure
glimpse(frs_data)
```

```
Rows: 44,145
Columns: 45
```

\$ household	<int> 6087, 6101, 6103, 6122, 6134, 6136, 6138, 6140, 6143,~
\$ family	<int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
\$ person	<int> 5, 3, 3, 3, 2, 4, 4, 3, 3, 4, 4, 3, 4, 2, 5, 3, 4, 3,~
\$ country	<chr> "England", "England", "England", "Northern Ireland", ~
\$ region	<chr> "London", "South East", "Yorks and the Humber", "Nort~
\$ age_group	<chr> "05-10", "05-10", "05-10", "05-10", "05-10", "05-10",~
\$ sex	<chr> "Female", "Male", "Male", "Female", "Female", "Female~
\$ marital_status	<chr> "Single", "Single", "Single", "Single", "Single", "Si~
\$ ethnicity	<chr> "Mixed / multiple ethnic groups", "White", "White", "~
\$ hrp	<chr> "Not HRP", "Not HRP", "Not HRP", "Not HRP", "Not HRP"~
\$ rel_to_hrp	<chr> "Son/daughter (incl. adopted)", "Son/daughter (incl. ~
\$ lifestage	<chr> "Child (0-17)", "Child (0-17)", "Child (0-17)", "Chil~
\$ dependent	<chr> "Dependent", "Dependent", "Dependent", "Dependent", "~
\$ arrival_year	<chr> "UK Born", "UK Born", "UK Born", "UK Born", "UK Born"~
\$ birth_country	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ care_hours	<chr> "0 hours per week", "0 hours per week", "0 hours per ~
\$ educ_age	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ educ_type	<chr> "School (full-time)", "School (full-time)", "School (~
\$ fam_youngest	<chr> "7", "4", "0", "7", "0", "9", "10", "0", "3", "10", "~
\$ fam_toddlers	<int> 0, 1, 1, 0, 2, 0, 0, 2, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1,~
\$ fam_size	<int> 4, 4, 4, 3, 4, 4, 3, 5, 4, 4, 3, 4, 4, 3, 5, 4, 4, 4,~
\$ happy	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ health	<chr> "Not known", "Not known", "Not known", "Not known", "~
\$ hh_accom_type	<chr> "Terraced house/bungalow", "Detached house/bungalow",~
\$ hh_benefits	<int> 10868, 0, 1768, 8632, 8372, 1768, 1768, 1768, 0, 0, 1~
\$ hh_composition	<chr> "Three or more adults, 1+ children", "One adult femal~
\$ hh_ctax_band	<chr> "Band D", "Band F", "Band A", "Band B", "Band A", "Ba~
\$ hh_housing_costs	<chr> "4316", "10296", "5408", "Northern Ireland", "5720", ~
\$ hh_income_gross	<int> 54236, 180804, 26936, 19968, 17992, 76596, 31564, 366~
\$ hh_income_net	<int> 44668, 120640, 23556, 19968, 17992, 62868, 29744, 287~
\$ hh_size	<int> 5, 4, 4, 3, 4, 4, 4, 5, 4, 4, 4, 4, 4, 5, 5, 4, 4, 4,~
\$ hh_tenure	<chr> "Mortgaged (including part rent / part own)", "Mortga~
\$ highest_qual	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ income_gross	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
\$ income_net	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
\$ jobs	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ life_satisf	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ nssec	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ sic_chapter	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ sic_division	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ soc2010	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ work_hours	<chr> "Dependent child", "Dependent child", "Dependent chil~
\$ workstatus	<chr> "Dependent Child", "Dependent Child", "Dependent Chil~

```
$ years_ft_work      <chr> "Dependent child", "Dependent child", "Dependent chil~
$ survey_weight      <int> 2315, 1317, 2449, 427, 1017, 1753, 1363, 1344, 828, 1~
```

and summary:

```
summary(frs_data)
```

household	family	person	country
Min. : 1	Min. :1.000	Min. :1.00	Length:44145
1st Qu.: 4816	1st Qu.:1.000	1st Qu.:1.00	Class :character
Median : 9673	Median :1.000	Median :2.00	Mode :character
Mean : 9677	Mean :1.106	Mean :1.98	
3rd Qu.:14553	3rd Qu.:1.000	3rd Qu.:3.00	
Max. :19380	Max. :6.000	Max. :9.00	
region	age_group	sex	marital_status
Length:44145	Length:44145	Length:44145	Length:44145
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character
ethnicity	hrp	rel_to_hrp	lifestage
Length:44145	Length:44145	Length:44145	Length:44145
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character
dependent	arrival_year	birth_country	care_hours
Length:44145	Length:44145	Length:44145	Length:44145
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character
educ_age	educ_type	fam_youngest	fam_toddlers
Length:44145	Length:44145	Length:44145	Min. :0.0000
Class :character	Class :character	Class :character	1st Qu.:0.0000
Mode :character	Mode :character	Mode :character	Median :0.0000
			Mean :0.2557
			3rd Qu.:0.0000
			Max. :4.0000

fam_size	happy	health	hh_accom_type
Min. :1.000	Length:44145	Length:44145	Length:44145
1st Qu.:2.000	Class :character	Class :character	Class :character
Median :2.000	Mode :character	Mode :character	Mode :character
Mean :2.599			
3rd Qu.:4.000			
Max. :9.000			
hh_benefits	hh_composition	hh_ctax_band	hh_housing_costs
Min. : 0	Length:44145	Length:44145	Length:44145
1st Qu.: 0	Class :character	Class :character	Class :character
Median : 1768	Mode :character	Mode :character	Mode :character
Mean : 5670			
3rd Qu.:10192			
Max. :54080			
hh_income_gross	hh_income_net	hh_size	hh_tenure
Min. : -326092	Min. : -334776	Min. : 1.00	Length:44145
1st Qu.: 22256	1st Qu.: 20748	1st Qu.:2.00	Class :character
Median : 35984	Median : 31512	Median :3.00	Mode :character
Mean : 46076	Mean : 37447	Mean :2.96	
3rd Qu.: 57252	3rd Qu.: 47008	3rd Qu.:4.00	
Max. :1165216	Max. :1116596	Max. :9.00	
highest_qual	income_gross	income_net	jobs
Length:44145	Min. : -354848	Min. : -358592	Length:44145
Class :character	1st Qu.: 52	1st Qu.: 0	Class :character
Mode :character	Median : 12740	Median : 12012	Mode :character
	Mean : 17305	Mean : 14204	
	3rd Qu.: 23712	3rd Qu.: 20384	
	Max. :1127360	Max. :1110928	
life_satisf	nssec	sic_chapter	sic_division
Length:44145	Length:44145	Length:44145	Length:44145
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character
soc2010	work_hours	workstatus	years_ft_work
Length:44145	Length:44145	Length:44145	Length:44145
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

survey_weight

Min. : 221
 1st Qu.: 1097
 Median : 1380
 Mean : 1459
 3rd Qu.: 1742
 Max. : 39675

2.2.2 Understanding the Structure of the FRS Datafile

In the FRS data structure, each row represents a person, but:

- Each person is nested within a family.
- Each family is nested within a household.

Below is an example dataset structure:

household	family	person	region	age_group	sex	marital_status	rel_to_hrp
1	1	1	London	40-44	Female	Married/Civil partner-ship	Spouse
1	1	2	London	40-44	Male	Married/Civil partner-ship	Household Representative
1	1	3	London	5-10	Male	Single	Son/daughter (incl. adopted)
1	1	4	London	5-10	Female	Single	Son/daughter (incl. adopted)
1	1	5	London	16-19	Male	Single	Step-son/daughter
2	1	1	Scotland	35-39	Male	Single	Household Representative
3	1	1	Yorks and the Humber	35-39	Female	Married/Civil partner-ship	Household Representative
3	1	2	Yorks and the Humber	35-39	Male	Married/Civil partner-ship	Spouse
3	1	3	Yorks and the Humber	5-10	Male	Single	Step-son/daughter

household	family	person	region	age_group	sex	marital_status	rel_to_hrp
4	1	1	Wales	0-4	Male	Single	Son/daughter (incl. adopted)
4	1	2	Wales	60-64	Male	Married/Civil partner-ship	Household Representative
4	1	3	Wales	55-59	Female	Married/Civil partner-ship	Spouse
4	2	3	Wales	30-34	Female	Single	Son/daughter (incl. adopted)

The first five people in the FRS all belong to the same household (household 1); they also all belong to the same family. This family comprises a married middle-aged couple plus their three children, one of whom is a stepson.

The second household (household 2) comprises only one person – a single middle-aged male. The third household comprises another married couple, this time with two children.

Superficially the fourth household looks similar to households 1 and 2: a married couple plus their daughter. The difference is that this particular married couple is nearing retirement age, and their daughter is middle-aged. Consequently, despite being a child of the married couple, the middle-aged daughter is treated as a separate ‘family’ (family 2 in the household). This is because the FRS (and Census) define a ‘family’ as a couple plus any ‘dependent’ children. A dependent child is defined as a child who is either aged 0-15 or aged 16-19, unmarried and in full-time education. All children aged 16-19 who are married or no longer in full-time education are regarded as ‘independent’ adults who form their own family unit, as are all children aged 20+.

The inclusion of all persons in a household allows us more flexibility in the types of research question we can answer. For example, we could explore how the likelihood of a woman being in paid employment `WorkStatus` is influenced by the age of the youngest child still living in her family (if any) `fam_youngest`.

In the FRS (and Census), a “family” is defined as a couple and any “dependent” children. Dependent children are defined as those aged 0–15, or aged 16–19 if unmarried and in full-time education.

2.2.3 Explore the Distribution of Your Outcome Variable

Before starting your analysis, it is critical to know the type of scale used to measure your outcome variable: is it categorical or continuous? Here we will start off by exploring a continuous variable which can then turn into a categorical variable (e.g. top earners: yes or no).

We explore the income distribution in the UK by first looking at the low and high end of the distribution ie. What sorts of people have high (or low) incomes?

In the FRS each person's annual income is recorded, both gross (pre-tax) and net (post-tax). This income includes all income sources, including earnings, profits, investment returns, state benefits, occupational pensions etc. As it is possible to make a loss on some of these activities, it is also possible (although unusual) for someone's gross or net annual income in a given year to be negative (representing an overall loss).

Task: Load the FRS dataset into your R environment, if it's not already loaded, and inspect the data.

Open the dataset in RStudio's **Data Viewer** to explore its structure, including the `income_gross` and `income_net` variables.

```
# Open the data in the RStudio Viewer
View(frs_data)
```

in the **Data Viewer** tab, scroll horizontally to locate the `income_gross` and `income_net` columns. If columns are listed alphabetically, they will appear near other attributes that start with "income."

You should notice two things:

- Incomes are recorded to the nearest £, NOT in income bands.
- Dependent children almost all have a recorded income of £0.

This second observation highlights the somewhat loose wording of our question above (*What sorts of people have high (or low) incomes?*). To avoid reaching the somewhat banal conclusion that those with the lowest of all incomes are almost all children, we should re-frame the question more precisely as *What sorts of people (excluding dependent children) have low incomes?*

Task: Determine the Scale of the Outcome Variable.

```
# Summarize income variables
summary(frs_data$income_gross)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-354848	52	12740	17305	23712	1127360

```
# Summarize income variables
summary(frs_data$income_net)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-358592	0	12012	14204	20384	1110928

Task: Exclude Dependent Children.

You need to select all cases (persons) that are independent, that is where the variable `dependent` has values different from `!= "Dependent"` or equal `== "Independent"`.

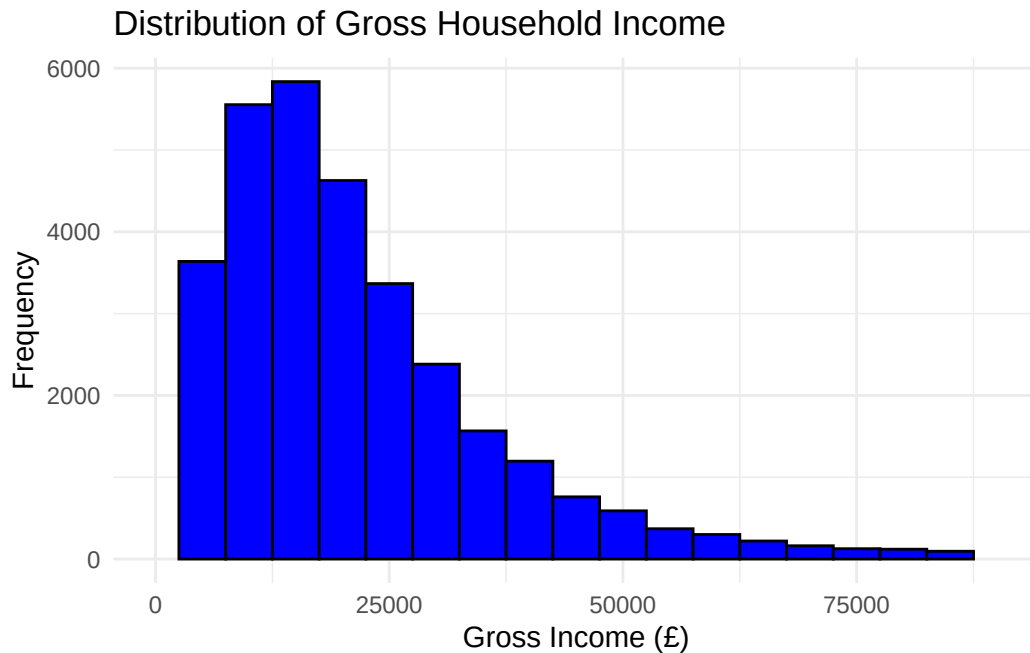
```
# Filter to include only independent persons
frs_independent <- frs_data %>% filter(dependent != "Dependent")
```

Task: Create a basic histogram (a visualisation lecture is scheduled later on).

The income variables in the FRS are all scale variables so a good starting point is to examine its distribution looking at a histogram of `income_gross`.

```
library(ggplot2)

ggplot(frs_independent, aes(x = income_gross)) +
  geom_histogram(binwidth = 5000, fill = "blue", color = "black") +
  labs(
    title = "Distribution of Gross Household Income",
    x = "Gross Income (£)",
    y = "Frequency"
  ) +
  xlim(0, 90000) +
  theme_minimal()
```



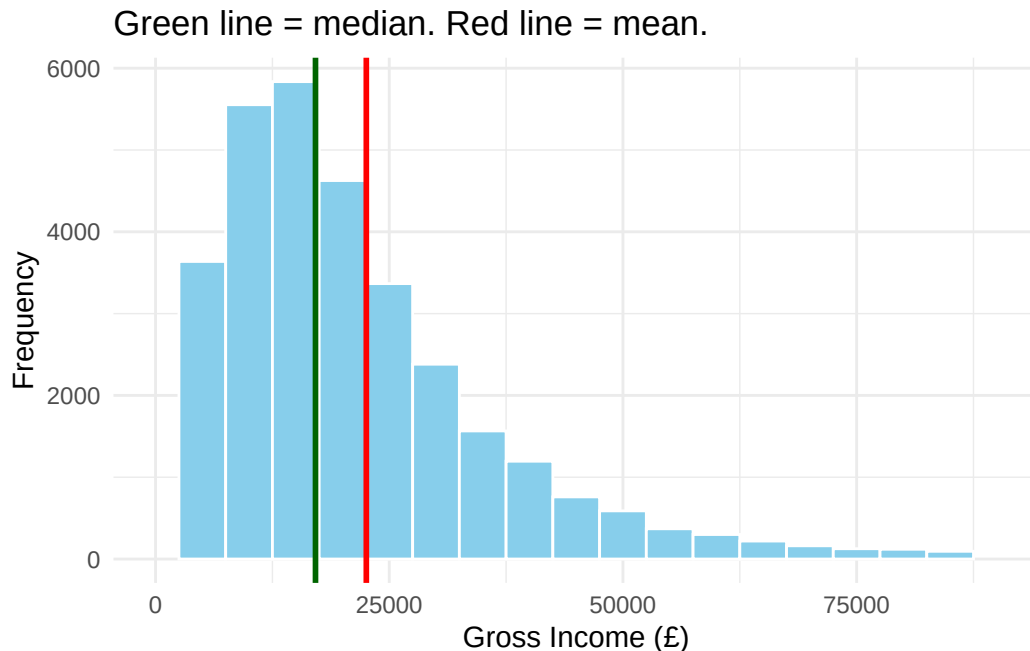
You should see the histogram below. It reveals that the income distribution is very skewed with few people earning high salaries and the majority earning just over or less 35,000 annually.

Why this matters: the mean and the median disagree.

Look back at the `summary()` output. The mean gross income is about £22,544, but the median is about £17,108 — a difference of more than £5,000. Both are “the average”. Both are correct. So which one should you report?

The histogram answers it. Let’s draw the two numbers onto it:

```
ggplot(frs_independent, aes(x = income_gross)) +
  geom_histogram(binwidth = 5000, fill = "skyblue", color = "white") +
  geom_vline(xintercept = median(frs_independent$income_gross), colour = "darkgreen", linewidth = 1) +
  geom_vline(xintercept = mean(frs_independent$income_gross), colour = "red", linewidth = 1) +
  labs(
    title = "Green line = median. Red line = mean.",
    x = "Gross Income (£)",
    y = "Frequency"
  ) +
  xlim(0, 90000) +
  theme_minimal()
```



`geom_vline()` simply draws a vertical line at the position you give it. Here we give it the median and the mean.

The green median line sits in the middle of the tall bars, where most people actually are. The red mean line sits noticeably to the right, in a region where comparatively few people are. Nobody moved it there deliberately: the mean is dragged upwards by the long tail of very high earners off to the right of the chart (remember, the highest income in this dataset is over £1,000,000). The median cannot be dragged like this, because it only cares about who is in the middle of the queue, not how rich the richest person is.

Task: Which of the two lines better describes “a typical person’s income” in this dataset? This is not a trick question — look at where the bars are.

This is why income, house prices and similar skewed variables are almost always reported as medians, and it is a point worth making in your assignment if you use one of them. It is also a warning: **whenever you report a mean, look at the histogram first.** If the distribution has a long tail, the mean is describing the tail as much as it is describing your typical case.

Task: Adopt a regrouping strategy.

You can also cross-tabulate gross (or net) income with any of the other variables in the FRS to your heart’s content – or can you?

Again, here is important to recall that the income variables in the FRS are all ‘scale’ variables; in other words, they are precise measures rather than broad categories. Consequently, every single person in the FRS potentially has their own unique income value. That could make for

a table c. 44,000 rows long (one row per person) if each person has their own unique value. The solution is to create a categorical version of the original income variable by assigning each person to one of a set of income categories (income bands). Having done this, cross-tabulation then becomes possible.

But which strategy to use? Equal intervals, percentiles or ‘ad hoc’. Here I would suggest that ‘ad hoc’ is best: all you want to do is to allocate each independent adult to one of three arbitrarily defined groups: ‘low’, ‘middle’ and ‘high’ income. **Define Low and High Income Thresholds**

Define thresholds for income categories:

- Low-income threshold: £_____
- High-income threshold: £_____

Task: Create a New Variable Based on Regrouping of Original Variable.

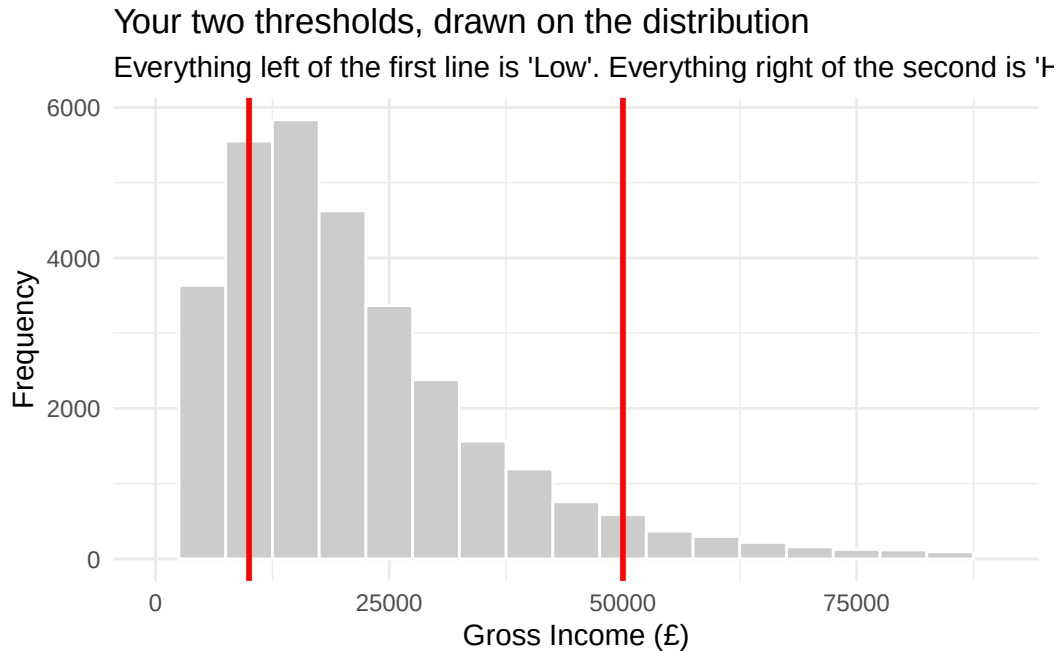
Recode `income_gross` into categories based on the chosen thresholds.

```
# Define thresholds for income categories
LOW_THRESHOLD <- 10000 # Replace with the upper limit for low income
HIGH_THRESHOLD <- 50000 # Replace with the lower limit for high income

# Define income categories based on thresholds
frs_independent <- frs_independent %>%
  mutate(income_category = case_when(
    income_gross <= LOW_THRESHOLD ~ "Low",
    income_gross >= HIGH_THRESHOLD ~ "High",
    TRUE ~ "Middle" ))
```

Before looking at the results, it is worth seeing what you have just done to the data. The two thresholds are simply two lines drawn across the distribution:

```
ggplot(frs_independent, aes(x = income_gross)) +
  geom_histogram(binwidth = 5000, fill = "grey80", color = "white") +
  geom_vline(xintercept = LOW_THRESHOLD, colour = "red", linewidth = 1) +
  geom_vline(xintercept = HIGH_THRESHOLD, colour = "red", linewidth = 1) +
  labs(
    title = "Your two thresholds, drawn on the distribution",
    subtitle = "Everything left of the first line is 'Low'. Everything right of the second is 'High'",
    x = "Gross Income (£)",
    y = "Frequency"
  ) +
  xlim(0, 90000) +
  theme_minimal()
```

Notice how much of the population falls between the lines, and how little falls beyond the right-hand one. Now change `LOW_THRESHOLD` to 15000, re-run both chunks, and watch the red line move and the “Low” group grow.

Task: There is no correct answer here, and that is the point. You chose those two numbers, and your choice decides who counts as poor and who counts as rich. If you recode a variable like this in your assignment, you must say what thresholds you used and why — the marking criteria ask for exactly that.

The `mutate()` function in R, from the **dplyr** package, is used to add or modify columns in a data frame. It allows you to create new variables or transform existing ones by applying calculations or conditional statements directly within the function.

Explanation of the code

- `frs_independent %>%`: The pipe operator `%>%` sends `frs_independent` into `mutate()`, allowing us to apply transformations without reassigning it repeatedly.
- `mutate()`: Starts the transformation process by defining new or modified columns.
- `income_category = case_when(...)`:
 - This creates a new column named `income_category`.
 - The `case_when()` function defines conditions for assigning values to this new column.
- `case_when()`:
 - `case_when()` is used here to assign categorical labels based on conditions.

- `income_gross <= LOW_THRESHOLD ~ "Low"`: If `income_gross` is less than or equal to `LOW_THRESHOLD`, `income_category` will be labeled “Low.”
- `income_gross >= HIGH_THRESHOLD ~ "High"`: If `income_gross` is greater than or equal to `HIGH_THRESHOLD`, `income_category` will be labeled “High.”
- `TRUE ~ "Middle"`: Any values not meeting the previous conditions are labeled “Middle.”

Task: Add some Metadata.

Define metadata for the new variable by labeling income categories.

```
# Add metadata by converting to a factor and defining labels

frs_independent$income_category <- factor(frs_independent$income_category,
                                           levels = c("Low", "Middle", "High"), labels = c("<= £10,000", "£10,001 - £49,999", ">= £50,000"))
```

Task: Check your work.

Examine the frequency distribution of the variable you have just created. Both variables should have the same number of missing cases, unless:

- Missing cases in the old variable have been intentionally converted into valid cases in the new variable.
- You forgot to allocate a new value to one of the old variable categories, in which case the new variable will have more missing cases than the old variable.

```
# Frequency distribution of income categories
table(frs_independent$income_category)
```

<= £10,000	£10,001 - £49,999	>= £50,000
8584	22981	2271

After preparing the data, use cross-tabulations to compare income levels across demographic groups.

```
# Cross-tabulate income category by age group, nationality, etc.
table(frs_independent$income_category, frs_independent$age_group)
```

	16-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59	60-64
<= £10,000	373	680	492	558	474	511	554	652	781	826
£10,001 - £49,999	263	1241	1802	2056	2052	1948	1995	1967	1749	1772
>= £50,000	1	8	59	186	314	331	334	356	237	177

	65-69	70-74	75+
<= £10,000	773	744	1166
£10,001 - £49,999	2073	1554	2509
>= £50,000	144	56	68

Explore income distribution across different regions.

```
# Cross-tabulate income category by region
table(frs_independent$income_category, frs_independent$region)
```

	East Midlands	East of England	London	North East	North West
<= £10,000	562	665	740	357	878
£10,001 - £49,999	1550	1855	1850	979	2347
>= £50,000	135	245	367	48	174

	Northern Ireland	Scotland	South East	South West	Wales
<= £10,000	874	1212	895	588	399
£10,001 - £49,999	2305	3234	2563	1707	971
>= £50,000	123	322	367	149	63

	West Midlands	Yorks and the Humber
<= £10,000	744	670
£10,001 - £49,999	1892	1728
>= £50,000	164	114

Tips for Cross-Tabulation

- Place the income variable in the columns.
- Add multiple variables in the rows to create simultaneous cross-tabulations.

2.3 Practice: From Sample to Population

Everything we have done so far describes **the people in our dataset**. But nobody actually cares about the 33,836 adults in the FRS file. We care about adults in the UK. The FRS is a

sample: the Department for Work and Pensions knocked on a set of randomly chosen doors, and these are the people who answered.

The key question: our sample says the mean gross income is about £22,544. If the DWP had knocked on a *different* set of doors, would they have got the same number?

2.3.1 Sampling Variability

Let's find out. For the next few minutes, pretend the 33,836 people in the FRS **are** the whole population. That lets us do something the DWP cannot: draw a small sample out of it, look at the answer, then put everybody back and draw another one.

```
mean(sample(frs_independent$income_gross, size = 500))
```

```
[1] 23852.61
```

Read that from the inside out: `sample(...)` picks 500 incomes at random, and `mean(...)` averages them.

Task: Run that line three or four more times. Write down the answer each time.

You will get a different number every time — usually somewhere between about £21,000 and £25,000, but essentially never exactly £22,544. Nobody made a mistake. This is **sampling variability**: the unavoidable wobble that comes from a sample being only a part of the whole.

Now let R do it a thousand times and collect all the answers:

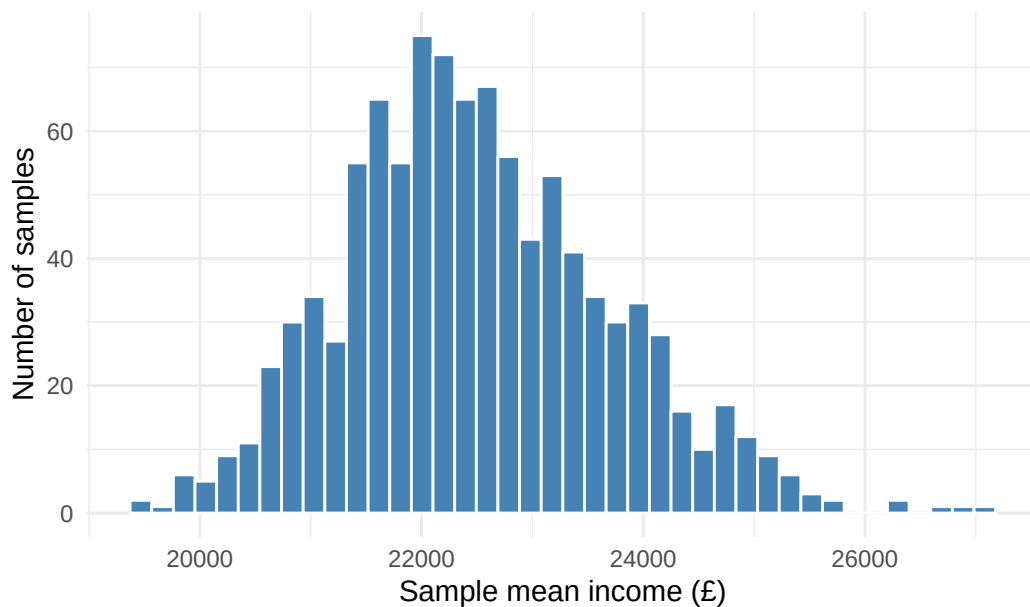
```
many_means <- replicate(1000, mean(sample(frs_independent$income_gross, size = 500)))
```

`replicate(1000, ...)` does what its name suggests: it runs whatever is in the brackets 1000 times over and keeps every result. `many_means` now holds 1000 numbers, each the average income of a different imaginary survey.

```
means_table <- data.frame(sample_mean = many_means)

ggplot(means_table, aes(x = sample_mean)) +
  geom_histogram(bins = 40, fill = "steelblue", color = "white") +
  labs(
    title = "1000 samples of 500 people: what mean income did each one report?",
    x = "Sample mean income (£)",
    y = "Number of samples"
  ) +
  theme_minimal()
```

1000 samples of 500 people: what mean income did each one have?



Two things are worth noticing. The histogram is centred on the true value of about £22,500 — samples are not systematically wrong, they are wrong in *both* directions roughly equally. And it is far narrower than the income distribution you plotted earlier: individual incomes ran from large losses to over a million, but *sample means* nearly all land within a few thousand pounds of each other.

2.3.2 The Standard Error

The width of that histogram is the thing we want, because it says how far a typical sample lands from the truth. It has a name: the **standard error**.

```
sd(many_means)
```

```
[1] 1196.565
```

About £1,200. In other words, a sample of 500 people typically misses the true mean income by around £1,200.

Be careful not to confuse this with the standard deviation of income, which was about £27,000. That one describes how much **people** differ from each other. The standard error describes how much **samples** differ from each other. Mixing up the two is one of the most common errors in student reports.

In real life you only ever have one sample, so you cannot build the histogram above. Happily you do not need to — the standard error can be worked out from your single sample, as the standard deviation divided by the square root of the sample size:

```
sd(frs_independent$income_gross) / sqrt(500)
```

```
[1] 1227.262
```

Compare that with the `sd(many_means)` value. They agree closely: the formula predicts the width of a histogram we never had to build.

2.3.3 What a P-value Means

From next week onwards, almost every result you produce will come with a **p-value** attached, so it is worth knowing now what one is.

Suppose we ask whether men and women have different incomes. In our data they clearly do:

```
frs_independent %>%  
  group_by(sex) %>%  
  summarise(mean_income = mean(income_gross, na.rm = TRUE), n = n())
```

```
# A tibble: 2 x 3  
  sex      mean_income      n  
  <chr>      <dbl> <int>  
1 Female    18134. 17858  
2 Male     27473. 15978
```

A gap of roughly £9,300. But we have just spent half an hour establishing that samples wobble, so before concluding anything about the UK we have to rule out the boring explanation: **could a gap this large have turned up purely through the luck of who got sampled, even if men and women earned the same across the country?**

That is the question a p-value answers, and it is the only question a p-value answers:

A p-value is: if there were genuinely no difference in the population, how often would sampling alone hand us a gap at least as large as the one we observed?

A small p-value means “hardly ever, so sampling luck is a poor explanation for what I am seeing”. A large p-value means “quite often, so I cannot rule out that this is just noise”. The statement being tested — that there is genuinely no difference in the population — is called the **null hypothesis**.

```
t.test(income_gross ~ sex, data = frs_independent)
```

Welch Two Sample t-test

```
data: income_gross by sex
t = -30.94, df = 26135, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Female and group Male is not 0
95 percent confidence interval:
 -9930.465 -8747.247
sample estimates:
mean in group Female    mean in group Male
      18134.42           27473.28
```

The squiggle `~` is read as “by” or “depends on”, so `income_gross ~ sex` means “income, broken down by sex”. You will meet this notation constantly from next week, because it is how R writes down a model.

The p-value is reported as `< 2.2e-16`, R’s way of writing “smaller than 0.00000000000000022”. Sampling variation is essentially impossible as an explanation for a gap this large in a sample this size, so we conclude that a gender income gap exists in the population — not just in our spreadsheet. By convention, a p-value below 0.05 is called **statistically significant**.

Two warnings to carry into your assignment:

- A p-value is **not** the probability that your finding is correct. It is calculated *assuming* there is no real difference, so it cannot tell you how likely that assumption is.
- A small p-value does **not** mean the effect is large or important. With 33,836 people, even a trivial £50 gap would come out “highly significant”. Always report the size of an effect alongside its p-value.

Going further (optional, in your own time): there is more to say about all of this — what a confidence interval really means, why “not significant” does not mean “no difference”, and why Census data needs handling differently. It is written up in [Samples, Uncertainty and P-values](#). You do not need it today, but it will help when you write your report.

3 Lab: Correlation, Simple, and Multiple Linear Regression

Open this lab in RStudio: in the Files pane, click labs and then 02.MultipleLinear.qmd. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk and adding your notes and answers in the file. (First time? See the [Setup](#) guide.)

Follow the practical below. You can describe what you are doing in normal text. See [here](#) for how to format normal text in Markdown documents.

In this week's practical, we will review how to calculate and visualise correlation coefficients between variables. This practical is split into four parts. The first focuses on measuring and visualising the relationship between **continuous variables**. The second goes through the implementation of a Linear Regression Model, again between continuous variables. The third checks whether the fitted model can be trusted, and the fourth is for practice.

Before getting into it, have a look at this [resource](#), it really helps understand how regression models work.

The lecture's slides can be found [here](#).

Before completing the practical, please take this [quiz](#).

Learning Objectives:

- Visualise the association between two continuous variables using a scatterplot.
- Measure the strength of the association between two variables by calculating their correlation coefficient.
- Build a formal regression model.
- Understand how to estimate and interpret a multiple linear regression model.
- Check whether a fitted model is trustworthy, using diagnostic plots.

Note on file paths: the Census data is already in your `data/Census2021` folder. As in the previous lab, the code reads it with `"../data/..."`: `..` tells R to go up from `labs` to the main folder (`stats/`) and then into `data/`.

3.1 Part I. Correlation

3.1.1 Data Overview: Descriptive Statistics

Let's start by picking **one dataset derived from the English-Wales 2021 Census data**. You can choose one dataset that aggregates data either at a) county, b) district, or c) ward-level. Lower Tier Local Authority-, Region-, and Country-level data is also available in the data folder.

see also: https://canvas.liverpool.ac.uk/courses/93312/pages/census-data?module_item_id=2727713

```
# Load necessary libraries
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
options(scipen = 999, digits = 4) # Avoid scientific notation and round to 4 decimals globally

# load data
census <- read.csv("../data/Census2021/EW_DistrictPercentages.csv") # District level
```

We're using a (district/ward/etc.) level census dataset that includes:

- % of population with poor health (variable name: `pct_Very_bad_health`).
- % of population with no qualifications (`pct_No_qualifications`).
- % of male population (`pct_Males`).
- % of population in a higher managerial/professional occupation (`pct_Higher_manager_prof`).

First, let's get some descriptive statistics that help identify general trends and distributions in the data.

```
# Summary statistics
summary_data <- census %>%
  select(pct_Very_bad_health, pct_No_qualifications, pct_Males, pct_Higher_manager_prof) %>%
  summarise_all(list(mean = mean, sd = sd))
summary_data
```

```
      pct_Very_bad_health_mean pct_No_qualifications_mean pct_Males_mean
1                1.173                17.9                48.97
      pct_Higher_manager_prof_mean pct_Very_bad_health_sd pct_No_qualifications_sd
1                13.22                0.3401                3.959
      pct_Males_sd pct_Higher_manager_prof_sd
1                0.6605                4.73
```

Q1. Complete the table below by specifying each variable type (continuous or categorical) and reporting its mean and standard deviation.

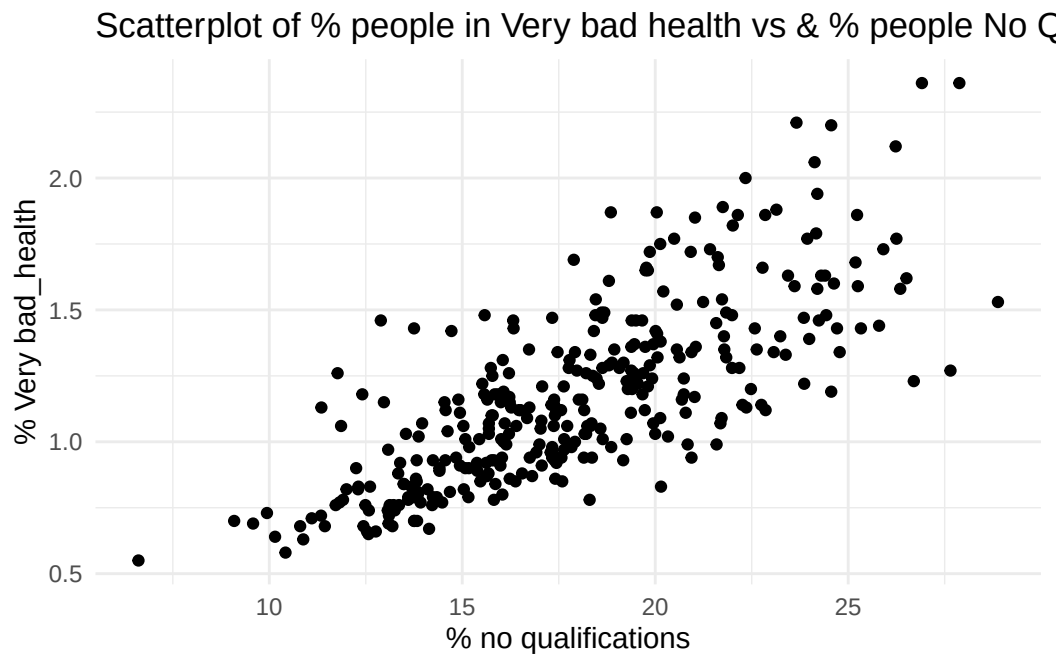
Variable Name	Type (Continuous or Categorical)	Mean	Standard Deviation
pct_Very_bad_health			
pct_No_qualifications			
pct_Males			
pct_Higher_manager_prof			

3.1.2 Simple visualisation for continuous data

You can visualise the relationship between two continuous variables using a scatter plot. Using the chosen census datasets, visualise the association between the % of population with bad health (`pct_Very_bad_health`) and each of the following:

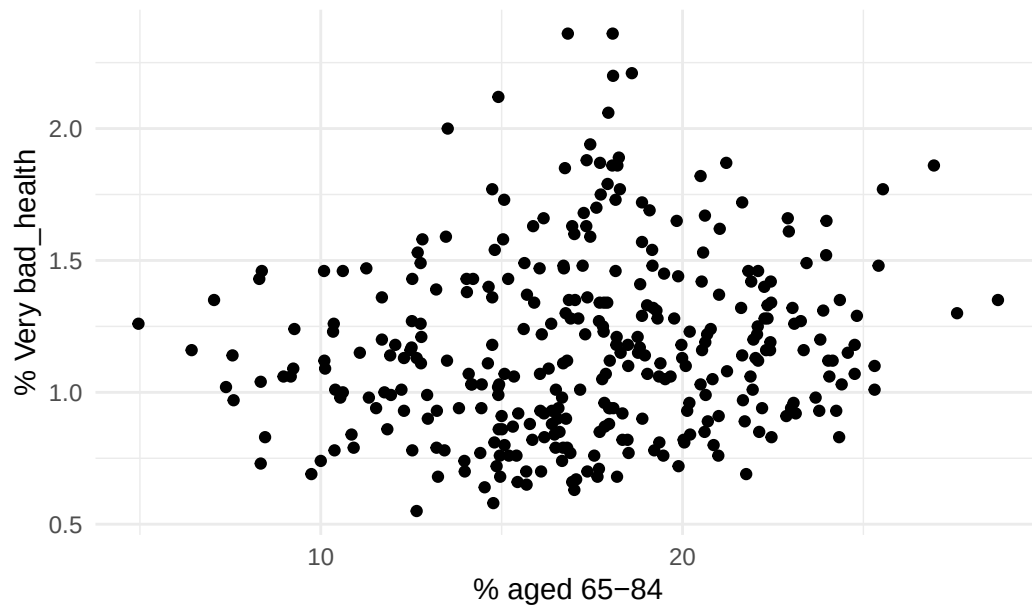
- the % of population with no qualifications (`pct_No_qualifications`);
- the % of population aged 65 to 84 (`pct_Age_65_to_84`);
- the % of population in a married couple (`pct_Married_opposite_sex_couple`);
- the % of population in a Higher Managerial or Professional occupation (`pct_Higher_manager_prof`).

```
# 1
ggplot(census, aes(x = pct_No_qualifications, y = pct_Very_bad_health)) +
  geom_point() +
  labs(title = paste("Scatterplot of % people in Very bad health vs & % people", "No Quali",
    x = "% no qualifications", y = "% Very bad_health") +
  theme_minimal()
```

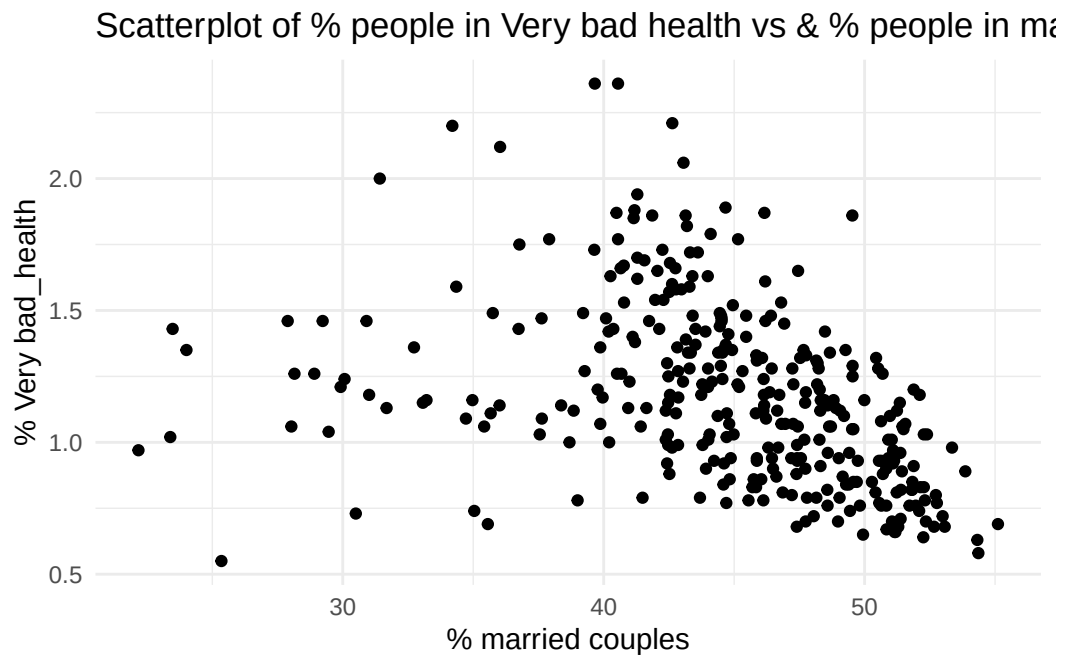


```
# 2
ggplot(census, aes(x = pct_Age_65_to_84, y = pct_Very_bad_health)) +
  geom_point() +
  labs(title = paste("Scatterplot of % people in Very bad health vs & % people", "aged betw",
    x = "% aged 65-84", y = "% Very bad_health") +
  theme_minimal()
```

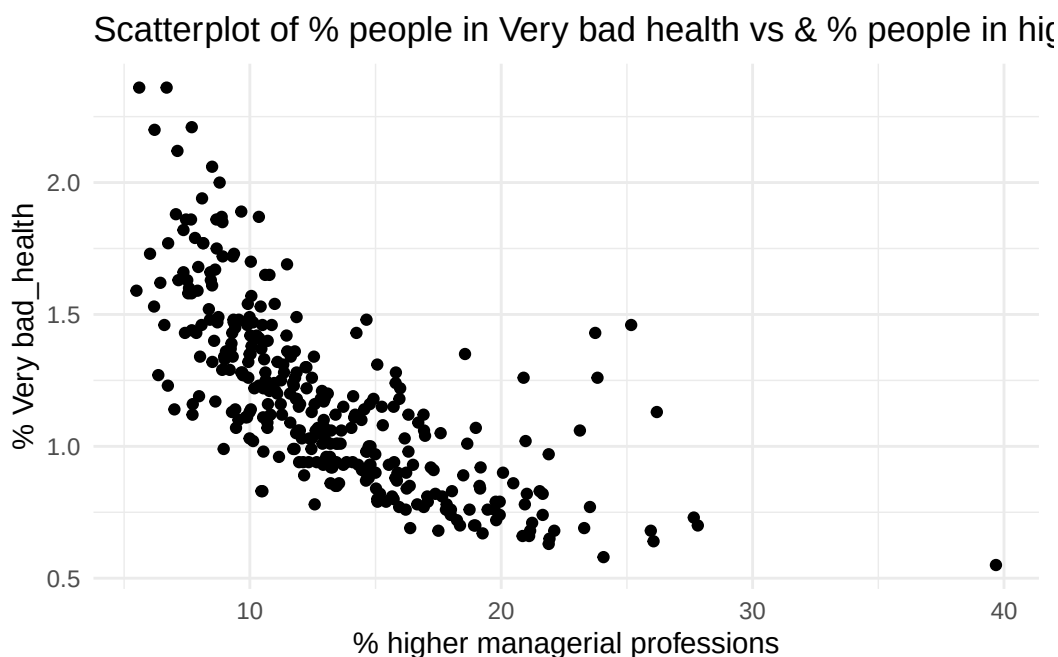
Scatterplot of % people in Very bad health vs & % people aged



```
# 3
ggplot(census, aes(x = pct_Married_opposite_sex_couple, y = pct_Very_bad_health)) +
  geom_point() +
  labs(title = paste("Scatterplot of % people in Very bad health vs & % people", "in marri",
    x = "% married couples", y = "% Very bad_health") +
  theme_minimal()
```



```
# 4
ggplot(census, aes(x = pct_Higher_manager_prof, y = pct_Very_bad_health)) +
  geom_point() +
  labs(title = paste("Scatterplot of % people in Very bad health vs & % people", "in higher",
    x = "% higher managerial professions", y = "% Very bad_health") +
  theme_minimal()
```



Q2. Which of the associations do you think is strongest, which one is the weakest?

As noted before, an observed association between two variables is no guarantee of causation. It could be that the observed association is:

- simply a chance one due to sampling uncertainty;
- caused by some third underlying variable which explains the spatial variation of both of the variables in the scatterplot;
- due to the inherent arbitrariness of the boundaries used to define the areas being analysed (the 'Modifiable Area Unit Problem').

Q3. Setting these caveats to one side, are the associations observed in the scatterplots suggestive of any causative mechanisms of bad health?

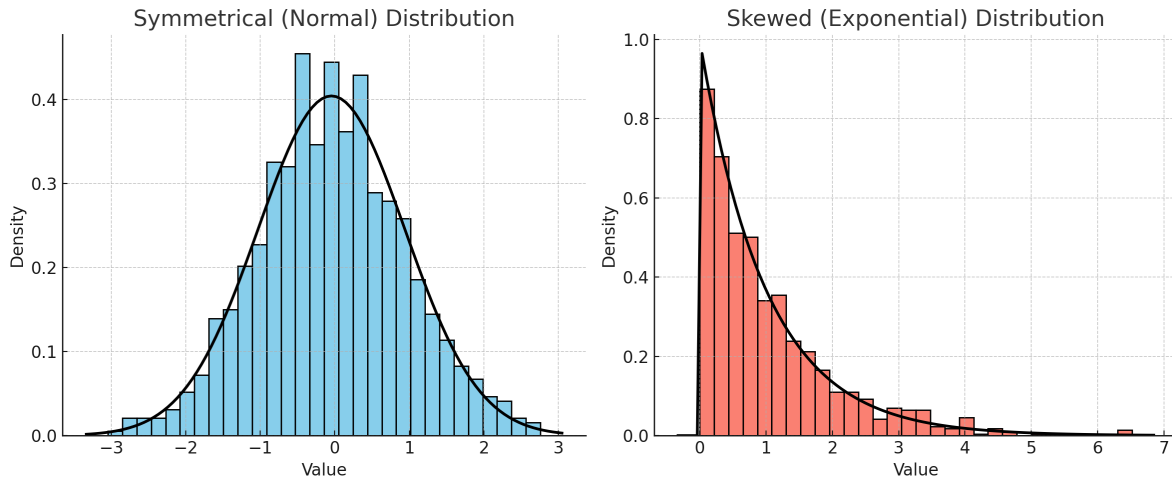
Rather than relying upon an impressionistic view of the strength of the association between two variables, we can measure that association by calculating the relevant correlation coefficient. The Table below identifies the statistically appropriate measure of correlation to use between two continuous variables.

Variable Data Type	Measure of Correlation	Range
Both symmetrically distributed	Pearson's	-1 to +1
One or both with a skewed distribution	Spearman's Rank	-1 to +1

The difference: Pearson's measures how well the two variables follow a *straight line*, and assumes both are reasonably symmetrically distributed. Spearman's works on ranks instead, so it copes with skewed variables and with relationships that are consistently increasing or decreasing without being straight.

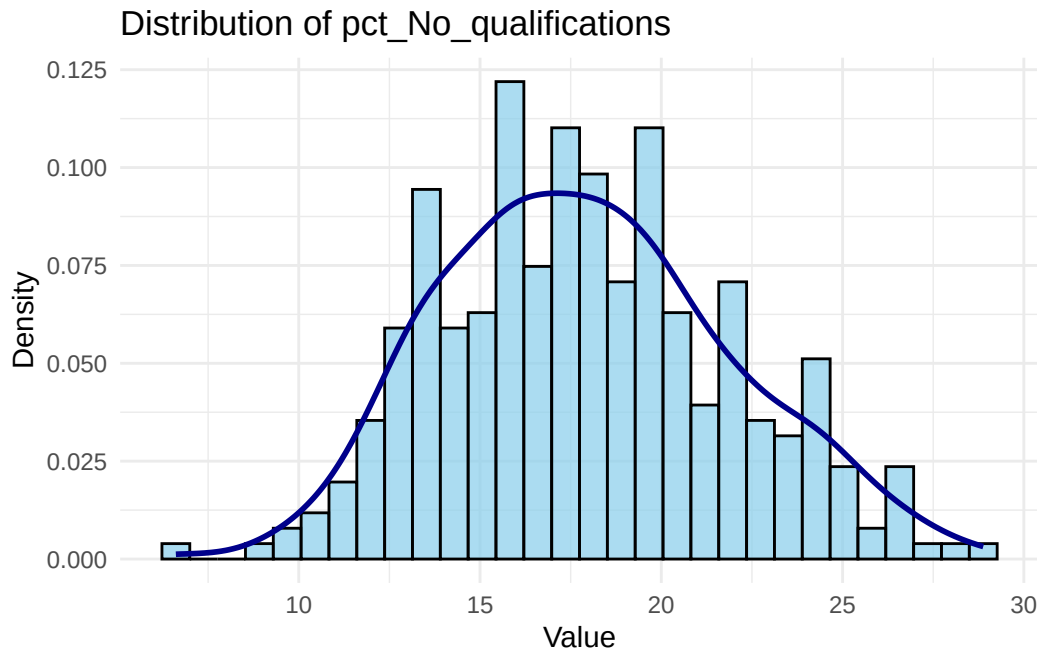
When calculating correlation for a single pair of variables, select the method that best fits their data distribution:

- Use **Pearson's** if both variables are symmetrically distributed.
- Use **Spearman's** if one or both variables are skewed.



You can check the distribution of a variable (e.g. `pct_No_qualifications` like this):

```
# Plot histogram with density overlay for a chosen variable (e.g., 'pct_No_qualifications')
ggplot(census, aes(x = pct_No_qualifications)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, color = "black", fill = "skyblue") +
  geom_density(color = "darkblue", linewidth = 1) +
  labs(title = "Distribution of pct_No_qualifications", x = "Value", y = "Density") +
  theme_minimal()
```



If you are comparing several pairs of variables, use the **same** measure for all of them, because Pearson and Spearman values are not directly comparable in size. Calculating both is a useful check: if the two broadly agree, your conclusion does not depend on which one you picked. In a report you would normally present just one, usually Pearson's.

Research Question 1: Which of our selected variables are most strongly correlated with % of population with bad health?

To answer this question, complete the Table below by editing/running this code:

Pearson correlations

```
pearson_correlation <- cor(census$pct_Very_bad_health,
  census$pct_No_qualifications, use = "complete.obs", method = "pearson")

# Display the results
cat("Pearson Correlation:", pearson_correlation, "\n")
```

Pearson Correlation: 0.7619

Spearman correlations:


```
spearman_correlation <- cor(census$pct_Very_bad_health,
  census$pct_No_qualifications, use = "complete.obs", method = "spearman")
cat("Spearman Correlation:", spearman_correlation, "\n")
```

Spearman Correlation: 0.7781

Covariates	Pearson	Spearman
pct_Very_bad_health - pct_No_qualifications		
pct_Very_bad_health - pct_Age_65_to_84		
pct_Very_bad_health - pct_Married_opposite_sex_couple		
pct_Very_bad_health - pct_Higher_manager_prof		

What can you make of these numbers?

If you think you have found a correlation between two variables in our dataset, this doesn't mean that the same association holds in general. As we saw last week, an association measured on a sample might not hold in the wider population, simply because of who happened to be sampled. (The Census data used here covers every district rather than a sample, so read the test below as asking whether a pattern this strong is likely to have arisen by chance.)

For this reason, we need to verify whether the correlation is statistically significant,

```
# significance test for pearson, for example
pearson_test <- cor.test(census$pct_Very_bad_health,
  census$pct_No_qualifications, method = "pearson", use = "complete.obs")
pearson_test
```

Pearson's product-moment correlation

```
data: census$pct_Very_bad_health and census$pct_No_qualifications
t = 21, df = 329, p-value <0.0000000000000002
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7127 0.8037
sample estimates:
 cor
0.7619
```

Look at <https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/cor.test> for details about the function. But in general, when calculating the correlation between two variables, a **p-value** accompanies the correlation coefficient to indicate the statistical significance of the observed association. This p-value tests the null hypothesis that there is no association between the two variables (i.e., that the correlation is zero).

As we saw last week, the p-value answers one specific question: *if there were genuinely no association between these two variables in the population, how often would sampling alone produce a correlation at least as strong as the one we observed?* A small p-value means that sampling luck is a poor explanation for what we are seeing, and we reject the null hypothesis.

By convention, a p-value below 0.05 is called statistically significant. Sometimes, in research papers or tables, significance levels are denoted with asterisks: one asterisk (*) indicates $p < 0.05$, two asterisks (**) indicate $p < 0.01$, and three asterisks (***) indicate $p < 0.001$.

Two reminders from last week, because both are easy to get wrong in a report (there is more on all of this in [Samples, Uncertainty and P-values](#), which is worth reading before you write up):

- A p-value is **not** the probability that the association is real, and *** does not mean “99.9% certain there is a relationship”. The p-value is calculated *assuming* there is no association, so it cannot tell you how likely that assumption is.
- A p-value tells you whether an association is **detectable**, not whether it is **large**. With enough observations, a negligible correlation will still come out significant. Always report the size of the correlation alongside its p-value, and interpret the size.
- A p-value says nothing about **causation**. It cannot tell a real causal relationship apart from a confounded one; that judgement comes from your literature review and research design, never from the output.

3.2 Part II. Implementing a Linear Regression Model

A key goal of data analysis is to explore the potential factors of health at the local district level. So far, we have used cross-tabulations and various bivariate correlation analysis methods to explore the relationships between variables. One key limitation of standard correlation analysis is that it remains hard to look at the associations of an outcome/dependent variable to multiple independent/explanatory variables at the same time. Regression analysis provides a very useful and flexible methodological framework for such a purpose. Therefore, we will investigate how various local factors impact residents’ health by building a multiple linear regression model in R.

We use `pct_Very_bad_health` as a proxy for residents’ health.

Research Question 2: How do local factors affect residents’ health?

Dependent (or Response) Variable:

- % of population with bad health (`pct_Very_bad_health`).

Independent (or Explanatory) Variables:

- % of population with no qualifications (`pct_No_qualifications`).
- % of male population (`pct_Males`).
- % of population in a higher managerial/professional occupation (`pct_Higher_manager_prof`).

Load some other Libraries

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats   1.0.0      v stringr   1.5.1
v lubridate 1.9.3      v tibble    3.2.1
v purrr     1.0.2      v tidyr     1.3.1
v readr     2.1.5
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(broom)
```

and the data (if not loaded):

```
# Load dataset
census <- read.csv("../data/Census2021/EW_DistrictPercentages.csv")
```

A regression model estimates how changes in the **dependent** variable Y are associated with changes in the **independent** variables X_1, X_2, X_3 . It does this by finding the line (or, with several predictors, the flat surface) that comes closest to all the observed points at once — “closest” meaning it makes the gaps between the observed values of Y and the values the model predicts as small as possible. This is why it is called *Ordinary Least Squares* (OLS) regression.

You met **Simple Linear Regression** in class, where one independent variable explains an outcome. Real-world phenomena usually have several causes at once, and multiple linear regression simply extends the same idea to several independent variables together.

Here, regression allows us to examine the relationship between people’s health rates and multiple independent variables.

Before starting, we define two hypotheses:

- **Null hypothesis** (H_0): For each variable X_n , there is no effect of X_n on Y .
- **Alternative hypothesis** (H_1): There is an effect of X_n on Y .

We will test if we can reject the null hypothesis.

3.2.1 Fitting the Model

```
# Linear regression model
model <- lm(pct_Very_bad_health ~ pct_No_qualifications + pct_Males + pct_Higher_manager_prof)
summary(model)
```

Call:

```
lm(formula = pct_Very_bad_health ~ pct_No_qualifications + pct_Males +
    pct_Higher_manager_prof, data = census)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.4911	-0.1357	-0.0368	0.0985	0.7669

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.00293	0.87981	4.55	0.0000076	***
pct_No_qualifications	0.05283	0.00591	8.94	< 0.00000000000000002	***
pct_Males	-0.07353	0.01785	-4.12	0.0000479	***
pct_Higher_manager_prof	-0.01318	0.00494	-2.67	0.008	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.213 on 327 degrees of freedom

Multiple R-squared: 0.61, Adjusted R-squared: 0.607

F-statistic: 171 on 3 and 327 DF, p-value: <0.00000000000000002

Code explanation

- **lm()** stands for “linear model”. The formula reads as “very bad health, explained by no qualifications, males and higher managerial/professional”: the dependent variable goes on the left of the `~`, and the independent variables on the right, joined by `+`.
- **model <-** stores the fitted model in an object called **model**, so we can ask it questions afterwards.

- `summary(model)` prints the results. It produces a lot at once, so we will take it apart one piece at a time below.

We can focus only on certain output metrics:

```
# Regression coefficients
coefficients <- tidy(model)
coefficients
```

```
# A tibble: 4 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)          4.00      0.880      4.55 7.58e- 6
2 pct_No_qualifications 0.0528    0.00591     8.94 2.99e-17
3 pct_Males            -0.0735    0.0178    -4.12 4.79e- 5
4 pct_Higher_manager_prof -0.0132    0.00494    -2.67 7.97e- 3
```

These are:

- **Regression Coefficient Estimates.**
- **P-values.**
- **Adjusted R-squared.**

3.2.2 How to interpret the output metrics

3.2.2.1 Regression Coefficient Estimates

The **Estimate** column tells us how Y changes as each independent variable changes.

Intercept (β_0): the value of Y when every independent variable is zero.

Slopes ($\beta_1, \beta_2, \dots$): the average change in Y for a one unit increase in that independent variable, *while the others stay where they are*. Two things to watch:

- **Know your units.** One “unit” could be a year for an age variable, or one percentage point for a percentage — which is the case for every variable this week.
- **“Holding the others constant”** is what makes multiple regression worth doing: each coefficient is the effect of that variable over and above whatever the other variables in the model already account for.

So for our model:

- The association of `pct_No_qualifications` is positive and strong: each increase of 1 percentage point in `pct_No_qualifications` is associated with an increase of 0.05 percentage points in the very bad health rate.
- The association of `pct_Males` is negative and strong: each increase of 1 percentage point in `pct_Males` is associated with a decrease of 0.07 percentage points in `pct_Very_bad_health` across England and Wales.
- The association of `pct_Higher_manager_prof` is negative but weak: each increase of 1 percentage point in `pct_Higher_manager_prof` is associated with a decrease of 0.013 percentage points in `pct_Very_bad_health`.

3.2.2.2 P-values and Significance

Each coefficient gets its own p-value, reported in the column `Pr(>|t|)` with the same asterisk conventions as in Part I. It asks the same question we met last week: *if X_n had no effect on Y in the population, how often would chance alone produce a coefficient at least as far from zero as the one we estimated?* Very small p-values mean chance is a poor explanation for the estimated relationship, so we reject H_0 for that variable.

In this case, we can say:

- Given that the p-value is indicated by *******, changes in `pct_No_qualifications` and `pct_Males` are significantly associated with changes in `pct_Very_bad_health` at the <0.001 level; the association is highly statistically significant; we can be confident that the observed relationship between these variables and `pct_Very_bad_health` is not due to chance.
- Given that the p-value is indicated by ******, changes in `pct_Higher_manager_prof` are significantly associated with changes in `pct_Very_bad_health` at the <0.01 level. Chance remains an unlikely explanation for this coefficient, though the evidence is weaker than for the other two variables.

In all three cases we can reject the **Null** hypothesis (H_0 : there is no association between the independent variable and Y).

Remember, if the *p-value* of a coefficient is smaller than 0.05, that coefficient is conventionally called statistically significant. If the *p-value* is larger than 0.05, you have **no evidence** of an association, which is not at all the same as evidence that there is **no** association. A real relationship can easily fail to reach significance when the sample is small or the data are noisy (there is a worked example of exactly this in [Samples, Uncertainty and P-values](#)). Phrase this carefully in your report: “no significant association was found” is defensible, “there is no association” is not.

Note also that significance and importance are separate questions. `pct_Higher_manager_prof` is statistically significant, but its coefficient of -0.013 means that a 1 percentage point change

in it is associated with a change of roughly one hundredth of a percentage point in very bad health. Report the size of an effect, not just its stars.

3.2.2.3 R-squared and Adjusted R-squared

R-squared tells you what share of the variation in Y your independent variables account for. An R-squared of 0.6 means the model explains 60% of the variation in Y ; the remaining 40% is down to things the model does not contain.

Adjusted R-squared does the same job, but penalises you for adding variables. This matters because plain R-squared can only ever go up when you add a predictor, even a useless one, so it is the adjusted version you should compare between models and the adjusted version you should report.

There is no threshold that makes a model “good”. What counts as a respectable R-squared depends entirely on what you are modelling: area-level Census data often gives high values because districts are averages of thousands of people, while a model of individual survey responses might explain 10% and still be a perfectly sound piece of work. Do not chase the number, and do not dismiss a model for having a low one — explain what it means in your context.

3.2.3 Interpreting the Results

`coefficients`

```
# A tibble: 4 x 5
  term                estimate std.error statistic  p.value
<chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         4.00      0.880      4.55 7.58e- 6
2 pct_No_qualifications 0.0528    0.00591     8.94 2.99e-17
3 pct_Males          -0.0735    0.0178    -4.12 4.79e- 5
4 pct_Higher_manager_prof -0.0132    0.00494    -2.67 7.97e- 3
```

Q4. Complete the table above by filling in the coefficients, t-values, p-values, and indicating if each variable is statistically significant.

Variable Name	Coefficients	t-values	p-values	Significant?
pct_No_qualifications				
pct_Males				
pct_Higher_manager_prof				

From the lecture notes, you know that the Intercept or Constant represents the estimated average value of the outcome variable when the values of all independent variables are equal to zero.

Q5. When values of `pct_Males`, `pct_No_qualifications` and `pct_Higher_manager_prof` are all *zero*, what is the % of population with very bad health? Is the intercept term meaningful? Are there any districts (or zones, depending on the dataset you chose) with zero percentages of persons with no qualification in your data set?

Q6. Interpret the regression coefficients of `pct_Males`, `pct_No_qualifications` and `pct_Higher_manager_prof`. Do they make sense?

3.2.4 Identify factors of % bad health

Now combine the above two sections and identify factors affecting the percentage of population with very bad health. Fill in each row for the direction (positive or negative) and significance level of each variable.

Variable Name	Positive or Negative	Statistical Significance
<code>pct_No_qualifications</code>		
<code>pct_Higher_manager_prof</code>		
<code>pct_Males</code>		

Q7. Think about the potential conclusions that can be drawn from the above analyses. Try to answer the research question of this practical: How do local factors affect residents' health? Think about causation *vs* association and consider potential confounders when interpreting the results. How could these findings influence local health policies?

3.3 Part III. Checking the Model

We have a model, an R-squared of 0.61, and three significant coefficients. It is tempting to stop here and start writing. Do not.

R-squared and p-values tell you how well the model fits **given that a straight line was the right thing to fit in the first place**. They cannot tell you whether that assumption was sensible. Checking the model is what separates a report in the 50s from one in the 60s and 70s, and the marking criteria explicitly reward it.

Here is a thirty-second demonstration of the problem. Let's invent some data where the relationship between *x* and *y* is definitely *not* a straight line, and fit a straight line to it anyway:


```
# invent 100 observations where y follows a curve, not a line
made_up <- data.frame(x = 1:100)
made_up$y <- made_up$x^2 + rnorm(100, sd = 500)

bad_model <- lm(y ~ x, data = made_up)
summary(bad_model)
```

Call:

```
lm(formula = y ~ x, data = made_up)
```

Residuals:

Min	1Q	Median	3Q	Max
-2132.1	-706.2	69.7	696.2	2211.7

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1591.97	193.07	-8.25	0.000000000000076 ***
x	100.14	3.32	30.17	< 0.0000000000000002 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 958 on 98 degrees of freedom

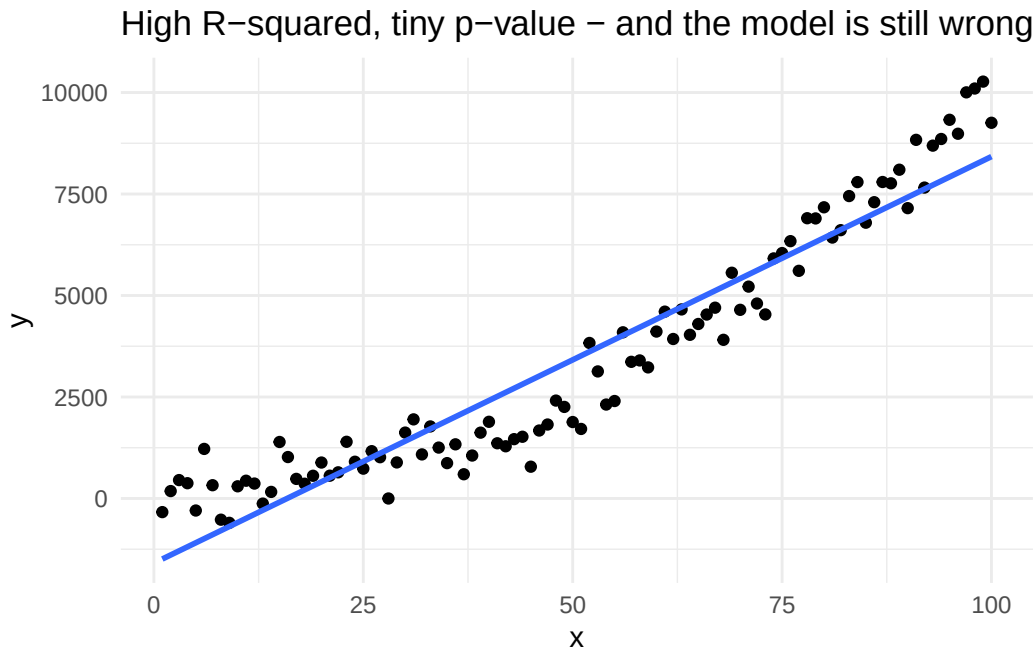
Multiple R-squared: 0.903, Adjusted R-squared: 0.902

F-statistic: 910 on 1 and 98 DF, p-value: <0.0000000000000002

Look at the output: an R-squared of around 0.9, and a p-value on `x` too small for R to print. (Your figure will differ a little from the one on this page, and will change again each time you run the chunk, because the data is invented fresh each time.) On the numbers alone this is a triumph. But we built the data ourselves, so we know for a fact that the straight line is the wrong shape. Plot it and you can see the problem immediately:

```
ggplot(made_up, aes(x = x, y = y)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "High R-squared, tiny p-value - and the model is still wrong") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



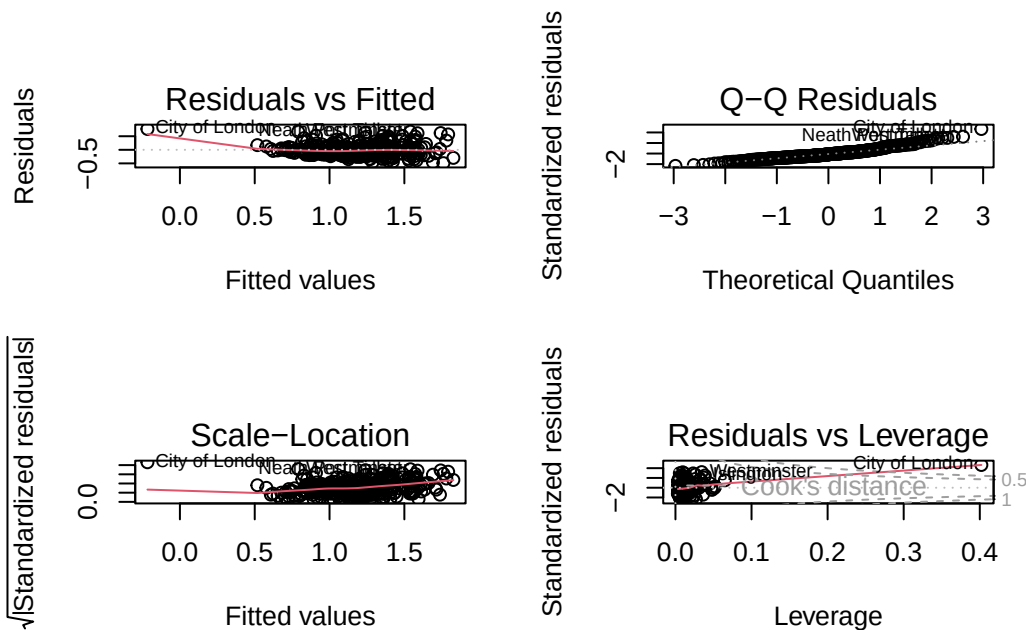
The line is too high at both ends and too low in the middle, and it will be wrong in the same direction for the same observations every time. **No number in the `summary()` output warned us about this.** Only the picture did.

That is the whole argument for what follows. With one predictor you could simply look at a scatterplot, as we did in Part I. With three predictors you cannot — there is no way to draw it on a page — so R gives us a different set of pictures instead.

3.3.1 The Four Diagnostic Plots

Calling `plot()` on a fitted model produces four diagnostic charts:

```
par(mfrow = c(2, 2)) # arrange the four plots in a 2x2 grid
plot(model, labels.id = census$District)
```



```
par(mfrow = c(1, 1)) # reset the plotting area afterwards
```

Code explanation

- `par(mfrow = c(2, 2))`: splits the plotting window into a 2-by-2 grid so all four charts appear together.
- `plot(model)`: applied to a regression model, this produces the standard diagnostic charts rather than a scatterplot.
- `labels.id = census$District`: labels unusual points with district names instead of row numbers, which is far more useful to a geographer.

Each panel answers a different question. You are reading these **visually**, looking for obvious patterns, not running further tests:

1. **Residuals vs Fitted**. Is a straight line the right shape? The residuals should scatter randomly around the horizontal line at zero. A clear curve means the relationship is not linear and your model is systematically wrong for some districts. A funnel or fan shape means the model predicts some districts far more precisely than others (*heteroscedasticity*), which makes the standard errors, and therefore the p-values, unreliable.
2. **Q-Q Residuals**. Are the residuals roughly normally distributed? Points should sit close to the diagonal line. Bending away at the ends usually signals a few extreme districts rather than a fatal problem.

3. **Scale-Location.** Another view of the same question as panel 1: the line should be roughly flat. A rising line confirms that the spread of errors grows with the predicted value.
4. **Residuals vs Leverage.** Which districts are unusual enough to be pulling the line around on their own? R labels the most extreme ones by name. If a single district sits far out on its own here, it is worth asking whether it belongs in the analysis at all — the City of London, for example, is a financial district with only about 8,500 residents, which makes it a very odd “place” to include alongside ordinary local authorities.

Q8. Look at each of the four panels in turn. Is there obvious curvature in the first panel? Is there a fan shape? Which districts are labelled as unusual, and does the fourth panel single any one of them out?

3.3.2 What to Report in Your Assignment

You do not need to include all four plots in your report, and you should not pad your word count with diagnostic output. What earns marks is a short, specific paragraph in your Methods or Results section showing that you looked. For example:

Diagnostic plots showed no substantial departure from linearity, though the residuals fanned slightly for districts with higher predicted values, suggesting the model predicts some districts less precisely than others. The City of London was flagged as an unusually influential observation; given its atypical residential population of roughly 8,500, its inclusion alongside ordinary local authorities should be treated with caution.

Two sentences, and it demonstrates exactly the critical engagement the higher marking bands ask for.

3.4 Part IV. Practice and Extension

If you haven’t understood something, if you have doubts, even if they seem silly, ask.

1. Finish working through the practical.
2. Revise the material.
3. Extension activities (optional): Think about other potential factors of very bad health and test your ideas with new linear regression models. Run `plot()` on them too, and see whether the same districts keep being flagged as unusual.

4 Lab: Correlation and Multiple Linear Regression with Qualitative Variables

Open this lab in RStudio: in the Files pane, click labs and then 03.QualitativeVariable.qmd. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk and adding your notes and answers in the file. (First time? See the [Setup](#) guide.)

The lecture's slides can be found [here](#).

Last week, we introduced Multiple Linear Regression (MLR) - a statistical method that models the relationship between a dependent variable and two or more independent variables, allowing researchers to examine how various predictors jointly influence an outcome. By using the following R code, we create and interpret the model:

```
model <- lm(pct_Very_bad_health ~ pct_No_qualifications + pct_Males + pct_Higher_manager_pro  
data = census)  
  
summary(model)
```

In a regression model, independent/predictor variables could be continuous or categorical (or qualitative). While continuous variables capture quantitative effects, **categorical variables provide insights into differences across groups**. When we say categorical variables, we normally mean:

- Nominal Data: categorical data without natural order. E.g. Gender, Colour, Country...
- Ordinal Data: categorical data with a meaningful order. E.g. Education level, Customer satisfaction, Grade...

By blending continuous and categorical predictors, MLR with categorical variables enhances the model's ability to reflect real-world complexities and improves interpretability, as it allows analysts to assess how each category or group within an independent variable influences the dependent variable.

For most categorical (especially the *nominal*) variables, they cannot be included in the regression model directly as a continuous independent variable. Instead, these qualitative independent variables should be included in regression models by using the **dummy variable** approach, transforming categorical information into a numerical format suitable for regression analysis.

However, R provides a powerful shortcut, by automatically handling this process when the categorical variable is designated as a factor and included in the regression model. This makes it much easier for you to use categorical variables in the regression model to assess the effects of categorical groupings on the dependent variable alongside continuous predictors.

Learning Objectives:

In this week's practical we are going to

- Analyse categorical/qualitative variables
- Estimate and interpret a multiple linear regression model with categorical variables
- Make predictions using a regression model

4.1 Analysing categorical variables

Recall that in Week 2 you became familiar with R by using the Census data. Today we will explore both the Family Resource Survey (FRS) and the Census data by using their categorical variables. The Census data is already in your 'data/Census2021/' folder, and you uploaded 'FRS16-17_labels.csv' to 'data/FRS/' in Week 1. If it is not there, download it from [Canvas](#) and upload it to 'data/FRS/' (use the **Upload** button in the Files pane: see [Setup](#), section 5). From the same Canvas page, also download 'FRS_dictionary.xlsx' to your computer: it explains what each variable means.

To start today's practical session, we first will use 'FRS16-17_labels.csv'. You can open the .csv file in Excel and see that, unlike the Census dataset, the FRS dataset contains many qualitative/categorical variables, such as hh_tenure (Housing Tenure), happy (How happy did you feel yesterday?), health (How is your health in general), etc. To know the meaning of all the column names and the values in cell, you may need to refer to the 'FRS_dictionary.xlsx' to help your understanding. Recall from our lecture that these categorical variables will need to be treated as dummy variables in the regression model. In R, the variable type of qualitative/categorical variables is called '**factor**'.

As usual we first load the necessary libraries.

If R returns a “cannot open file” error, check in the **Files** pane that the file is really in the folder named in the code (e.g. `data/FRS/`).

Libraries used today:

- **dplyr**: a basic library provides a suite of functions for data manipulation
- **ggplot2**: a widely-used data visualisation library to help you create nice plots through layered plotting.

- **tidyverse**: a collection of R packages designed for data science, offering a cohesive framework for data manipulation, visualization, and analysis. Containing dplyr, ggplot2 and other basic libraries.
- **broom**: a part of the tidyverse and is designed to convert statistical analysis results into tidy data frames.
- **forcats**: designed to work with factors, which are used to represent categorical data. It simplifies the process of creating, modifying, and ordering factors.
- **vcd**: visualise and analyse categorical data.

A useful shortcut to format your code: select all your code lines, use Ctrl+Shift+A to automatically format them in a tidy way.

4.1.1 Data overview

All these packages are already installed in your codespace, so we just load them:

```
# Load necessary libraries
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(broom)
```

Or we can use library tidyverse which includes ggplot2, dplyr and other fundamental libraries together already.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.0      v stringr 1.5.1
v lubridate 1.9.3    v tibble 3.2.1
v purrr 1.0.2       v tidyr 1.3.1
v readr 2.1.5

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

We will also use the `forcats` library, so

```
library(forcats)
```

Exactly as you did in previous weeks, we first load in the dataset:

```
frs_data <- read.csv("../data/FRS/FRS16-17_labels.csv")
```

Recall that in previous weeks we used the following code to get an overview of the dataset. Familiarise yourself with it again:

```
View(frs_data)
```

and also `summary()` to produce summaries of each variable

```
summary(frs_data)
```

You may notice that for the numeric variables such as *hh_income_gross* (household gross income) and *work_hours* (worked hours per week), the `summary()` offers useful descriptive statistics. While for the qualitative information, such as *age_group* (age group), *highest_qual* (Highest educational qualification), *marital_status* (Marital status) and *nssec* (Socio-economic status), the `summary()` function is not as useful, since reporting a mean or median makes little sense for categories.

When performing descriptive analysis for categorical or qualitative variables, we focus on summarising the frequency and distribution of categories within the variable. This analysis helps understand the composition and diversity of categories in the data, which is especially useful for identifying patterns, common categories, or potential data imbalances.


```
# Frequency count
table(frs_data$age_group)
```

```

  0-4  05-10  11-15  16-19  20-24  25-29  30-34  35-39  40-44  45-49  50-54  55-59  60-64
2914  3575  2599  1858  1929  2353  2800  2840  2790  2883  2975  2767  2775
65-69 70-74  75+
2990  2354  3743
```

```
table(frs_data$highest_qual)
```

```

A-level or equivalent      Degree or above      Dependent child
              5260                  9156              10298
GCSE or equivalent        Not known              Other
              9729                  6820              2882
```

```
table(frs_data$marital_status)
```

```

Cohabiting Divorced/civil partnership dissolved
              4015                  2199
Married/Civil partnership      Separated
              18195              747
Single      Widowed
              16663              2326
```

```
table(frs_data$nssec)
```

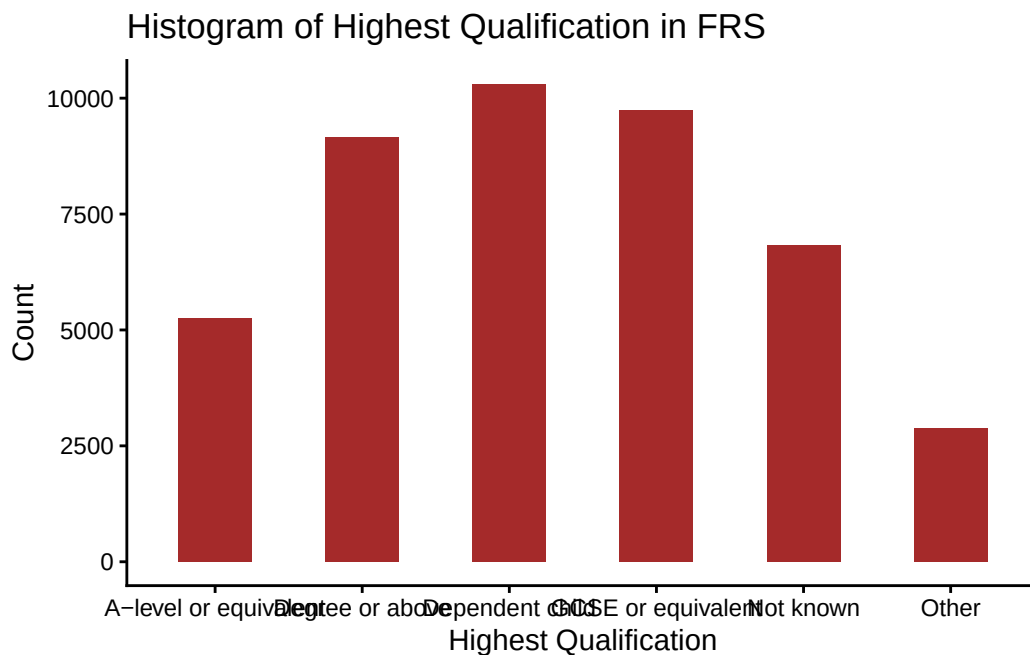
```

Dependent child
              10299
Full-time student
              963
Higher professional occupations
              3004
Intermediate occupations
              4372
Large employers and higher managers
```

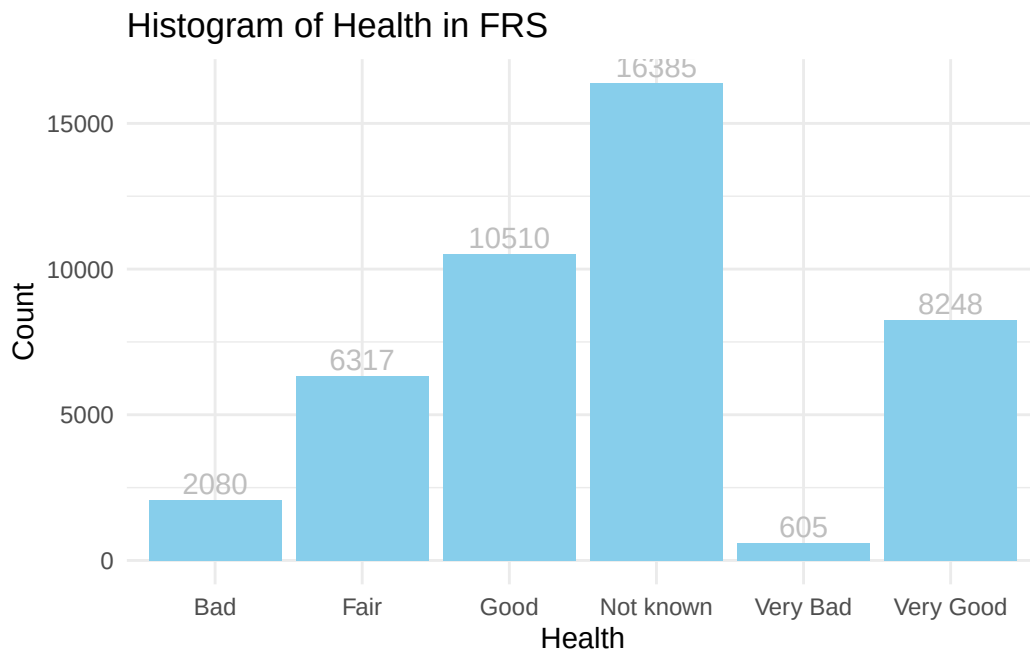
Lower managerial and professional occupations	1025
Lower supervisory and technical occupations	8129
Never worked or long-term unemployed	2400
Not classifiable	1516
Routine occupations	107
Semi-routine occupations	4205
Small employers and own account workers	5226
	2899

By using ggplot2, it is easy to create some nice descriptive charts for the categorical variables, just like you did for the continuous variables last week.

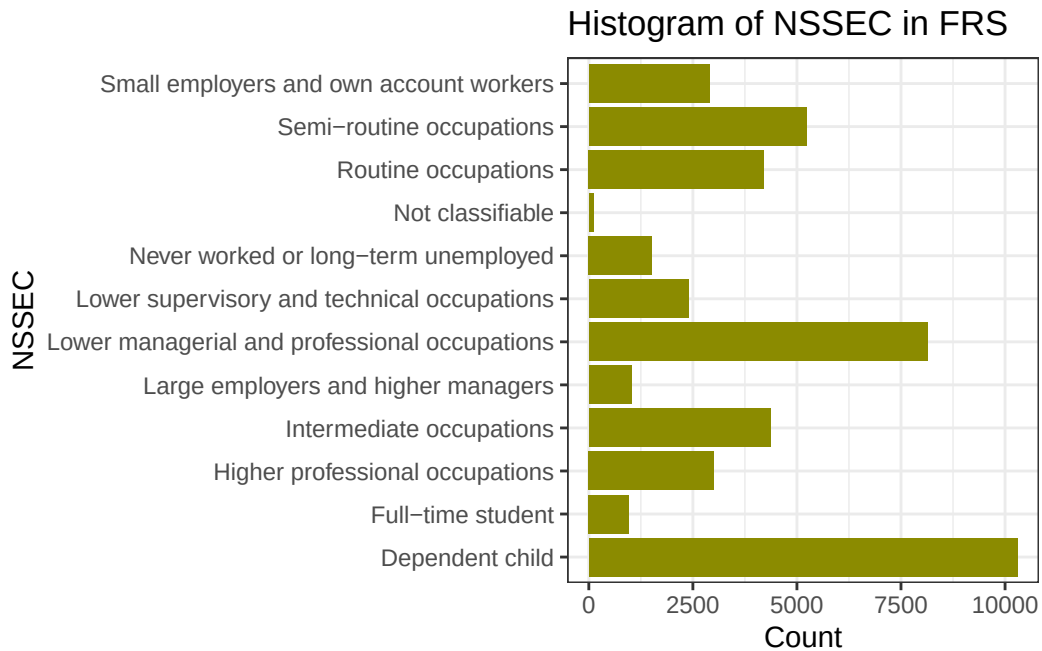
```
ggplot(frs_data, aes(x = highest_qual)) +
  geom_bar(fill="brown",width=0.5) +
  labs(title = "Histogram of Highest Qualification in FRS", x = "Highest Qualification", y =
  theme_classic())#choose theme type, try theme_bw(), theme_minimal() see differences
```



```
ggplot(frs_data, aes(x = health)) +
  geom_bar(fill="skyblue") +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.3, colour = "grey")+ #add text
  labs(title = "Histogram of Health in FRS", x = "Health", y = "Count")+ #set text info
  theme_minimal()
```

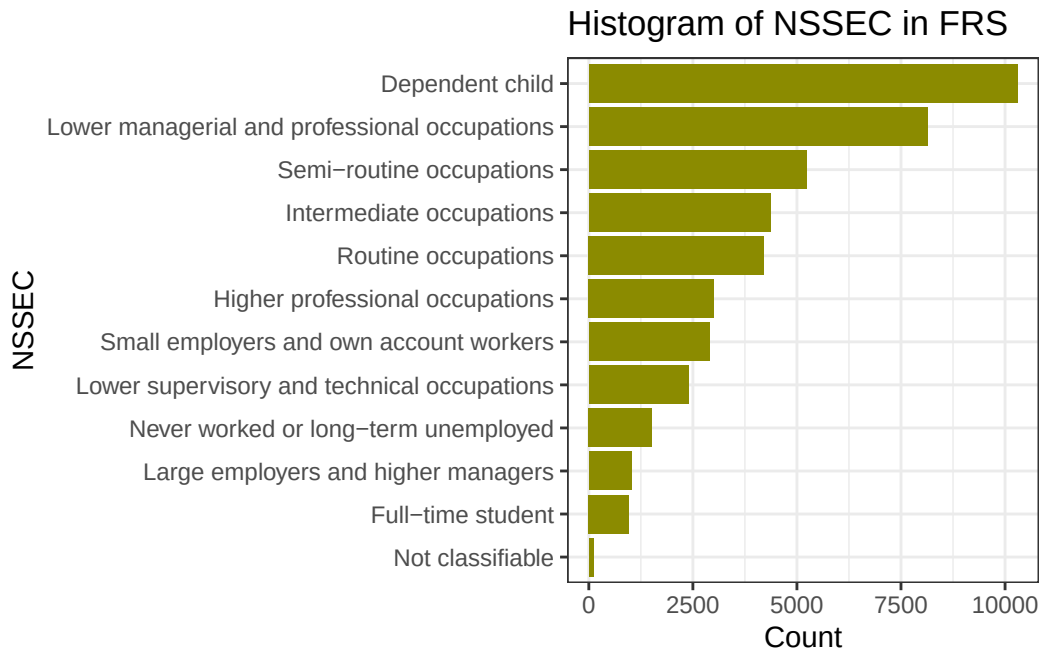


```
ggplot(frs_data, aes(x = nssec)) +
  geom_bar(fill = "yellow4") +
  labs(title = "Histogram of NSSEC in FRS", x = "NSSEC", y = "Count") +
  coord_flip()+ #Flip the Axes, add a # in front of this line, to make the code in gray and y
  theme_bw()
```



If we want to reorder the Y axis by from highest to lowest, we use the functions in `forcats` library. `fct_infreq()`: orders by the value's frequency of the variable `nssec`. `fct_rev()`: reverses the order to go from highest to lowest.

```
ggplot(frs_data, aes(x = fct_rev(fct_infreq(nssec)))) +
  geom_bar(fill = "yellow4") +
  labs(title = "Histogram of NSSEC in FRS", x = "NSSEC", y = "Count") +
  coord_flip()+ #Flip the Axes, add a # in front of this line, to make the code in gray and y
  theme_bw()
```



You can change the variables in `ggplot()` to make your own histogram chart for the variables you are interested in. You will learn more of visualisation methods in Week 5's practical.

4.1.2 Correlation

As before, rather than relying upon an impressionistic view of the strength of the association between two variables, we can measure that association by calculating the relevant correlation coefficients.

To calculate the correlation between categorical data, we first use Chi-squared test to assess the independence between pairs of categorical variables, then we use Cramér's V to measure the strength of association - the correlation coefficient in R.

Pearson's chi-squared test (2) is a statistical test applied to sets of categorical data to evaluate how likely it is that any observed difference between the sets arose by chance. If the p-value is low (typically < 0.05), it suggests a significant association between the two variables.

```
chisq.test(frs_data$health, frs_data$happy)
```

```
Warning in chisq.test(frs_data$health, frs_data$happy): Chi-squared
approximation may be incorrect
```

Pearson's Chi-squared test

```
data: frs_data$health and frs_data$happy  
X-squared = 45594, df = 60, p-value < 2.2e-16
```

If you see a warning message of Chi-squared approximation may be incorrect. This is because some expected frequencies in one or more cells of the cross-tabulation (health * happy) are too low. The df means degrees of freedom and it is related to the size of the table and the number of categories in each variable. The most important message from the output is the estimated p-value, which shows as p-value < 2.2e-16 (2.2 with the decimal point moved 16 places to the left; it is a very small number, so it is written in scientific notation). P-value of the chi-squared test is far smaller than 0.05, so we can say the correlation is statistically significant.

Cramér's V is a measure of association for categorical (nominal or ordinal) data. It ranges from 0 (no association) to 1 (strong association). The main downside of using Cramer's V is that no information is provided on whether the correlation is positive or negative. This is not a problem if the variable pair includes a nominal variable but represents an information loss if both variables being correlated are ordinal.

```
# Load the 'vcd' package  
library(vcd)
```

Warning: package 'vcd' was built under R version 4.4.2

Loading required package: grid

```
# create the crosstable  
crosstab <- table(frs_data$health, frs_data$happy)  
  
# Calculate Cramér's V  
assocstats(crosstab)
```

	X ²	df	P(> X ²)
Likelihood Ratio	54036	60	0
Pearson	45594	60	0

Phi-Coefficient	:	NA
Contingency Coeff.:	:	0.713
Cramer's V	:	0.454

```
#you can also directly calculate the association between variables
assocstats(table(frs_data$health, frs_data$age_group))
```

```

                X^2 df P(> X^2)
Likelihood Ratio 26557 75      0
Pearson          23854 75      0

```

```

Phi-Coefficient   : NA
Contingency Coeff.: 0.592
Cramer's V        : 0.329

```

Q1: What are strongest and weakest associations health status?

Which of our selected person-level variables is most strongly/weakly correlated with an individual's health status? Use the codes of Chi-square test and Cramer's V to answer this question by completing Table 1.

Table 1 Person-level correlations with health status

Covariates		Correlation Coefficient <i>Cramer's V</i>	Statistical Significance <i>p-value</i>
<i>health</i>	<i>age_group</i>		
<i>health</i>	<i>highest_qual</i>		
<i>health</i>	<i>marital_status</i>		
<i>health</i>	<i>nssec</i>		

4.2 Income inequality with respect to gender and health status

In this section, we will work with individual-level data ("FRS16-17_label.csv") to explore income inequality with respect to gender and health status.

To explore income inequality, we need to work with a data set excluding dependent children. In addition, we look at individuals who are the representative persons of households. Therefore, we will select cases (or samples) that meet both conditions.

We want R to select persons only if they are the representative persons of households and they are not dependent children. The variables involved are `hrp` and `dependent`; using the categories "HRP" and "Independent", you can select the appropriate cases. We also want to exclude the health variable reported as "Not known".

```
frs_df <- frs_data %>% filter(hrp == "HRP" &
                             dependent == "Independent" &
                             health != "Not known")
```

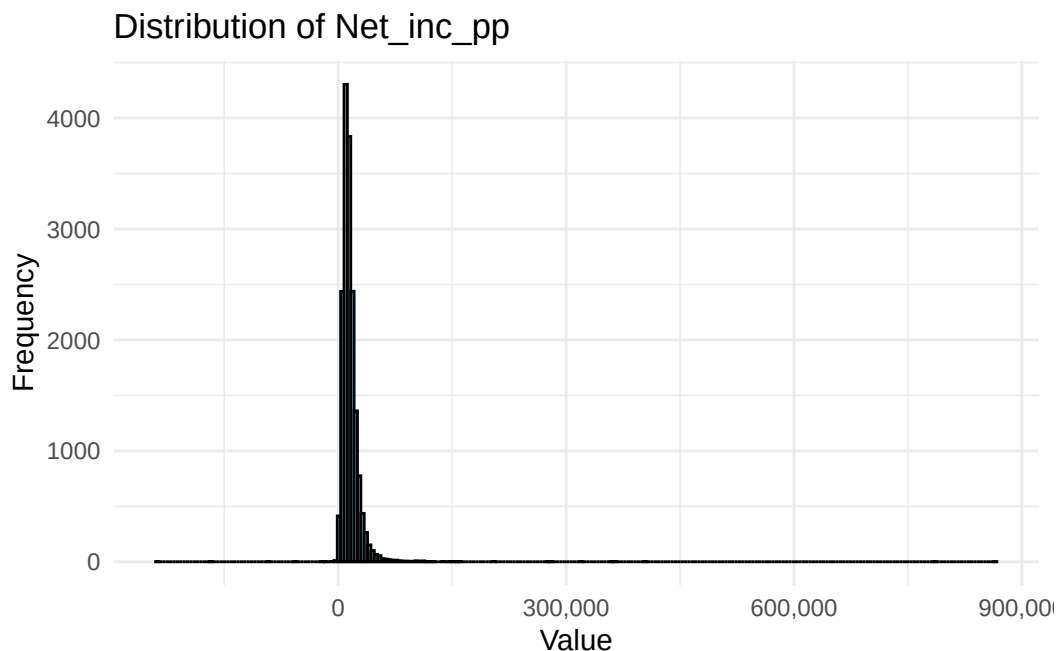
Then, we create a new numeric variable `Net_inc_pp` indicating net income per capita, which is our dependent variable:

```
frs_df$Net_inc_pp = frs_df$hh_income_net / frs_df$hh_size
summary(frs_df$Net_inc_pp)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-238160   9074   13347   15834   19136  864812
```

The distribution of the net household income per capita can be visualised by using `ggplot()`

```
ggplot(frs_df, aes(x = Net_inc_pp)) +
  geom_histogram(
    bins = 250, #A higher bins value means more, narrower bars, covers smaller range of values
    color = "black",
    fill = "skyblue",
    alpha = 0.7
  ) + labs(title = "Distribution of Net_inc_pp", x = "Value", y = "Frequency") +
  theme_minimal()
```



The histogram shows that `Net_inc_pp` is **highly right-skewed**. Most households are concentrated in a narrow band between roughly £0 and £40,000 per person, with the peak around £10,000–£15,000, while a long tail stretches to the right up to a maximum of £864,812. This skewness is also visible in the `summary()` output: the mean (£15,834) is noticeably higher than the median (£13,347), because a small number of very high incomes pull the mean upwards. For skewed data like this, the median is a better measure of a “typical” value. There are also a few **negative values** (down to −£238,160), which can occur when households report losses, for example from self-employment. These extreme values are rare, but they stretch the x-axis so much that the main body of the distribution looks very thin. Keep them in mind when interpreting the regression results, as outliers can influence the estimated coefficients.

Now that we understand how net income per capita is distributed, we can ask **why** it varies so much between households. In this section, we explore whether income differs by **gender** and **health status**. If men and women, or people in better and worse health, have systematically different incomes, this would point to income inequality between these groups. Since both are categorical variables, we first need to check their categories before including them in a regression model.

Our two qualitative independent variables “**sex**” and “**health**”. Let’s first know what they look like:

```
table(frs_df$sex)
```

Female	Male
7647	9180

```
table(frs_df$health)
```

Bad	Fair	Good	Very Bad	Very Good
1472	4253	6277	426	4399

Remember from the lecture what we did with `Region` in the long-term illness example before adding it to the regression model? Two steps are needed before including a categorical variable in a regression model:

1. **Convert it to a factor**, so that R treats it as categorical rather than as text or numbers.
2. **Choose the reference category** (baseline), as the model’s coefficients will show how each of the other categories differs from it.

Here, we use **Female** and **Very Bad** health as the reference categories, but you can choose others if you wish. The code below combines both steps in one line: `as.factor()` converts the variable to a factor, and `fct_relevel()` sets the reference category.

```
frs_df$sex <- fct_relevel(as.factor(frs_df$sex), "Female")
frs_df$health <- fct_relevel(as.factor(frs_df$health), "Very Bad")
```

Implement the regression model with the two qualitative independent variables.

```
model_frs <- lm(Net_inc_pp ~ sex + health, data = frs_df)
summary(model_frs)
```

Call:

```
lm(formula = Net_inc_pp ~ sex + health, data = frs_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-255133	-6547	-2213	3515	845673

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12115.5	762.9	15.881	< 2e-16 ***
sexMale	2091.2	240.6	8.691	< 2e-16 ***
healthBad	-102.8	854.3	-0.120	0.904205
healthFair	1051.3	789.0	1.332	0.182751
healthGood	2766.0	777.4	3.558	0.000375 ***
healthVery Good	4931.8	787.8	6.260	3.95e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15530 on 16821 degrees of freedom

Multiple R-squared: 0.01646, Adjusted R-squared: 0.01616

F-statistic: 56.29 on 5 and 16821 DF, p-value: < 2.2e-16

As in Week 2, here is what the code does:

lm() Function:

- `lm()` stands for “linear model” and is used to fit a linear regression model in R.
- The formula syntax `Net_inc_pp ~ sex + health` specifies a relationship between:
 - **Dependent Variable:** `Net_inc_pp`.

- **Independent Variables:** `sex` and `health`. The model is fitted using the `frs_df` dataset (`data = frs_df`).

Storing the Model: The `model_frs <-` syntax stores the fitted model in an object called `model_frs`.

`summary(model_frs)` provides a detailed output of the model's results, including:

- **Coefficients:** Estimates of how each category differs from its reference category in `Net_inc_pp` (i.e., how each independent variable is associated with `Net_inc_pp`).
- **Standard Errors:** How precise each coefficient estimate is (smaller = more precise).
- **t-values** and **p-values:** Indicate the statistical significance of the effect of each independent (explanatory) variable.
- **R-squared** and **Adjusted R-squared:** Show how well the independent variables explain the variance in the dependent variable.
- **F-statistic:** Tests the overall significance of the model.

The result can be formatted by:

```
library(broom)
tidy(model_frs)
```

```
# A tibble: 6 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)   12116.      763.     15.9 2.21e-56
2 sexMale       2091.      241.      8.69 3.90e-18
3 healthBad     -103.      854.     -0.120 9.04e- 1
4 healthFair    1051.      789.      1.33 1.83e- 1
5 healthGood    2766.      777.      3.56 3.75e- 4
6 healthVery Good 4932.      788.      6.26 3.95e-10
```

Q2. Income inequality by gender and health

What conclusions could be drawn in terms of income inequalities with respect to gender and health status? Also think about the statistical significance of these differences.

The interpretation of this table is again similar to what you have learnt in Week 2, but you may find that R has automatically treated the categorical/qualitative variables `sex` and `health` as dummy variables. Each category gets a coefficient except the reference category set by `fct_relevel()`. That is why the output has `sexMale` but no `sexFemale`, and no `healthVery Bad`.

Each coefficient is therefore interpreted as a difference from the reference category, **Female** with **Very Bad** health, holding the other variable constant:

- The **Intercept** (£12,116) is the predicted net income per capita for the reference group: females with Very Bad health.
- The association of **sexMale** is positive and statistically significant (+£2,091, $p < 0.001$): men have £2,091 more net income per capita than women with the same health status, indicating a gender gap in income.
- The associations of **healthBad** and **healthFair** are not statistically significant ($p = 0.90$ and 0.18): there is no evidence that their net income per capita differs from people in Very Bad health.
- The associations of **healthGood** and **healthVery Good** are positive and statistically significant (+£2,766 and +£4,932, both $p < 0.001$): compared with people in Very Bad health, people in Good and Very Good health have £2,766 and £4,932 more net income per capita respectively, suggesting that income rises with better health.

We then read R-squared and Adjusted R-squared to evaluate the model fit. Both are only around **0.016**, meaning **sex** and **health** together explain only 1.6% of the variation in **Net_inc_pp**. The model is therefore poor for prediction, and more variables (e.g. education level, occupation, age) would be needed to explain income better. Even so, the F-test is highly significant ($p < 2.2e-16$), so the income gaps by gender and health identified above are real, even though they explain only a small part of why incomes differ.

4.3 Implementing a linear regression model with a qualitative independent variable

To gain further practice on using linear regression models with qualitative variables as independent variables, we extend the model you created last week by adding the qualitative/categorical variable **Region** from the Census 2021 dataset. This allows us to answer **Research Question 2: How does health vary across regions in England and Wales?** We use the dataset **EW_DistrictPercentages.csv**, which is already in your 'data/Census2021' folder.

The practical is split into two main parts. The first focuses on implementing a linear regression model with a qualitative independent variable. **Note that** you first need to set the reference category (baseline), as the outcomes of the model reflect the **differences** between each category and the baseline. The second part focuses on prediction based on the estimated linear regression model.

First we load the UK district-level census dataset.

```
# load data
LAcensus <- read.csv("../data/Census2021/EW_DistrictPercentages.csv") # Local authority level
```

Using the district-level census dataset “**EW_DistrictPercentages.csv**”, the variable “Region” (labelled as Government Office Region) is used to explore regional inequality in health.

Familiarise yourself with the dataset by using the same code as last week:

```
#view the data
View(LAcensus)
glimpse(LAcensus)
```

The `names()` function returns all the column names.

```
names(LAcensus)
```

The `dim()` function simply returns the number of rows and number of columns.

```
dim(LAcensus)
```

```
[1] 331 161
```

There are 331 rows and 151 columns in the dataset. It would be very hard to scan through the data if we use so many variables altogether. Therefore, we can select several columns to tailor for this practical. You can of course include other variables you are interested in also by their names:

```
df <- LAcensus %>% select(c("pct_Longterm_sick_or_disabled",
                           "pct_No_qualifications",
                           "pct_Males",
                           "pct_Higher_manager_prof",
                           "Region"))
```

Simple descriptives of this new data

```
summary(df)
```

```

pct_Longterm_sick_or_disabled pct_No_qualifications pct_Males
Min.      :1.330                Min.      : 6.61                Min.      :46.77
1st Qu.:2.865                1st Qu.:15.06                1st Qu.:48.62
Median :3.810                Median :17.63                Median :48.98
Mean    :4.013                Mean    :17.90                Mean    :48.97
3rd Qu.:4.800                3rd Qu.:20.41                3rd Qu.:49.30
Max.     :9.110                Max.     :28.88                Max.     :55.02

pct_Higher_manager_prof      Region
Min.      : 5.49              Length:331
1st Qu.: 9.79              Class :character
Median :12.34              Mode  :character
Mean    :13.22
3rd Qu.:15.80
Max.     :39.68

```

Now we can retrieve the “Region” column from the data frame by simply using `df$Region`. But what if we want to understand the data better, like the following questions?

Q3. Categories of Region

How many categories does the variable “Region” entail? How many local authority districts does each region include?

Simply using the function `table()` will return the answer.

```
table(df$Region)
```

```

                East                East Midlands                London
                45                  35                  33
North East                North West                South East
                12                  39                  64
South West                Wales                West Midlands
                30                  22                  30
Yorkshire and The Humber
                21

```

The `table()` function tells us that this data frame contains 10 regions, and the number of LAs belonging to each region.

R can only include categorical variables as the **factor** type, so we set the column *Region* to `factor()`

```
df$Region<- factor(df$Region)
```

4.3.1 Include the categorical variables into a regression model

We will continue with a very similar regression model fitted in last week that relates percentage long-term illness (*pct_Longterm_sick_or_disabled*) to percentage no-qualification (*pct_No_qualifications*), Percentage Males (*pct_Males*) and Percentages Higher Managerial or Professional occupation (*pct_Higher_manager_prof*).

Decide which region to be set as the baseline category. The principle is that if you want to compare the (average) long term illness outcome of Region A to those of other regions, Region A should be chosen as the baseline category. For example, if you want to compare the (average) long term illness outcome of London to rest of regions in the England and Wales, London should be selected as the baseline category.

Implement the regression model with the categorical variables - *Region* in our case. R will automatically handle the qualitative variable as dummy variables, so you don't need to worry about any of that. But you need to let R know which category of your qualitative variable is your reference category or the baseline. Here we will use London as our first go. **Note:** We choose London as the baseline category so the London region will be excluded in the independent variable list.

Therefore, first, we set London as the reference:

```
df$Region <- fct_relevel(df$Region, "London")
```

Similar to last week, we build our linear regression model, but also include the *Region* variable into the model.

```
model <- lm(pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications + pct_Higher_m  
summary(model)
```

Call:

```
lm(formula = pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications +  
    pct_Higher_manager_prof + Region, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.73799	-0.42623	-0.08528	0.41308	2.20676

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.615375	3.018425	2.523	0.01212	*
pct_Males	-0.167291	0.061638	-2.714	0.00701	**
pct_No_qualifications	0.251334	0.023179	10.843	< 2e-16	***
pct_Higher_manager_prof	0.007145	0.020244	0.353	0.72438	
RegionEast	-0.760329	0.173260	-4.388	1.56e-05	***
RegionEast Midlands	-0.254826	0.194419	-1.311	0.19090	
RegionNorth East	1.164153	0.260808	4.464	1.12e-05	***
RegionNorth West	0.784897	0.192449	4.078	5.74e-05	***
RegionSouth East	-0.328575	0.165728	-1.983	0.04827	*
RegionSouth West	0.096484	0.214507	0.450	0.65317	
RegionWales	1.382217	0.220643	6.264	1.21e-09	***
RegionWest Midlands	-0.415570	0.196345	-2.117	0.03508	*
RegionYorkshire and The Humber	-0.118795	0.217473	-0.546	0.58528	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7086 on 318 degrees of freedom

Multiple R-squared: 0.7593, Adjusted R-squared: 0.7502

F-statistic: 83.6 on 12 and 318 DF, p-value: < 2.2e-16

You have already learnt how to interpret the output of regression model last week: **Significance** (p-value), **Coefficient Estimates**, and **Model fit** (R squared and Adjusted R-squared).

Q4. Regional differences compared with London

Relating back to this week's lecture notes, indicate what regions have statistically significant differences in the percentage of long-term illness, compared to London?

First, the **Significance** and the **Coefficient Estimates**. The p-values in the last column of the output table show that most of the independent variables are significant predictors of `pct_Longterm_sick_or_disabled`. All coefficients below are interpreted **when the other predictors remain the same**.

- The **Intercept (7.62)** is the predicted long-term sick/disabled rate for a **London** district (the reference category) when `pct_Males`, `pct_No_qualifications` and `pct_Higher_manager_prof` are all 0%. Since no real district has 0% males, we do not interpret this value directly; it is the baseline value to which the other coefficients are added when making predictions.

- **Continuous variables.** Similar to last week, `pct_No_qualifications` is significant at the <0.001 level (three asterisks ***), indicating a highly statistically significant relationship. We are slightly less confident about `pct_Males`, which is significant at the <0.01 level (two asterisks **). Their coefficient estimates tell us that:
 - The association of `pct_No_qualifications` is positive and statistically significant (**+0.25, $p < 0.001$**): each 1% increase in residents with no qualifications is associated with a 0.25% increase in the long-term sick/disabled rate, when the other predictors remain the same.
 - The association of `pct_Males` is negative and statistically significant (**−0.17, $p < 0.01$**): each 1% increase in males is associated with a 0.17% decrease in the long-term sick/disabled rate, when the other predictors remain the same.
 - The association of `pct_Higher_manager_prof` is not statistically significant (**$p = 0.72$**).
- **Region dummy variables.** R has created a dummy variable (the items starting with `Region`) for every region except the reference category, **London**. Each coefficient therefore shows how much higher (positive) or lower (negative) a region's long-term sick/disabled rate is **compared with London**, when the other predictors remain the same:
 - **Significant at the <0.001 level (***):** `RegionEast`, `RegionNorth East`, `RegionNorth West` and `RegionWales`. For example, the North East has a **1.16%** higher rate than London, and Wales a **1.38%** higher rate, while the East has a **0.76%** lower rate.
 - **Significant at the 0.05 level (*):** `RegionSouth East` and `RegionWest Midlands`. This is weaker evidence than ***, so they are only mildly statistically significant.
 - **Not significant:** `RegionEast Midlands`, `RegionSouth West` and `RegionYorkshire and The Humber`, meaning their rates are not significantly different from London's.

You can interpret the other regions in the same way using their coefficient estimates.

Reminder: You **cannot** tell from this output whether North East and Wales, or any two regions other than London, are significantly different from each other, because each region's coefficient only compares that region with London. To compare North East and Wales, set either of them as the reference category, using `df$Region <- fct_relevel(df$Region, "North East")` or `df$Region <- fct_relevel(df$Region, "Wales")`, and re-run the model.

Last but not least, the **Model Fit**. R-squared (**0.759**) and Adjusted R-squared (**0.750**) are both around 0.75: the model explains 75.9% of the variation in the long-term sick/disabled rate across districts, or 75.0% after adjusting for the number of independent variables. Together, they suggest a strong fit of the model.

4.3.2 Change the baseline category

If you would like to learn about differences in long-term illness between the North West and other regions in England and Wales, you need to change the baseline category from London to the North West region.

```
df$Region <- fct_relevel(df$Region, "North West")
```

The regression model is specified again as follows:

```
model1 <- lm(
  pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications + pct_Higher_manager_prof
  data = df
)

summary(model1)
```

Call:

```
lm(formula = pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications +
    pct_Higher_manager_prof + Region, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.73799	-0.42623	-0.08528	0.41308	2.20676

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.400272	3.042017	2.761	0.006090	**
pct_Males	-0.167291	0.061638	-2.714	0.007009	**
pct_No_qualifications	0.251334	0.023179	10.843	< 2e-16	***
pct_Higher_manager_prof	0.007145	0.020244	0.353	0.724376	
RegionLondon	-0.784897	0.192449	-4.078	5.74e-05	***
RegionEast	-1.545226	0.158845	-9.728	< 2e-16	***
RegionEast Midlands	-1.039723	0.165225	-6.293	1.03e-09	***
RegionNorth East	0.379256	0.235090	1.613	0.107685	

```

RegionSouth East      -1.113472    0.152463   -7.303 2.27e-12 ***
RegionSouth West      -0.688413    0.182460   -3.773 0.000192 ***
RegionWales            0.597320    0.189607    3.150 0.001786 **
RegionWest Midlands   -1.200467    0.174076   -6.896 2.89e-11 ***
RegionYorkshire and The Humber -0.903692    0.191943   -4.708 3.74e-06 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7086 on 318 degrees of freedom

Multiple R-squared: 0.7593, Adjusted R-squared: 0.7502

F-statistic: 83.6 on 12 and 318 DF, p-value: < 2.2e-16

Examining R-squared to measure the model fit. How robust is the new model? What % of the variance in the dependent variable has been explained?

Now, complete the following table.

Region names	Higher or lower than North West	Whether the difference is statistically significant (Yes or No)
East Midlands		
East		
North East		
South East		
London		
West Midlands		
Yorkshire and The Humber		
Wales		

Check the answer at the end of this practical page.

Now it is very easy to use your model to estimate the results Y (dependent variable - pct_Longterm_sick_or_disabled) by setting all the input independent variable X (pct_Males pct_No_qualifications and pct_Higher_manager_prof).

```

obj_London <- data.frame(
  pct_Males = 49.7,
  pct_No_qualifications = 24.3,
  pct_Higher_manager_prof = 14.7,
  Region = "London"
)
obj_WM <- data.frame(

```

```

    pct_Males = 49.8,
    pct_No_qualifications = 23.3,
    pct_Higher_manager_prof = 11.2,
    Region = "West Midlands"
  )
obj_NE <- data.frame(
  pct_Males = 49.8,
  pct_No_qualifications = 23.3,
  pct_Higher_manager_prof = 11.2,
  Region = "North East"
)

predict(model1, obj_London)

```

```

      1
5.513467

```

```
predict(model1, obj_WM)
```

```

      1
4.804829

```

```
predict(model1, obj_NE)
```

```

      1
6.384552

```

4.3.3 Recode the Region variable and explore regional inequality in health

In many real-world studies, we might not be interested in health inequality across all regions. For example, in this case study, we are interested in health inequality between *London, South, North, Midland and Wales*. We can achieve this by re-grouping regions in England and Wales based on the variable “Region”. That said, we need to have a new grouping of regions as follows:

Original region labels	New region labels
East Midlands	Midlands
East	South

London	London
North East	North
North West	North
South East	South
South West	South
West Midlands	Midlands
Yorkshire and The Humber	North
Wales	Wales

Here we use `mutate()` function in R to make it happen:

```
df <- df %>%
  mutate(
    New_region_label = fct_collapse(
      Region,
      North = c("North East", "North West", "Yorkshire and The Humber"),
      Midlands = c("East Midlands", "West Midlands"),
      South = c("East", "South East", "South West"),
      London = "London",
      Wales = "Wales"
    )
  )
```

This code may look a bit complex. You can simply type `?mutate` in your console. In the Help window on the right, RStudio gives you the explanation of the `mutate` function. This is a common way to use RStudio to help you learn what a function does. `mutate()` creates new columns that are functions of existing variables. Therefore, the `df %>% mutate()` means add a new column into the current dataframe `df`; the `New_region_label` in the `mutate()` function indicates the name of this new column is `New_region_label`. The right side of the `New_region_label =` indicates the value we want to assign to the `New_region_label` in each row.

The right side of `New_region_label` is

```
fct_collapse(Region, North = c("North East", "North West", "Yorkshire and The
Humber"), Midlands = c("East Midlands", "West Midlands"), South = c("East",
"South East", "South West"), London = "London", Wales = "Wales")
```

By using the code, the `fct_collapse()` function recodes or groups each value in the `Region` column into one of the broader categories: “London”, “South”, “North”, “Midlands” or “Wales”. Specifically, it takes the existing factor levels in `Region` and collapses multiple detailed regions into fewer, aggregated groups. For example, regions such as “*North East*”, “*North West*”, and

“Yorkshire and The Humber” are all grouped under “**North**”, while “East Midlands” and “West Midlands” are combined into “**Midlands**”, and so on.

Now we use the same way to examine our new column `New_region_label`:

```
table(df$New_region_label)
```

North	London	South	Midlands	Wales
72	33	139	65	22

Comparing with `Region`, we can now see that the mutate worked:

```
df[,c("Region", "New_region_label")]
```

Now you will have a new qualitative variable named `New_region_label` in which England and Wales are divided into five groups: London, South, North, Midlands and Wales.

Based on the newly generated qualitative variable `New_region_label`, we can now build our new linear regression model. Don't forget:

(1) R needs the categorical variables in a regression model to be of the factor type;

```
class(df$New_region_label)
```

```
[1] "factor"
```

The `class()` returns the type of the variable. The `New_region_label` is already a factor variable. If not, we need to convert it by the `as.factor()`, as we used above.

```
df$New_region_label = as.factor(df$New_region_label)
```

(1) Let R know which region you want to use as the baseline category. Here I will use London again, but of course you can choose other regions.

```
df$New_region_label <- fct_relevel(df$New_region_label, "London")
```

The linear regression model is set up below. This time we include `New_region_label` rather than `Region` as the region variable:

```

model2 <- lm(
  pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications + pct_Higher_manager_prof
  data = df
)

summary(model2)

```

Call:

```
lm(formula = pct_Longterm_sick_or_disabled ~ pct_Males + pct_No_qualifications +
    pct_Higher_manager_prof + New_region_label, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.83271	-0.48362	-0.06293	0.40581	2.11961

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.98363	3.20535	3.115	0.00201	**
pct_Males	-0.18418	0.06585	-2.797	0.00547	**
pct_No_qualifications	0.20143	0.02257	8.924	< 2e-16	***
pct_Higher_manager_prof	-0.03410	0.01986	-1.718	0.08684	.
New_region_labelNorth	0.47416	0.18706	2.535	0.01172	*
New_region_labelSouth	-0.50635	0.16433	-3.081	0.00224	**
New_region_labelMidlands	-0.41031	0.18517	-2.216	0.02740	*
New_region_labelWales	1.25183	0.23431	5.343	1.73e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.76 on 323 degrees of freedom

Multiple R-squared: 0.7188, Adjusted R-squared: 0.7127

F-statistic: 118 on 7 and 323 DF, p-value: < 2.2e-16

Q5. London vs North, and South vs Wales

Are there statistically significant differences in the percentage of people with long-term sick/disable between London and North, and between South and Wales, controlling for other variables? What conclusions could be drawn in terms of regional differences in health outcome?

4.4 Predictions using fitted regression model

Write down the % long term sick or disabled regression model with the new region label categorical variables.

Relating to this week's lecture, the % `pct_Longterm_sick_or_disabled` is equal to:

[write down the model]

Q6. Predicting long-term illness in Wales and London

Now imagine that the values of variables *pct_Males*, *pct_No_qualifications*, and *pct_Higher_manager_prof* are 49, 23 and 11, respectively, what would the percentage of persons with long-term sick/disable in Wales and London be?

Check the answer at the end of this practical page.

4.5 Answers to the Questions

4.5.1 Q1 associations with health status

Using the Chi-squared test and Cramér's V, Table 1 can be completed as follows:

Covariates		Cramér's V	p-value
<i>health</i>	<i>age_group</i>	0.329	< 0.001
<i>health</i>	<i>highest_qual</i>	0.287	< 0.001
<i>health</i>	<i>nssec</i>	0.280	< 0.001
<i>health</i>	<i>marital_status</i>	0.229	< 0.001

All four associations are statistically significant ($p < 0.001$). **Age group** has the strongest association with health status (Cramér's $V = 0.329$), and **marital status** the weakest (0.229). Remember that Cramér's V tells us the strength of an association, but not its direction.

4.5.2 Q2: income inequality by gender and health status

- **Gender:** men have £2,091 more net income per capita than women with the same health status ($p < 0.001$), indicating a significant gender gap in income.
- **Health:** compared with people in Very Bad health, people in Good and Very Good health have £2,766 and £4,932 more net income per capita respectively (both $p < 0.001$). The differences for Bad and Fair health are not statistically significant.
- **Model fit:** sex and health explain only 1.6% of the variation in income ($R^2 = 0.016$), so other factors matter much more. However, the income gaps by gender and health are statistically significant.

4.5.3 Q3: categories of Region

`table(df$Region)` shows **10 regions** with 331 local authority districts in total: East (45), East Midlands (35), London (33), North East (12), North West (39), South East (64), South West (30), Wales (22), West Midlands (30) and Yorkshire and The Humber (21).

4.5.4 Q4: regions that differ significantly from London

Holding the other predictors constant, compared with London:

- **Significantly higher:** Wales (+1.38%), North East (+1.16%) and North West (+0.78%), all $p < 0.001$.
- **Significantly lower:** East (−0.76%, $p < 0.001$), West Midlands (−0.42%, $p < 0.05$) and South East (−0.33%, $p < 0.05$).
- **Not significantly different:** East Midlands, South West, and Yorkshire and The Humber.

4.5.5 Changing the baseline to North West

R-squared (0.759) and Adjusted R-squared (0.750) are **exactly the same** as in the model with London as the baseline. Changing the reference category does not change the model fit; it only changes which comparisons the Region coefficients show.

Region names	Higher or lower than North	
	West	Statistically significant?
East Midlands	Lower (−1.04%)	Yes
East	Lower (−1.55%)	Yes
North East	Higher (+0.38%)	No ($p = 0.11$)
South East	Lower (−1.11%)	Yes

Region names	Higher or lower than North	
	West	Statistically significant?
South West	Lower (−0.69%)	Yes
London	Lower (−0.78%)	Yes
West Midlands	Lower (−1.20%)	Yes
Yorkshire and The Humber	Lower (−0.90%)	Yes
Wales	Higher (+0.60%)	Yes

Note that London is 0.78% lower than North West, which is the same size as the North West coefficient in the London-baseline model (+0.78%), with the opposite sign.

4.5.6 Q5: London vs North, and South vs Wales

- **London vs North:** yes. With London as the baseline, the North has a 0.47% higher long-term sick/disabled rate than London ($p = 0.012$), when the other predictors remain the same.
- **South vs Wales:** this **cannot** be answered from `model2`, because both South and Wales are compared only with London. To test it, set one of them as the reference category, e.g. `df$New_region_label <- fct_relevel(df$New_region_label, "South")`, and re-run the model. Doing so shows that Wales has a 1.76% higher rate than the South ($p < 0.001$).
- **Conclusion:** after controlling for sex, qualifications and occupation, `model2` tells us how each region compares with **London:** Wales (+1.25%) and the North (+0.47%) have significantly higher long-term sick/disabled rates than London, while the South (−0.51%) and the Midlands (−0.41%) have significantly lower rates. This suggests clear regional inequalities in health across England and Wales.

Reminder: these conclusions are all *relative to London*. Although the coefficients suggest an ordering (e.g. Wales appears higher than the North, and the North higher than the South), `model2` **cannot** tell us whether the differences between two non-London regions are statistically significant. To test any such comparison, the model must be **re-built** with one of those regions as the reference category, as we did above for South vs Wales.

4.5.7 The written-down model and Q6

The model of the new region label is: $\text{pct_Longterm_sick_or_disabled (\%)} = 9.98 + (-0.1842) * \text{pct_Males (\%)} + 0.2014 * \text{pct_No_qualifications (\%)} + (-0.0341) * \text{pct_Higher_manager_prof} + (0.4742) * \text{North} + (-0.5063) * \text{South} + (-0.4103) * \text{Midlands} + 1.2518 * \text{Wales}$, where

North, South, Midlands and Wales are dummy variables (1 if the district is in that region, 0 otherwise). London is the reference category, so all four dummies are 0 for London. Note that `pct_Higher_manager_prof` is not statistically significant at the 0.05 level ($p = 0.087$).

If `pct_Males`, `pct_No_qualifications` and `pct_Higher_manager_prof` are 49, 23 and 11:

- **Wales** (Wales = 1, other dummies = 0): $9.98 + (-0.1842)*49 + 0.2014*23 + (-0.0341)*11 + 1.2518*1$
- **London** (all dummies = 0): $9.98 + (-0.1842)*49 + 0.2014*23 + (-0.0341)*11$

You can paste these directly into your RStudio Console to get the results. You can also make a new object and use `predict()`:

```
obj_Wales <- data.frame(  
  pct_Males = 49,  
  pct_No_qualifications = 23,  
  pct_Higher_manager_prof = 11,  
  New_region_label = "Wales"  
)  
  
obj_London <- data.frame(  
  pct_Males = 49,  
  pct_No_qualifications = 23,  
  pct_Higher_manager_prof = 11,  
  New_region_label = "London"  
)  
  
predict(model2, obj_Wales)
```

```
1  
6.468668
```

```
predict(model2, obj_London)
```

```
1  
5.216834
```

Therefore, the predicted percentages of persons with long-term illness in Wales and London are 6.47% and 5.22% respectively. The difference (1.25%) is exactly the Wales coefficient, because the two districts have the same values for all the other predictors. If you got the right answers, congratulations: you can now use a regression model to make predictions!

5 Lab: Logistic Regression

Open this lab in RStudio: in the Files pane, click labs and then 04.LogisticRegression.qmd. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk and adding your notes and answers in the file. (First time? See the [Setup](#) guide.)

Last week, we learnt how to use qualitative variables in a multiple linear regression model to understand the relationship between independent variables $X_1 \dots X_n$ and the dependent variable Y . Today we will learn to use logistic regression for a binary dependent variable. A **logistic regression model** is a type of regression analysis used when the dependent variable is binary (e.g., “success/failure” or “0/1”). It estimates the probability of one outcome relative to the other using a logistic function. This model is commonly used in situations like predicting disease presence (yes/no) or customer churn (stay/leave). The independent variables can be continuous, categorical, or a mix of both. The model’s output is in the form of odds ratios, showing how predictors affect the likelihood of the outcome.

The lecture’s slides can be found [here](#).

Learning objectives

An understanding of, and ability to:

- Estimate and interpret a logistic regression model
- Assess the model fit

The application context of a binomial logistic regression model is when the dependent variable under investigation is a binary variable. Usually, a value of 1 for this dependent variable means the occurrence of an event; and, 0 otherwise. For example, the dependent variable for this practical is whether a person is a long-distance commuter i.e. 1, and 0 otherwise.

In this week’s practical, we are going to apply logistic regression analysis in an attempt to answer the following research question:

RESEARCH QUESTION: Who is willing to commute long distances?

The practical is split into two main parts. The first focuses on implementing a binary logistic regression model with R. The second part focuses on the interpretation of the resulting estimates.

5.1 Knowing the dataset and descriptive analysis

Prepare the data for implementing a logistic regression model. The data set used in this practical is the “sar_sample_label.csv” and “sar_sample_code.csv”. The SAR is a snapshot of census microdata, which are individual level data. The data sample has been drawn and anonymised from census and known as the Samples of Anonymised Records (SARs).

Both datasets are already in your ‘data/SAR/’ folder. Download the data dictionary (‘SAR_dictionary’) from [Canvas](#) to your computer. If you look at both .csv files you may find they have exactly same column names - yes, they are in fact the same spreadsheet but ‘sar_sample_label.csv’ is more readable for the cell contents as they are in text format but cells in the ‘sar_sample_code.csv’ is in coding format. The ‘SAR_dictionary’ is created to help you understand what the columns mean, just like the FRS dictionary for the FRS datasets last week. Yes, the two SAR csv files are actually the same dataframe, only one uses the label as the value but the other uses the code. This is quite normal: when we do surveys, we use text as the options of multiple choice questions for respondents to choose from, and therefore the collected survey results are in the ‘sar_sample_label’ format. The label format is usually more readable and is good to do descriptive analysis as we will show in today’s content. However, we may find the coding format is more concise for writing the code and thus will be used during regression analysis. **Please notice that coding the labels into numbers, doesn’t mean the categorical/qualitative variable has been converted into numeric/continuous numbers.**

Let’s first read in both for the data overview.

```
# already installed in your codespace
library(tidyverse)
```

Warning: package ‘ggplot2’ was built under R version 4.4.3

```
library(broom)
library(forcats)
```

```
sar_label <- read.csv("../data/SAR/sar_sample_label.csv")
sar_code <- read.csv("../data/SAR/sar_sample_code.csv")
```

Run the following codes to view both dataframes:

```
#View the sar_label dataset
View(sar_label)
```

```
#View the sar_code dataset  
View(sar_code)
```

To know the dataset better, we can run `dim()` to know how many individuals (rows) in this sample data and how many variables (columns) have been recorded.

```
dim(sar_label)
```

```
[1] 50000    39
```

```
#or  
dim(sar_code)
```

```
[1] 50000    39
```

So there are 50,000 respondents in the sample dataset and 39 variables are included.

After browsing both data sheets and also the data dictionary (sometimes we call it meta data), you may notice that all the columns in the SAR dataset are of categorical/qualitative type. Therefore, if we run the `summary()` function for the dataframes, as we did in previous weeks for the Census data or the FRS dataset, the outputs of the `summary()` may not be very useful. Again, please notice that the figures in `sar_code.csv` are not actual numbers, but codes for the different categories of the qualitative variables. The `summary()` results here are therefore not useful. Think back to what we learnt last week: when the variables are of categorical/qualitative type, how do we carry out the descriptive analysis for them?

```
summary(sar_code)  
summary(sar_label)
```

So, the answer is we focus on summarising the frequency and distribution of categories within the variable and the function in R for doing this is `table()`. For example, the variable “`work_distance`” captures a person’s commuting distance. We can run the `table()` for both label and code csv to understand the distribution, composition and diversity of categories of “`work_distance`” in the data. Try to repeat what we have learnt in Week 3 to finish the **descriptive analysis** for the categorical variables in SAR (by `ggplot2`).

```
table(sar_label$work_distance)
```

```

10 to <20 km
3650
2 to <5 km
4414
20 to <40 km
2014
40 to <60km
572
5 to <10 km
4190
60km or more
703
Age<16 or not working
25975
At home
2427
Less than 2 km
4028
No fixed place
1943
Work outside England and Wales but within UK
29
Work outside UK
21
Works at offshore installation (within UK)
34

```

```
table(sar_code$work_distance)
```

```

-9      1      2      3      4      5      6      7      8      9     10     11     12
25975  4028  4414  4190  3650  2014   572   703  2427  1943   29   21   34

```

To create a chart as part of the descriptive analysis, exactly as what we did in Week 3, we use the library `ggplot2` and the code:

```

library(ggplot2)
ggplot(sar_label, aes(x = work_distance)) +
  geom_bar(fill = "rosybrown") +
  labs(title = "Histogram of work distance in SAR sample", x = "Work Distance", y = "Count")

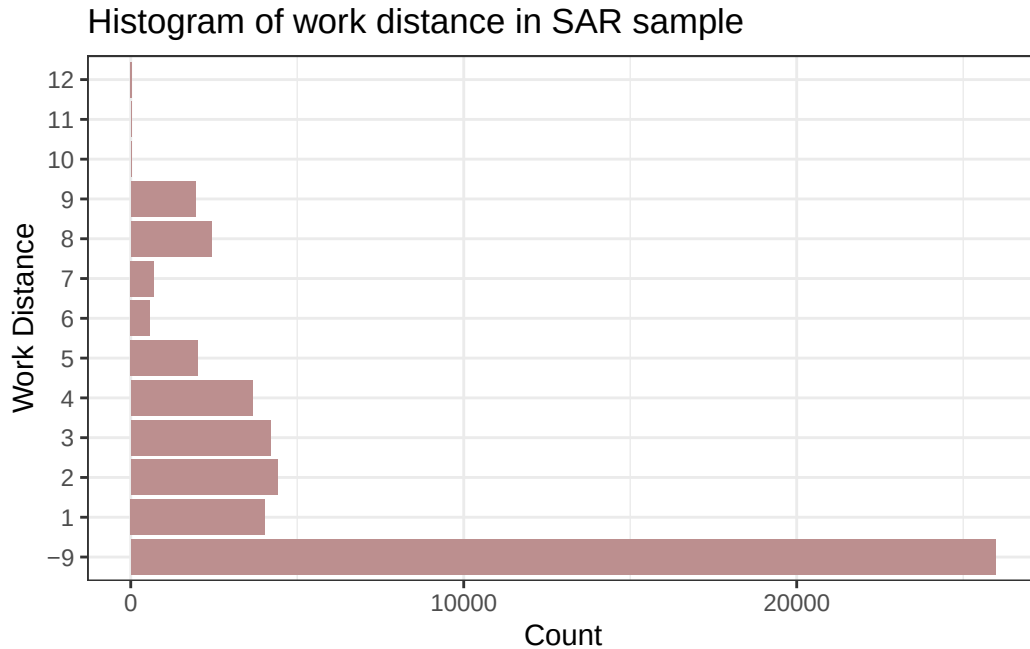
```

```
coord_flip()+ #Flip the Axes, add a # in front of this line, to make the code in gray and y
theme_bw()
```



We have learnt that for the categorical/qualitative variables, we need R to treat them as **factor** type. Therefore, if we want to create descriptive analysis charts by using `sar_code` , we first need to convert the code numbers from numeric to factor type.

```
library(ggplot2)
ggplot(sar_code, aes(x = as.factor(work_distance))) +
  geom_bar(fill = "rosybrown") +
  labs(title = "Histogram of work distance in SAR sample", x = "Work Distance", y = "Count")
coord_flip()+ #Flip the Axes, add a # in front of this line, to make the code in gray and y
theme_bw()
```

You may also realise that this chart is not that readable to be used as a descriptive analysis chart to put in the report as the Y-axis labels are all codes, therefore the first chart created by labels is a better choice. Based on ‘SAR_dictionary.xlsx’, we learn how the codes map to the categories for the variable work_distance in both csv.

Code for Work_distance	Categories
-9	Age<16 or not working
1	Less than 2 km
2	2 to <5 km
3	5 to <10 km
4	10 to <20 km
5	20 to <40 km
6	40 to <60 km
7	60km or more
8	At home
9	No fixed place
10	Work outside England and Wales but within UK
11	Work outside UK
12	Works at offshore installation (within UK)

There are a variety of categories in the variable, however, we are only interested in commuting distance and therefore in people reporting their commuting distance. **Thus, we will explore**

the numeric codes of the variable ranging from 1 to 8.

As we are also interested in exploring whether people with different socio-economic statuses (or occupations) tend to be associated with varying probabilities of commuting over long distances, we further filter or select cases.

```
table(sar_label$nssec)
```

```

Child aged 0-15
9698
Full-time student
3041
Higher professional occupations
3162
Intermediate occupations
5288
Large employers and higher managers
909
Lower managerial and professional occupations
8345
Lower supervisory and technical occupations
2924
Never worked or long-term unemployed
2261
Routine occupations
4660
Semi-routine occupations
5893
Small employers and own account workers
3819
```

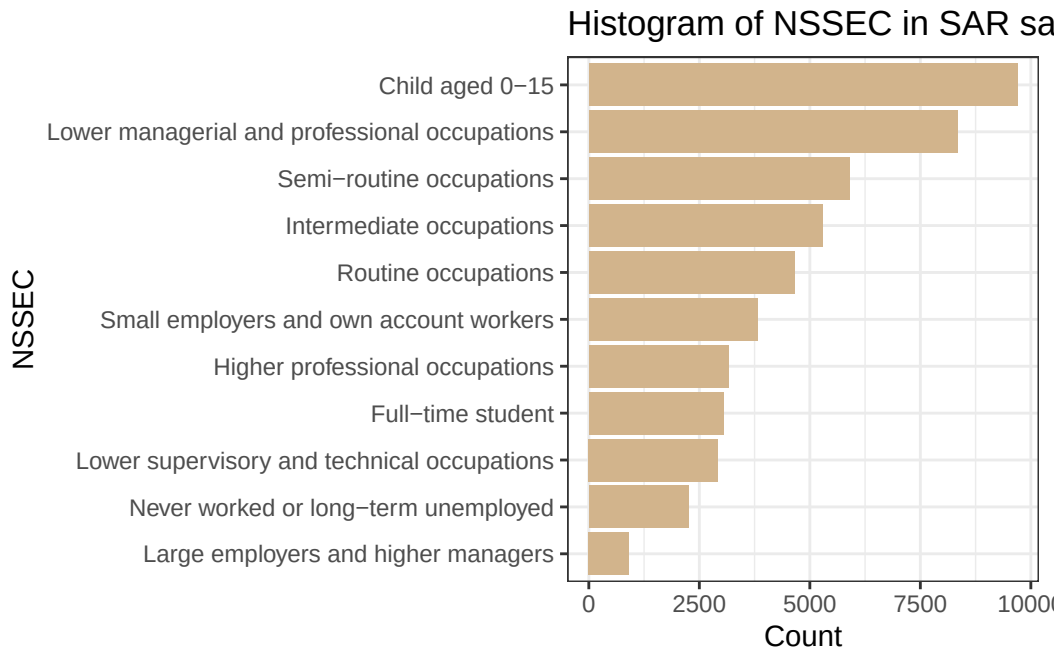
```
table(sar_code$nssec)
```

```

 1    2    3    4    5    6    7    8    9   10   12
909 3162 8345 5288 3819 2924 5893 4660 2261 3041 9698
```

```
ggplot(sar_label, aes(x = fct_rev(fct_infreq(nssec)))) +
  geom_bar(fill = "tan") +
  labs(title = "Histogram of NSSEC in SAR sample", x = "NSSEC", y = "Count") +
```

```
coord_flip()+ #Flip the Axes, add a # in front of this line, to make the code in gray and y
theme_bw()
```



From ‘SAR_dictionary.xlsx’, we learn what each code indicate corresponding category for variable **nssec**. For the following regression model, we select people who reported an occupation, and delete cases with numeric codes from 9 to 12, who are *unemployed*, *full-time students*, *children* and *not classifiable*.

Code for nssec	Category labels
1	Large employers and higher managers
2	Higher professional occupations
3	Lower managerial and professional occupations
4	Intermediate occupations
5	Small employers and own account workers
6	Lower supervisory and technical occupations
7	Semi-routine occupations
8	Routine occupations
9	Never worked or long-term unemployed
10	Full-time student
11	Not classifiable
12	Child aged 0-15

Now, as we did last week, we use `filter()` to prepare our dataframe. You may already realise that using `sar_code` is easier to do the filtering.

```
sar_df <- sar_code %>% filter(work_distance<=8 & nssec <=8 )
```

Q1. Create descriptive analysis for the two variables “work_distance”, “nssec” and “sex” with the new data, including (1) summarise the frequencies and (2) create histogram charts for the three variables.

5.2 Preparing the input variables

For a logistic regression model, we first need to **recode the “work_distance” variable into a binary dependent variable.**

A simple way to create a binary dependent variable representing long-distance commuting is to use the `mutate()` function as discussed in last week’s practical session. Before creating the binary variables from the “work_distance” variable, we need to define *what counts as a long-distance commuting move*. Such definition can vary. Here we define a long-distance commuting move as any commuting move over a distance above 60km (the category of “60km or more”).

```
sar_df <- sar_df %>% mutate(  
  New_work_distance = if_else(work_distance >6, 1,0))
```

Q2. Do descriptive analysis on the new `sar_df` dataframe with new column named `New_work_distance` by using the codes you have learnt.

The independent variables are gender and socio-economic status. The gender variable `sex` is a categorical variable. Therefore, as we’ve learnt last week, before adding the categorical variables into the regression model, we first need to make it a factor and then identify the reference category. For gender, we use male as the baseline. Prepare your “sex” variable before the regression model:

```
sar_df$sex <- relevel(as.factor(sar_df$sex),ref="1")
```

Then, we prepare socio-economic status variable `nssec` for the regression model. We are interested in whether people with occupations being “Higher professional occupations” are associated with a lower probability of commuting over long distances when comparing to people in other occupations. Therefore by using the “Higher professional occupations” code 2, we run the code as below:

```
sar_df$nssec <- relevel(as.factor(sar_df$nssec), ref = "2")
```

5.3 Implementing a logistic regression model

The binary dependent variable is long-distance commuting, variable name `New_work_distance`.

```
#create the model
m.glm = glm(New_work_distance~sex + nssec,
            data = sar_df,
            family= "binomial")
# inspect the results
summary(m.glm)
```

Call:

```
glm(formula = New_work_distance ~ sex + nssec, family = "binomial",
    data = sar_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.67337	0.05329	-31.401	< 2e-16 ***
sex2	-0.36678	0.04196	-8.742	< 2e-16 ***
nssec1	-0.12881	0.11306	-1.139	0.255
nssec3	-0.38761	0.06467	-5.994	2.05e-09 ***
nssec4	-1.03079	0.08439	-12.214	< 2e-16 ***
nssec5	1.22639	0.06489	18.898	< 2e-16 ***
nssec6	-1.38992	0.10919	-12.730	< 2e-16 ***
nssec7	-1.43909	0.09002	-15.986	< 2e-16 ***
nssec8	-1.48534	0.09646	-15.398	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 20441 on 33025 degrees of freedom
Residual deviance: 17968 on 33017 degrees of freedom
AIC: 17986

Number of Fisher Scoring iterations: 6

```
# odds ratios
exp(coef(m.glm))
```

```
(Intercept)      sex2      nssec1      nssec3      nssec4      nssec5
  0.1876138  0.6929649  0.8791416  0.6786766  0.3567267  3.4088847
      nssec6      nssec7      nssec8
  0.2490946  0.2371432  0.2264258
```

```
# confidence intervals
exp(confint(m.glm, level = 0.95))
```

Waiting for profiling to be done...

```
              2.5 %    97.5 %
(Intercept) 0.1688060 0.2080319
sex2        0.6381810 0.7522773
nssec1      0.7017990 1.0935602
nssec3      0.5981911 0.7708192
nssec4      0.3020431 0.4205270
nssec5      3.0037298 3.8739884
nssec6      0.2002766 0.3073830
nssec7      0.1984396 0.2824629
nssec8      0.1869397 0.2729172
```

5.3.1 Interpreting estimated regression coefficients

Compared to statistical inference in a multiple linear regression model (which you did in Weeks 2 and 3), the interpretation of coefficients (B) and odds ratios (Exp(B)) for the logistic regression model has some similarities and some differences:

It is the same that we read p-values of regression coefficients to assess significances of coefficients; for instance, by comparing p-values to the conventional level of significance of 0.05:

- If the p-value of a coefficient is smaller than 0.05, the coefficient is statistically significant. In this case, you can say that the relationship between an independent variable and the outcome variable is *statistically* significant.
- If the p-value of a coefficient is larger than 0.05, the coefficient is statistically insignificant. In this case, you can say or conclude that there is no statistically significant association or relationship between an independent variable and the outcome variable.

It is different in the way we interpret the regression coefficients (B) as we need to read the odds ratios (Exp(B)) from `exp(coef(m.glm))`

o For the variable **sex**, the negative coefficient and the odds ratio indicate that the odds of commuting over long distances for a female are 0.693 times those of a male (the reference group), with a confidence interval (CI), or likely range, of 0.63 to 0.75, holding all other variables constant (the socio-economic classification variable). Put differently, being female reduces the odds of long-distance commuting by 30.7% (1-0.693).

o For the variable **nssec**, the coefficients and odds ratios indicate the following for those whose socio-economic classification is:

- the p-value of Large employers and higher managers (nssec=1) is > 0.05 , so there is no statistically significant relationship between large employers and higher managers and long-distance commuting;
- Lower managerial and professional occupations (nssec=3) have 0.679 times the odds of long-distance commuting compared to our reference category (nssec=2, higher professional occupations), holding all other variables constant (the sex variable), with a likely range (CI) of 0.60 to 0.77. Therefore we can also say that workers in lower managerial and professional occupations have 32.1% (1-0.679) lower odds than higher professional workers of travelling more than 60km for work, for the same gender.
- Small employers and own account workers (nssec=5) have 3.409 times the odds of the reference category (nssec=2, higher professional occupations), holding the gender variable constant, with a likely range (CI) of 3.00 to 3.87.
- Compared to higher professional occupations (nssec=2), workers in routine occupations (nssec=8) have 0.226 times the odds of travelling more than 60km to work, with a CI of 0.19 to 0.27, when the other variables are held constant. Equivalently, being in a routine occupation reduces the odds of long-distance commuting by 77.4% (1-0.226).

Q3. Can you write down the findings you learnt from the model outcomes to other occupation groups (nssec = 4, nssec = 6 and nssec =7)?

5.3.2 Model fit

We include the R library **pscl** to calculate the measures of fit.

```
library(pscl)
```

Warning: package 'pscl' was built under R version 4.4.2

Classes and Methods for R originally developed in the
Political Science Computational Laboratory
Department of Political Science
Stanford University (2002-2015),
by and under the direction of Simon Jackman.
hurdle and zeroinfl functions by Achim Zeileis.

Relating back to this week's lecture notes, what is the Pseudo R^2 of the fitted logistic model (from the Model Summary table below)?

```
# Pseudo R-squared  
pR2(m.glm)
```

fitting null model for pseudo-r2

	llh	llhNull	G2	McFadden	r2ML
	-8.983928e+03	-1.022037e+04	2.472890e+03	1.209785e-01	7.214246e-02
r2CU					
	1.563288e-01				

```
# or in better format  
pR2(m.glm) %>% round(4) %>% tidy()
```

fitting null model for pseudo-r2

```
# A tibble: 6 x 2  
  names      x  
  <chr>    <dbl>  
1 llh      -8984.  
2 llhNull -10220.  
3 G2        2473.  
4 McFadden  0.121  
5 r2ML      0.0721  
6 r2CU      0.156
```

- **llh**: The log-likelihood of the fitted model.
- **llhNull**: The log-likelihood of the null model (without predictors).
- **G2**: The likelihood ratio statistic, showing the model's improvement over the null model.
- **McFadden**: McFadden's pseudo R-squared (a common measure of model fit).

- **r2ML**: Maximum likelihood pseudo R-squared.
- **r2CU**: Cox & Snell pseudo R-squared.

In multiple linear regression, R-squared indicates the % of the variance in the dependent variable that is explained by the independent variables. In a logistic regression model, R-squared is not directly applicable. Instead, we use pseudo R-squared measures, such as McFadden's pseudo R-squared, or Cox & Snell pseudo R-squared to provide an indication of model fit. For an individual-level dataset like the SAR, a value around 0.3 is considered a good fit. Our model is therefore not that robust for prediction, but it still describes the association between the variables and the categories. To improve the model, we may need more variables as the independent variables, you may identify some based on the related literature or debates on this topic.

5.3.3 Recode Socio-economic status variable and explore commuting differences

This time, we want to know whether “Lower supervisory and technical occupations”, “Semi-routine occupations” and “Routine occupations” are associated with a higher probability of commuting over long distances compared to people in other occupations.

We can use `mutate()` to create a new column, set the value of “Lower supervisory and technical occupations”, “Semi-routine occupations” and “Routine occupations” as original, while the rest as “Other occupations”. Here, by using the SAR in code format, we can make this much easier by using:

```
sar_df <- sar_df %>% mutate(New_nssec = fct_other(nssec, keep = c("6", "7", "8"), other_)
```

Or by using `if_else` and `%in%` in R, we can achieve the same result. `%in%` is an operator used to test if elements of one vector are present in another. It returns `TRUE` for elements found and `FALSE` otherwise.

```
sar_df <- sar_df %>% mutate(New_nssec = if_else(!nssec %in% c(6,7,8), "0", nssec))
```

Use “Other occupations” (code: 0) as the reference category by `relevel(as.factor())` and then create the regression model: `glm(New_work_distance~sex + New_nssec, data = sar_df, family= "binomial")`. Can you now run the model by yourself? Find the answer at the end of the practical.

Q4. If we want to explore whether people as independent employers show lower probability of commuting longer than 60km compared with other occupations, how will we prepare the input independent variables and what will be the specified regression model?

5.3.4 Prediction using fitted regression model

Relating to this week's lecture, we can write down the fitted model. **Note** that the numbers below are the *odds ratios* returned by `exp(coef(m.glm))`, not the log-odds coefficients shown by `summary(m.glm)`, so they multiply together rather than add up:

Odds of long-distance commuting = $0.188 \times (0.693 \text{ if female}) \times (0.679 \text{ if nssec3}) \times (0.357 \text{ if nssec4}) \times (3.409 \text{ if nssec5}) \times (0.249 \text{ if nssec6}) \times (0.237 \text{ if nssec7}) \times (0.226 \text{ if nssec8})$. We don't include nssec1, as the previous result shows it is not statistically significant, so we won't use the model to predict individuals in the nssec=1 occupation group.

By using R, you can create the object you would like to predict. Here we create three people; see whether you can interpret their gender and socio-economic classification.

```
objs <- data.frame(sex=c("1","2","1"),nssec=c("7","3","5"))
```

Then we can predict by using our model `m.glm`:

```
predict(m.glm, objs,type = "response")
```

```
      1      2      3  
0.04259618 0.08108050 0.39007797
```

So let us look at these three people. For the first, a male classified as Semi-routine occupation in NSSEC, the probability that he travels over 60km to work is only 4.26%. For the second, a female in a Lower managerial and professional occupation, the probability of long-distance commuting is 8.11%. Now work out the prediction for our last person. Remember: the model fit results show that the model is not very robust, so the predictions may not be very solid either.

5.4 Extension activities

The extension activities are designed to help you prepare for the assignment. For this week, try whether you can:

- Select a regression strategy and explain why a linear or logistic model is appropriate
- Perform one or a series of regression models, including different combinations of your chosen independent variables to explain and/or predict your dependent variable

5.5 Answers for Qs

Answer for Q1

For Q1, we have already had the codes for `work_distance` and `nssec`, so the only missing variable is `sex`:

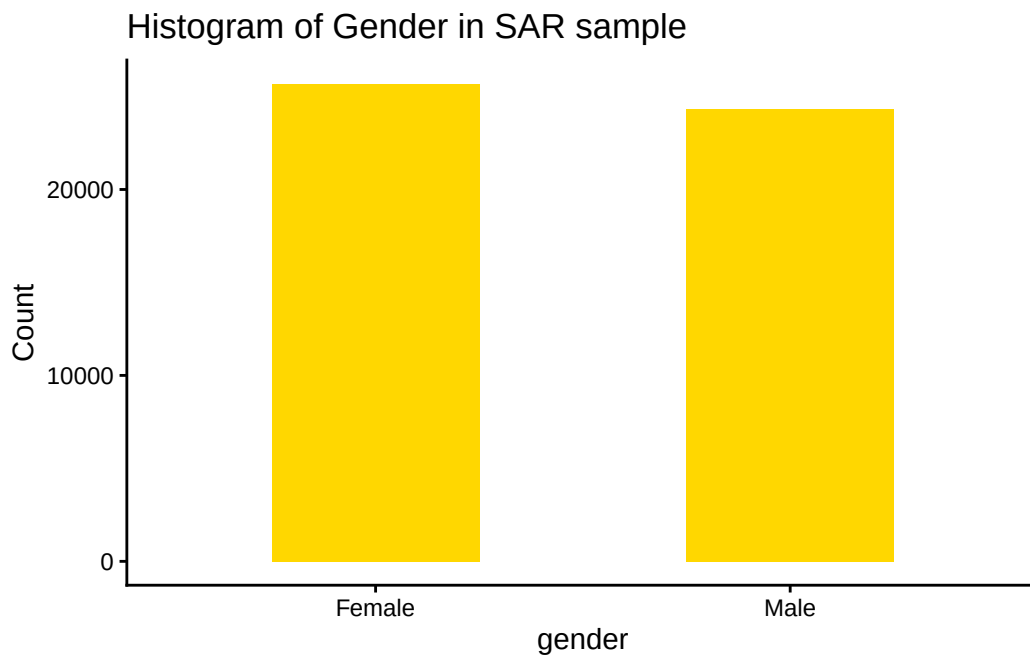
To summarise the frequency of each category, we use the code:

```
table(sar_label$sex)
```

```
Female    Male  
 25677   24323
```

To create bar charts for the distribution and composition of the gender variable in the SAR sample, we can run:

```
ggplot(sar_label, aes(x = sex)) +  
  geom_bar(fill="gold",width=0.5) +  
  labs(title = "Histogram of Gender in SAR sample", x = "gender", y = "Count")+ #set text info  
  theme_classic() #choose theme type, try theme_bw(), theme_minimal() see differences
```



Answer for Q2

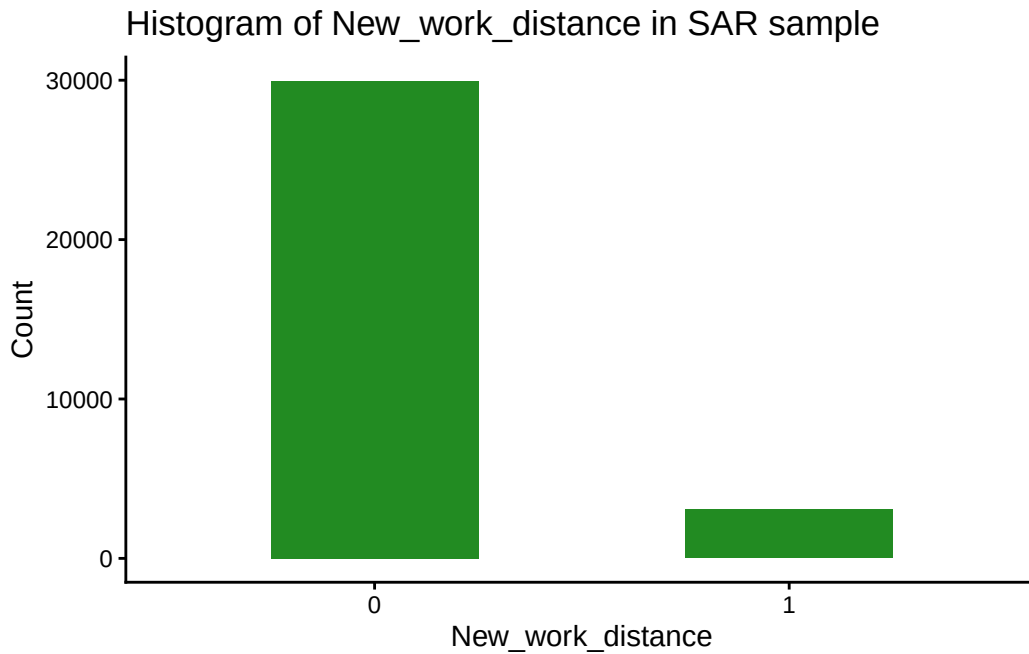
Similarly, the descriptive analysis for the newly created variable `New_work_distance` in `sar_df` should include:

```
table(sar_df$New_work_distance)
```

```
0      1
29954 3072
```

Here we see the categories of `New_work_distance` are in numeric type 0 or 1 (as we used code `sar_df <- sar_df %>% mutate( New_work_distance = if_else(work_distance >6, 1,0))` to do so), therefore in the histogram code, we use `as.factor()` to convert it as factor type.

```
ggplot(sar_df, aes(x = as.factor(New_work_distance))) +  
  geom_bar(fill="forestgreen",width=0.5) +  
  labs(title = "Histogram of New_work_distance in SAR sample", x = "New_work_distance", y = "Count") +  
  theme_classic()#choose theme type, try theme_bw(), theme_minimal() see differences
```



Now we can use the frequency and the bar chart to report that according to our definition of long-distance travelling to work (over 60 km), there are 3,072 individuals who are long-distance commuters in the SAR sample, which makes up 6.14% (3072/50000) of the full sample.

Answer for the model in Q4

In Q4, we want to explore whether people whose occupation is independent employer are associated with a lower probability of commuting over long distances compared to people in other occupations. So we set nssec = 5 (Small employers and own account workers) as the baseline category.

```
table(sar_df$New_nssec)
```

```

      0      6      7      8
20063 2790 5704 4469
```

Then we set the reference categories: sex as 1 (male) and nssec as 5, which is 'Small employers and own account workers':

```
sar_df$sex <- relevel(as.factor(sar_df$sex),ref="1")
sar_df$nssec <- relevel(as.factor(sar_df$nssec),ref="5")
```

Now, we build the logistic regression model and check out the outcomes:

```
model_new = glm(New_work_distance~sex + nssec, data = sar_df, family= "binomial")
summary(model_new)
```

Call:

```
glm(formula = New_work_distance ~ sex + nssec, family = "binomial",
    data = sar_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.44698	0.04138	-10.801	<2e-16 ***
sex2	-0.36678	0.04196	-8.742	<2e-16 ***
nssec2	-1.22639	0.06489	-18.898	<2e-16 ***
nssec1	-1.35519	0.10782	-12.569	<2e-16 ***
nssec3	-1.61400	0.05482	-29.442	<2e-16 ***
nssec4	-2.25717	0.07696	-29.329	<2e-16 ***

```

nssec6      -2.61631      0.10377 -25.212    <2e-16 ***
nssec7      -2.66548      0.08317 -32.047    <2e-16 ***
nssec8      -2.71172      0.09021 -30.059    <2e-16 ***
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 20441  on 33025  degrees of freedom
Residual deviance: 17968  on 33017  degrees of freedom
AIC: 17986

```

Number of Fisher Scoring iterations: 6

For the model interpretation, we need:

```

# odds ratios
exp(coef(model_new))

```

```

(Intercept)      sex2      nssec2      nssec1      nssec3      nssec4
0.63955383 0.69296493 0.29335107 0.25789714 0.19909052 0.10464615
      nssec6      nssec7      nssec8
0.07307217 0.06956621 0.06642224

```

```

# confidence intervals
exp(confint(model_new, level = 0.95))

```

Waiting for profiling to be done...

```

                2.5 %      97.5 %
(Intercept) 0.58959085 0.69343772
sex2        0.63818099 0.75227729
nssec2      0.25813190 0.33291942
nssec1      0.20787318 0.31733101
nssec3      0.17876868 0.22162844
nssec4      0.08984430 0.12149367
nssec6      0.05933796 0.08915692
nssec7      0.05895858 0.08169756
nssec8      0.05547661 0.07902917

```

And don't forget to measure the model fit:

```
# model fit
pR2(model_new) %>% round(4) %>% tidy()
```

fitting null model for pseudo-r2

Warning in tidy.numeric(.): 'tidy.numeric' is deprecated.
See help("Deprecated")

```
# A tibble: 6 x 2
  names      x
  <chr>    <dbl>
1 llh      -8984.
2 llhNull -10220.
3 G2        2473.
4 McFadden  0.121
5 r2ML      0.0721
6 r2CU      0.156
```

Q5. Now what conclusions can you draw when comparing the middle NSSEC groups to the higher groups?

6 Lab: Data Visualisation with ggplot

Open this lab in RStudio: in the Files pane, click labs and then 05.Visualisation.qmd. Save your own copy in myLabs (File -> Save As...) and work on the copy. Work through it from top to bottom, running each code chunk and adding your notes and answers in the file. (First time? See the [Setup](#) guide.)

The lecture's slides can be found [here](#).

6.1 Part 1: Towards the Assignment (30 min, or till when you feel you are good to go)

1. Identify a possible topic:
 - Load your dataset(s) in RStudio.
 - Define your Research Question.
 - Identify the related variables to analyse.
2. Discuss ideas:
 - Get feedback on feasibility and the clarity of research questions.

6.2 Part 2 - Visualisation: ggplot2 Functions and Arguments

For this session you need the following libraries: `ggplot2`, `dplyr`, `tidyr`, `kableExtra`, `ggridges`, `RColorBrewer`, `broom`, `scales`, `corrplot`, `vtable`. They are all already installed in your codespace; the code below loads each one when it is needed.

The `ggplot2` package in R is one of the most powerful tools for creating publication-quality visualisations. It uses a layered approach to building plots, starting with data, then adding mappings, geometries, and other components.

A `ggplot2` plot is built step by step:


```
ggplot(data, aes(x = <X-axis variable>, y = <Y-axis variable>, <other aesthetics>)) +  
  <geom_function()> +  
  <scales/themes/other layers>
```

6.2.1 ggplot()

- Initializes the plotting system.
- Main arguments:
- **data**: A data frame containing the variables to be plotted.
- **aes()**: Aesthetic mappings to connect data variables to visual properties like **x**, **y**, **color**, **fill**, **size**, etc.

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

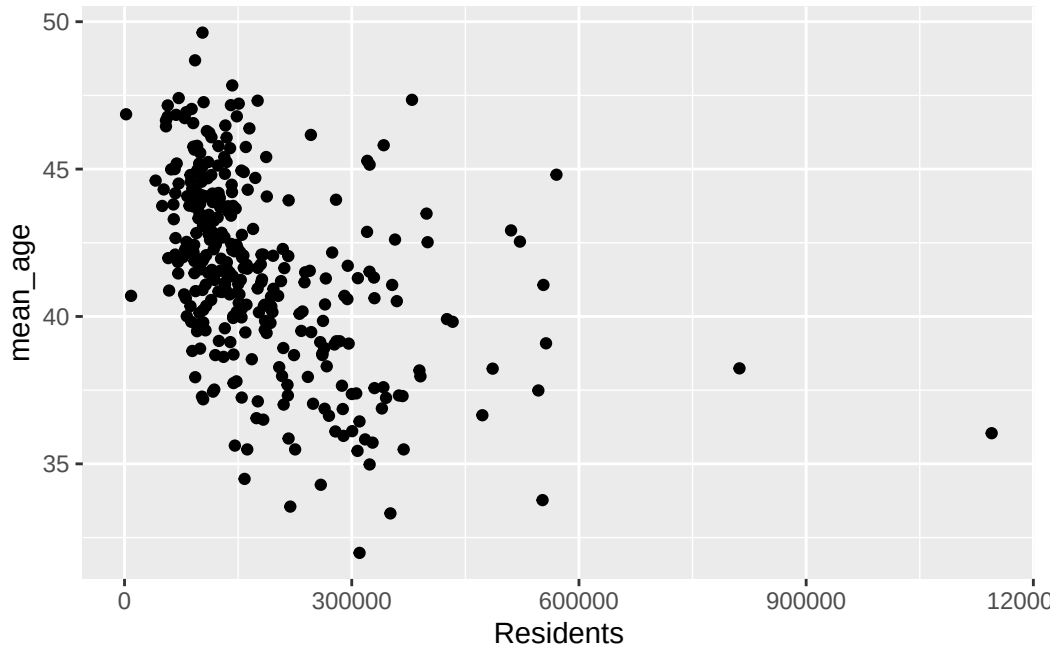
Example:

Let's load the data first:

```
census_data <- read.csv("../data/Census2021/EW_DistrictPercentages.csv")
```

Plotting Number of residents vs age (mean).

```
ggplot(data = census_data, aes(x = Residents, y = mean_age)) +  
  geom_point()
```



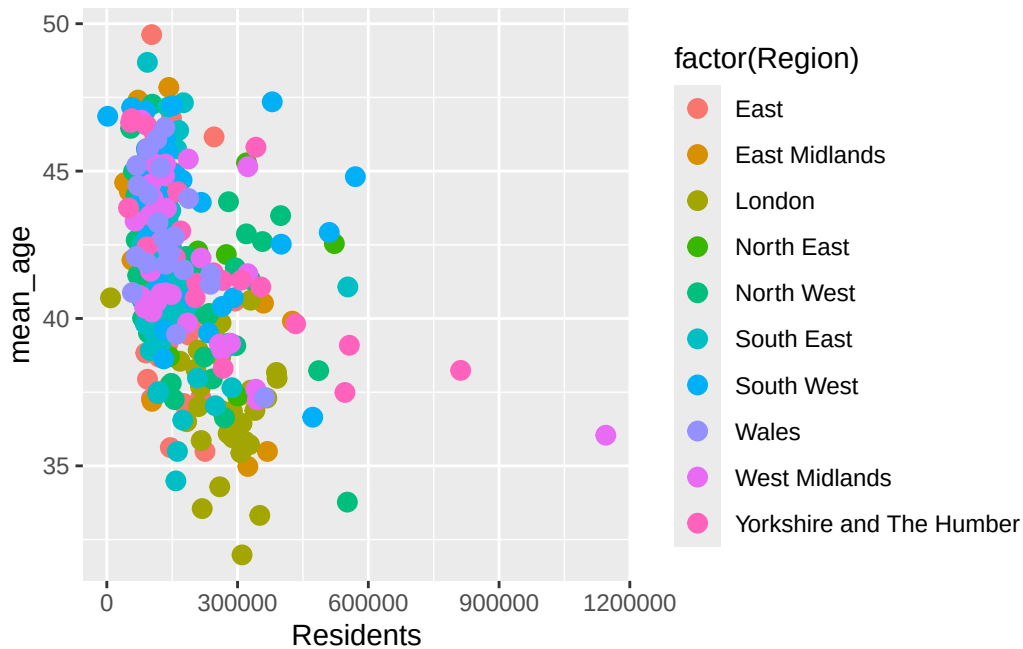
6.2.2 Geometries in ggplot2

You can plot the following geometries in ggplot:

Geometry	Function	Use Case
Point	<code>geom_point()</code>	Scatterplots
Line	<code>geom_line()</code>	Line plots
Bar	<code>geom_bar()</code>	Bar charts
Histogram	<code>geom_histogram()</code>	Histograms
Boxplot	<code>geom_boxplot()</code>	Boxplots
Density	<code>geom_density()</code>	Density plots
Smooth	<code>geom_smooth()</code>	Add regression or smoothing lines

Example: Scatterplot with `geom_point()`

```
ggplot(census_data, aes(x = Residents, y = mean_age, color = factor(Region))) +
  geom_point(size = 3)
```

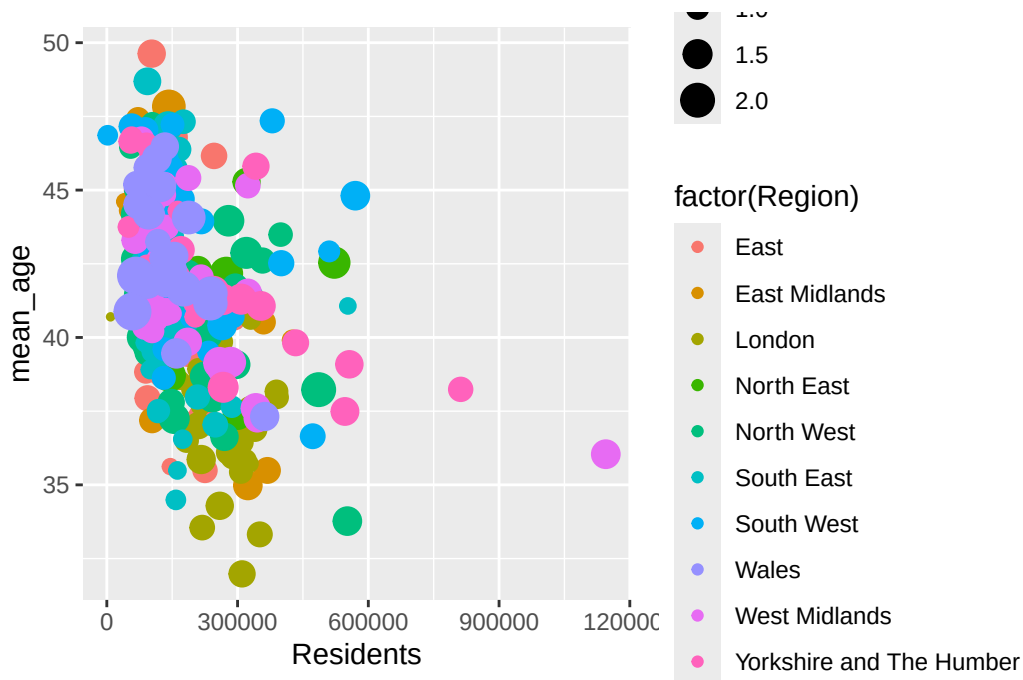


6.2.3 Aesthetics: `aes()`

- Maps data variables to visual properties.
- Common aesthetics:
 - `x`: X-axis variable.
 - `y`: Y-axis variable.
 - `color`: Changes point/line colors based on a variable.
 - `fill`: Fills bars/areas based on a variable.
 - `size`: Controls the size of points/lines.

Example: Adding color and size aesthetics

```
ggplot(census_data, aes(x = Residents, y = mean_age, color = factor(Region), size = pct_Very.
geom_point()
```



What's missing in the plot above? How can this be improved?

6.2.4 Faceting: `facet_*`

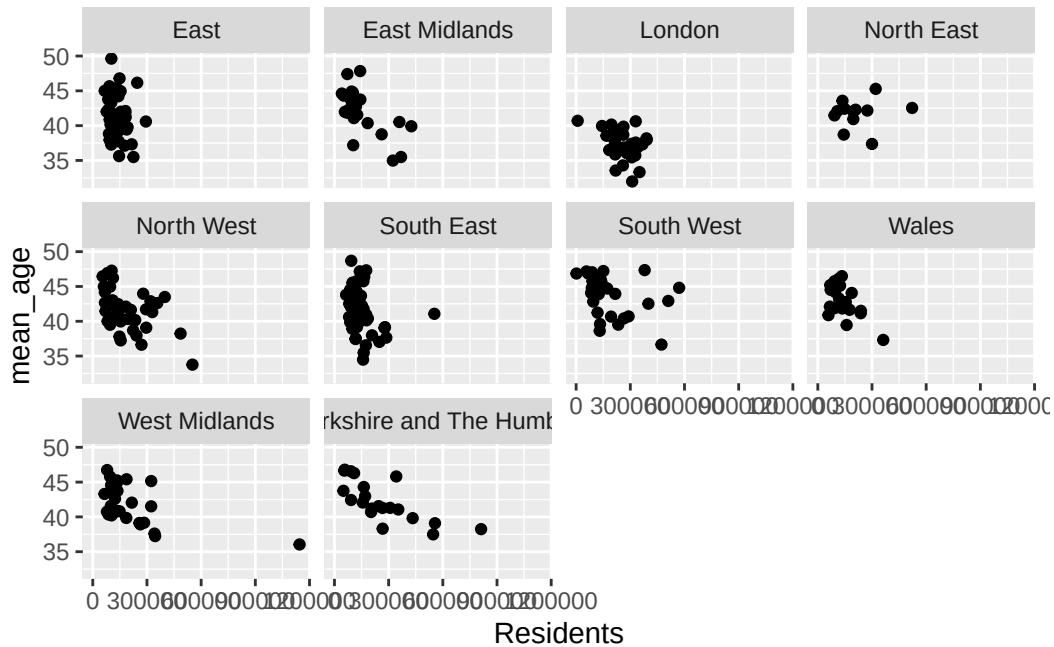
Faceting splits data into subsets and creates multiple small plots.

Function	Description
<code>facet_wrap(~var)</code>	Wraps plots across rows/columns.
<code>facet_grid(row ~ col)</code>	Creates a grid layout for plots.

Faceting is a technique to create multiple plots based on subsets of your data. It is particularly useful for comparing relationships or distributions across categories. By breaking down a dataset into smaller pieces according to a categorical variable, faceting helps visualise data trends or differences within subgroups. For instance, `facet_wrap(~ Region)` creates individual plots for each category in the `Region` variable, aligning them into rows and columns. Alternatively, `facet_grid(row ~ col)` creates a more structured layout, where two categorical variables determine the rows and columns of the grid.

Faceting is essential when dealing with data that needs to be compared across multiple dimensions. For example, a scatterplot faceted by sex can show how a relationship differs for males and females. Similarly, faceted histograms can reveal differences in the distribution of a variable across categories.

```
ggplot(census_data, aes(x = Residents, y = mean_age)) +
  geom_point() +
  facet_wrap(~Region)
```



What are the issues here?

6.2.5 Themes (`theme_*`)

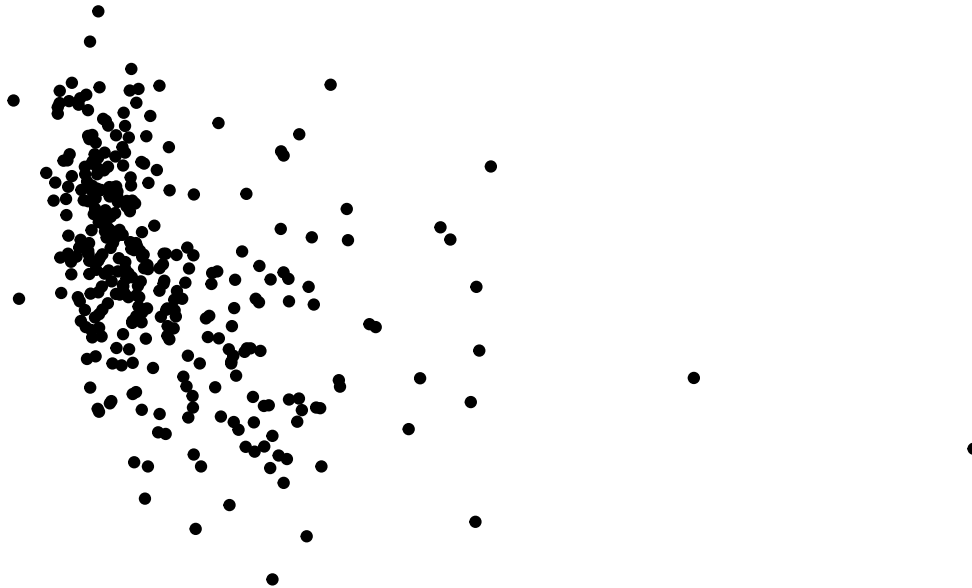
Themes control the overall appearance of the plot.

Function	Effect
<code>theme_minimal()</code>	Simple and clean theme.
<code>theme_classic()</code>	Classic-style plots.
<code>theme_dark()</code>	Dark background.
<code>theme_void()</code>	Minimal with no axes or grids.

Applying a theme

```
ggplot(data = census_data, aes(x = Residents, y = mean_age)) +
  geom_point() +
  theme_void() +
  labs(title = "Visualisation with Void Theme")
```

Visualisation with Void Theme



What do you think? Is this any good?

6.2.6 Scales

Scales adjust color, size, and axis properties.

Scale	Description
<code>scale_color_manual()</code>	Customizes colors for lines/points.
<code>scale_fill_brewer()</code>	Predefined color palettes for fills.
<code>scale_x_continuous()</code>	Modifies X-axis properties.
<code>scale_y_continuous()</code>	Modifies Y-axis properties.

`scale_color` functions modify the outline color of elements like points, lines, or the border of shapes, while `scale_fill` functions modify the interior fill color of shapes like bars, tiles, or boxes. Both are used to control aesthetics based on a variable mapped in `aes()` but apply to different visual aspects of the plot.

Example: Customizing axis and colors

The `RColorBrewer` library provides pre-defined color palettes specifically designed for data visualisation, ensuring clarity and accessibility. It includes sequential, diverging, and qualitative palettes suitable for various data types and visualisation needs.

Have a look at: <https://r-graph-gallery.com/38-rcolorbrewers-palettes.html>

```
# Automatically determine the number of colors needed

num_levels <- length(unique(census_data$Region))
```

The `num_levels` variable is a dynamic way to determine the number of unique levels in a categorical variable, such as `Region`. This approach is particularly helpful when you don't know in advance how many categories exist in the dataset.

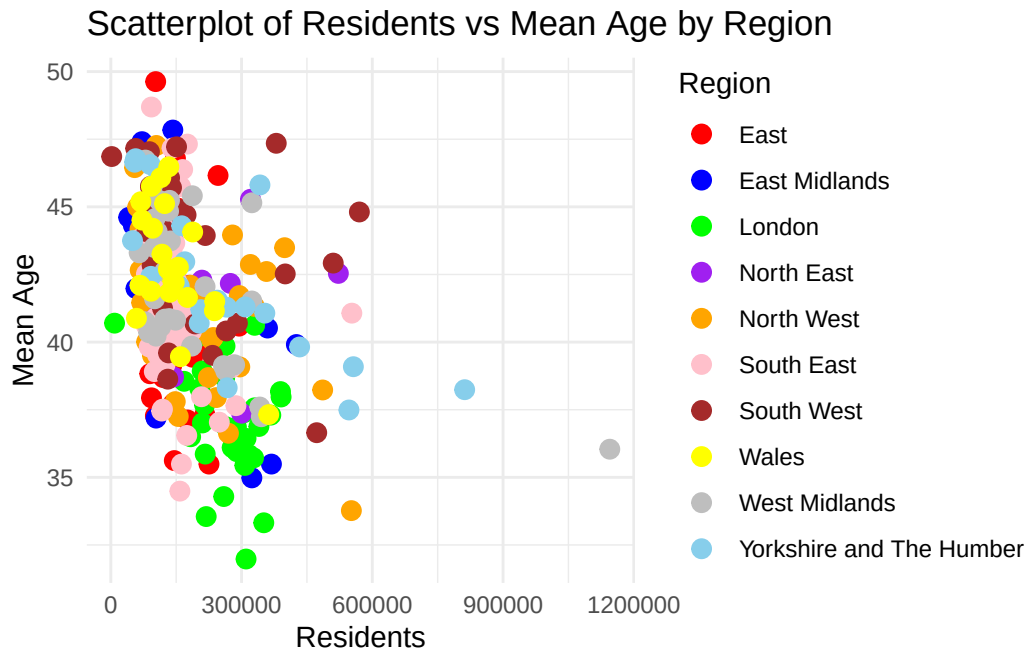
The code above calculates the number of unique categories in the `Region` variable, ensuring that your plot's color palette matches the data's requirements. Based on the value of `num_levels`, you can choose an appropriate color palette from libraries like `RColorBrewer`. For example:

```
library(RColorBrewer)
# Use an appropriate Brewer palette based on the number of levels
palette <- if (num_levels <= 8) "Set2" else "Paired" # Choose a palette with enough colors
```

This dynamically selects a palette suitable for the number of categories. If there are 8 or fewer levels, the `Set2` palette is used; for more levels, `Paired` ensures enough distinct colors (there's several palettes, check the link above for more).

When applying the palette in `ggplot2`, the function `scale_color_brewer(palette = palette)` ensures that the plot assigns colors based on the selected scheme. This process avoids hardcoding colors or palettes and ensures the plot adapts to changes in the data, such as when categories are added or removed.

```
ggplot(census_data, aes(x = Residents, y = mean_age, color = factor(Region))) +
  geom_point(size = 3) +
  scale_color_brewer(palette = palette) + # Dynamically chosen palette
  scale_x_continuous(breaks = seq(50, 350, 50)) +
  labs(title = "Write a title")
```

Consider in general, that there's several functions associated to scales

Category	Function	Description	Example Usage
Discrete Color Scales	<code>scale_color_brewer()</code>	Use pre-defined discrete palettes from RColorBrewer.	<code>scale_color_brewer(palette = "Set1")</code>
	<code>scale_fill_brewer()</code>	Fill colors for discrete variables from RColorBrewer.	<code>scale_fill_brewer(palette = "Pastel2")</code>
	<code>scale_color_manual()</code>	Manually assign colors for discrete variables.	<code>scale_color_manual(values = c("red", "blue", "green"))</code>
	<code>scale_fill_manual()</code>	Manually assign fill colors for discrete variables.	<code>scale_fill_manual(values = c("purple", "orange", "yellow"))</code>
	<code>scale_color_viridis_d()</code>	Use discrete palettes from the viridis package (colorblind-friendly).	<code>scale_color_viridis_d(option = "plasma")</code>
	<code>scale_fill_viridis_d()</code>	Fill discrete variables with viridis palettes.	<code>scale_fill_viridis_d(option = "cividis")</code>

Category	Function	Description	Example Usage
Continuous Color Scales	<code>scale_color_gradient()</code>	Two-color gradient for continuous variables.	<code>scale_color_gradient(low = "blue", high = "red")</code>
	<code>scale_fill_gradient()</code>	Two-color gradient for continuous fill variables.	<code>scale_fill_gradient(low = "green", high = "yellow")</code>
	<code>scale_color_gradient2()</code>	Diverging gradient with a midpoint for continuous variables.	<code>scale_color_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0)</code>
	<code>scale_fill_gradient2()</code>	Diverging gradient with a midpoint for continuous fill variables.	<code>scale_fill_gradient2(low = "purple", mid = "gray", high = "orange", midpoint = 50)</code>
	<code>scale_color_gradientn()</code>	Multi-color gradient for continuous variables.	<code>scale_color_gradientn(colors = c("blue", "green", "yellow", "red"))</code>
	<code>scale_fill_gradientn()</code>	Multi-color gradient for continuous fill variables.	<code>scale_fill_gradientn(colors = c("lightblue", "white", "pink"))</code>
	<code>scale_color_viridis_c()</code>	Continuous color scales from viridis.	<code>scale_color_viridis_c(option = "magma")</code>
	<code>scale_fill_viridis_c()</code>	Continuous fill scales from viridis.	<code>scale_fill_viridis_c(option = "inferno")</code>
Axis Scales	<code>scale_x_continuous()</code>	Customize numeric X-axis with limits and breaks.	<code>scale_x_continuous(limits = c(0, 100), breaks = seq(0, 100, 10))</code>
	<code>scale_y_continuous()</code>	Customize numeric Y-axis with limits and labels.	<code>scale_y_continuous(labels = scales::percent_format())</code>
	<code>scale_x_log10()</code>	Logarithmic transformation for X-axis.	<code>scale_x_log10()</code>
	<code>scale_y_log10()</code>	Logarithmic transformation for Y-axis.	<code>scale_y_log10()</code>

Category	Function	Description	Example Usage
Shape and Size Scales	<code>scale_x_reverse()</code>	Reverse the X-axis direction.	<code>scale_x_reverse()</code>
	<code>scale_y_reverse()</code>	Reverse the Y-axis direction.	<code>scale_y_reverse()</code>
	<code>scale_shape_manual()</code>	Manually assign shapes to categories for points.	<code>scale_shape_manual(values = c(19, 17, 15))</code>
	<code>scale_size_continuous()</code>	Scale the size of points based on a continuous variable.	<code>scale_size_continuous(range = c(1, 10))</code>
	<code>scale_size_manual()</code>	Manually specify point sizes for categorical variables.	<code>scale_size_manual(values = c(2, 4, 6))</code>

6.2.7 Additional Functions for Customization

- `labs()`: Adds labels for axes, title, and legend.
- `coord_flip()`: Flips X and Y axes for horizontal plots.

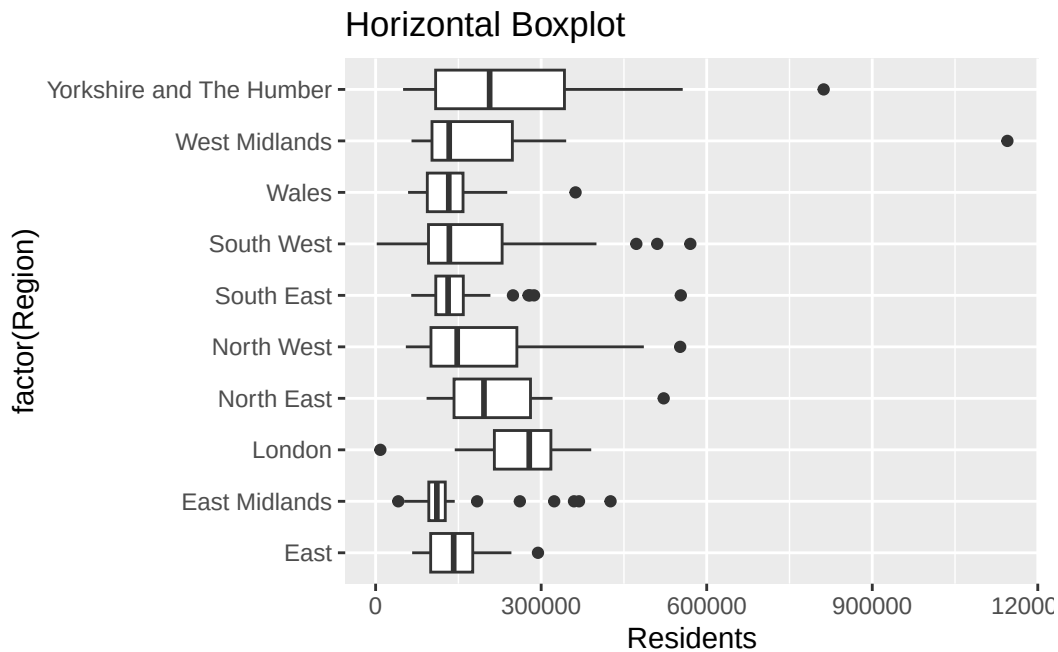
Labeling:

```
# labels
labs(
  title = "Plot Title",
  x = "X-Axis Label",
  y = "Y-Axis Label",
  color = "Legend Title"
)
```

```
<ggplot2::labels> List of 4
 $ x      : chr "X-Axis Label"
 $ y      : chr "Y-Axis Label"
 $ colour: chr "Legend Title"
 $ title  : chr "Plot Title"
```

Flipping:

```
# flipping
ggplot(census_data, aes(x = factor(Region), y = Residents)) +
  geom_boxplot() +
  coord_flip() +
  labs(title = "Horizontal Boxplot")
```



Key Tips for ggplot2:

- Start with simple plots and incrementally add layers (+).
- Use themes (`theme_minimal()`, `theme_classic()`) to clean up your plots.
- Explore palettes with `RColorBrewer` or `viridis` for colorblind-friendly options.

6.3 Part 3 - Visualisation: Making decent graphs (1h)

This section demonstrates how to create **publication-quality visualisations** in R using `ggplot2`.

Learning goals Developing an understanding of, and ability to create academic standard data visualisations.

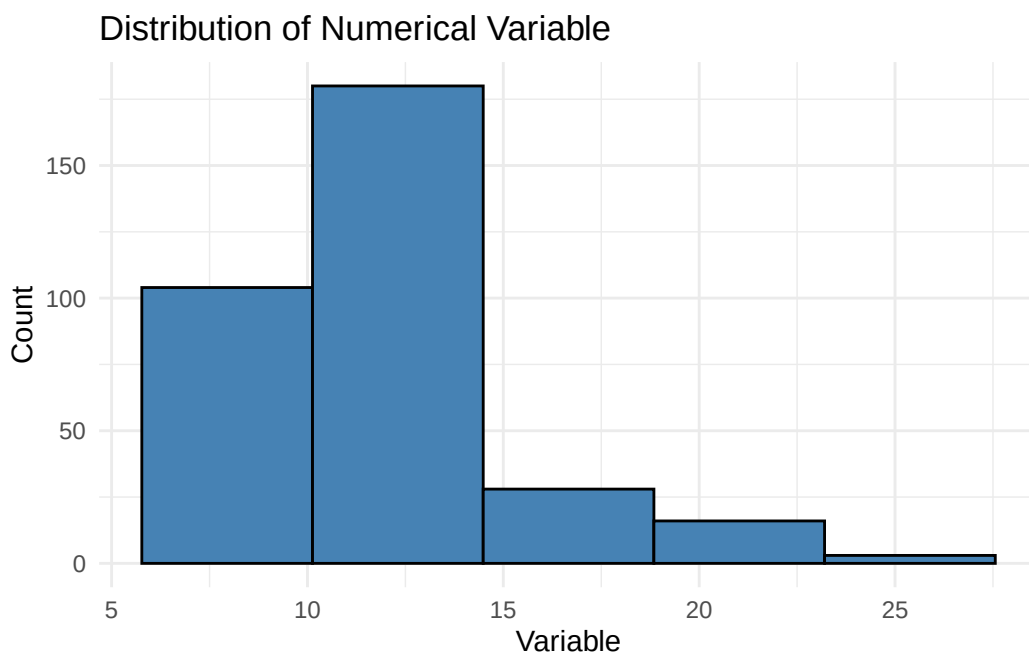
Let's load the datasets first. Check the paths, you may need to remove the “..”

```
frs_data <- read.csv("../data/FRS/FRS16-17_labels.csv")
census_data <- read.csv("../data/Census2021/EW_DistrictPercentages.csv")
```

6.3.1 Distribution of 1 Numerical variable:

Histogram

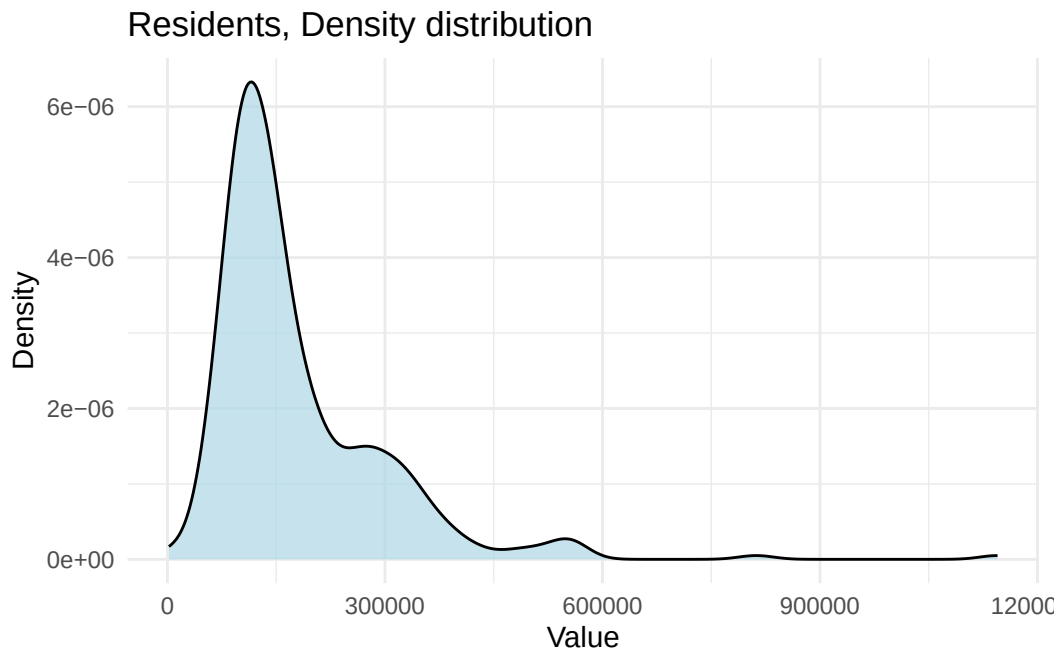
```
ggplot(census_data, aes(x = pct_Age_20_to_29)) +
  geom_histogram(bins = 5, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Numerical Variable", x = "Variable", y = "Count") +
  theme_minimal()
```



The `bins` parameter in a histogram determines the number of intervals (bins) into which the data range is divided. Each bin represents a range of values along the x-axis, and the height of each bar shows the count of observations within that range. For example, if `bins = 5`, the data is divided into five equal-width intervals. Using fewer bins results in a coarser view of the data, while more bins provide a finer, more detailed representation. However, too many bins can make the histogram appear cluttered and difficult to interpret. The width of each bin is automatically calculated by dividing the range of the data by the number of bins.

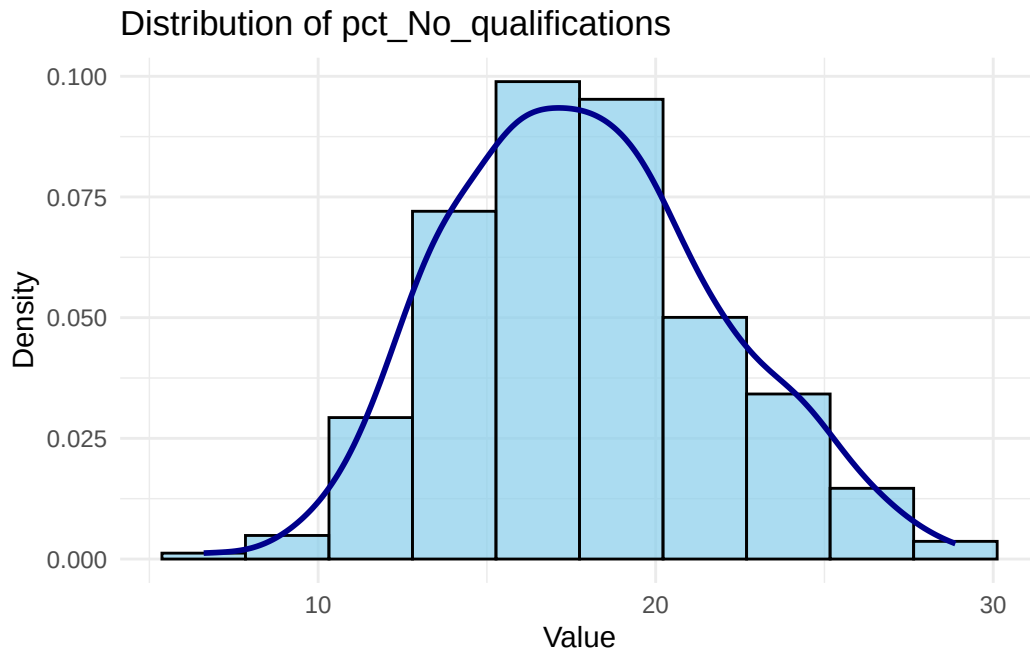
Density Plot

```
ggplot(census_data, aes(x = Residents)) +
  geom_density(fill = "lightblue", alpha = 0.7) +
  labs(title = "Residents, Density distribution", x = "Value", y = "Density") +
  theme_minimal()
```



Histogram + Density Distribution

```
# Plot histogram with density overlay for a chosen variable (e.g., 'pct_No_qualifications')
ggplot(census_data, aes(x = pct_No_qualifications)) +
  geom_histogram(aes(y = after_stat(density)), bins = 10, color = "black", fill = "skyblue") +
  geom_density(color = "darkblue", linewidth = 1) +
  labs(title = "Distribution of pct_No_qualifications", x = "Value", y = "Density") +
  theme_minimal()
```



Box Plot

A box plot, also known as a box-and-whisker plot, is a graphical representation of the distribution of a dataset. It displays the minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum values, with the central box representing the interquartile range (IQR) from Q1 to Q3. The whiskers extend from the box to show the range of the data, excluding outliers. Outliers are typically represented as individual points outside the whiskers.

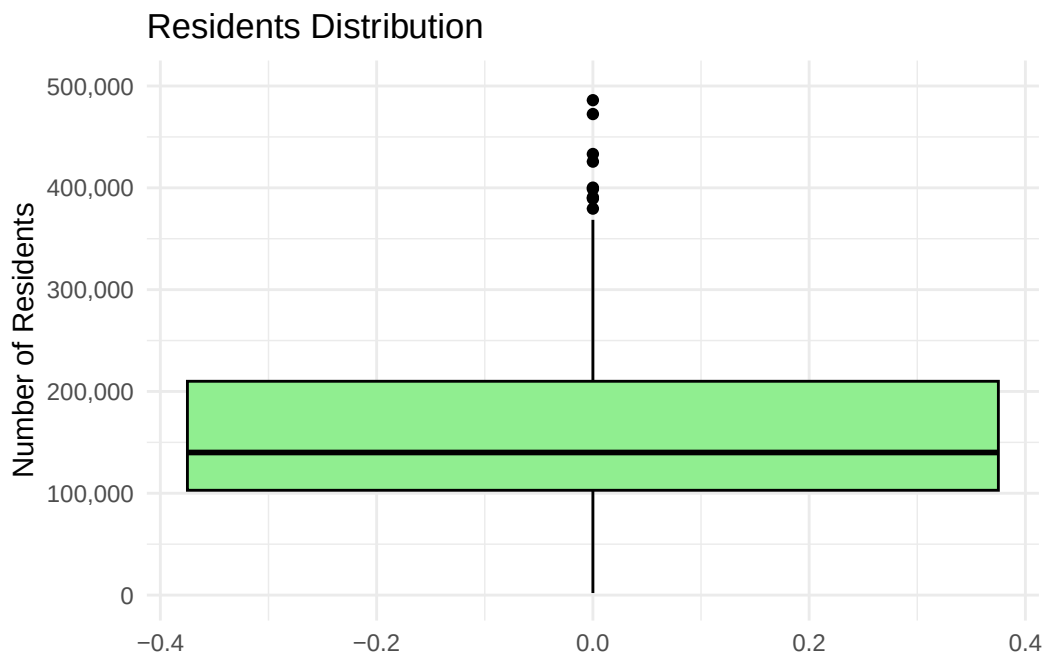
The box plot provides a concise summary of the data distribution, highlighting the spread, center, and potential skewness.

```
library(scales) # For label_comma()
```

Warning: package 'scales' was built under R version 4.4.3

```
ggplot(census_data, aes(y = Residents)) +
  geom_boxplot(fill = "lightgreen", color = "black") +
  labs(title = "Residents Distribution", y = "Number of Residents") +
  theme_minimal() +
  scale_y_continuous(
    limits = c(0, 500000),
    labels = label_comma() # Properly placed within scale_y_continuous()
  )
```

Warning: Removed 9 rows containing non-finite outside the scale range (``stat_boxplot()``).



```
# Setting axis limits for better readability and applying comma-separated format to large numbers
```

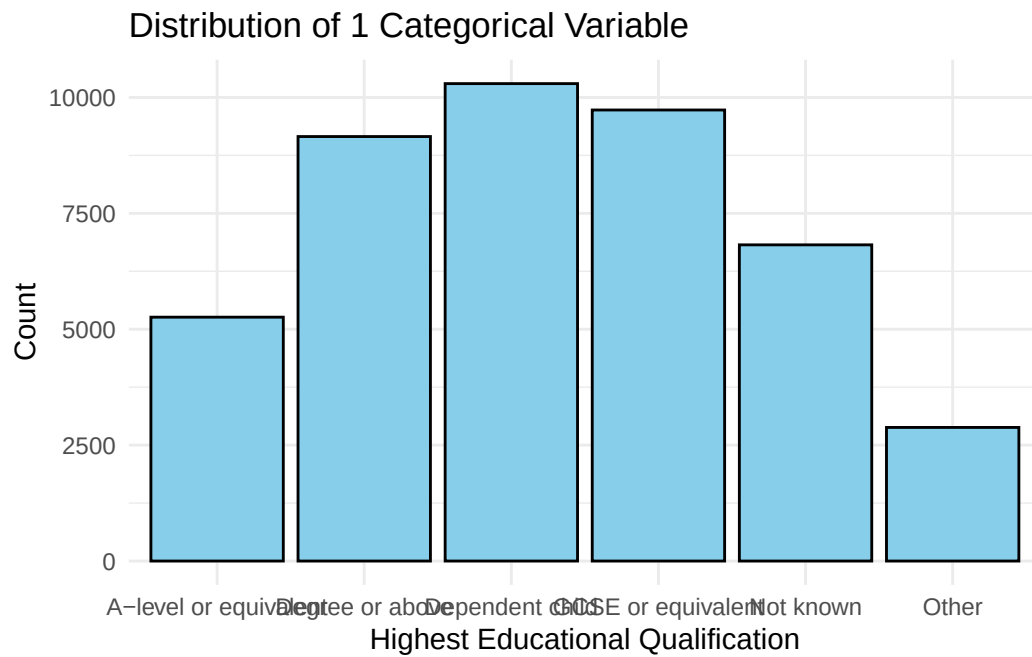
This looks kind of sad with just one variable, but it's the best way, usually, to plot distributions for numerical variables.

Notice the **outliers**. Outliers are data points that deviate significantly from the overall pattern of a dataset. They can arise due to measurement errors, variability in the data, or the presence of extreme values. Outliers can heavily influence statistical analyses, such as means or regression models, potentially leading to misleading conclusions. Identifying and addressing outliers is crucial to ensure robust and accurate results. Boxplots allow spotting them, along with interquartile range (IQR) analysis.

6.3.2 Distribution of 1 Categorical variable:

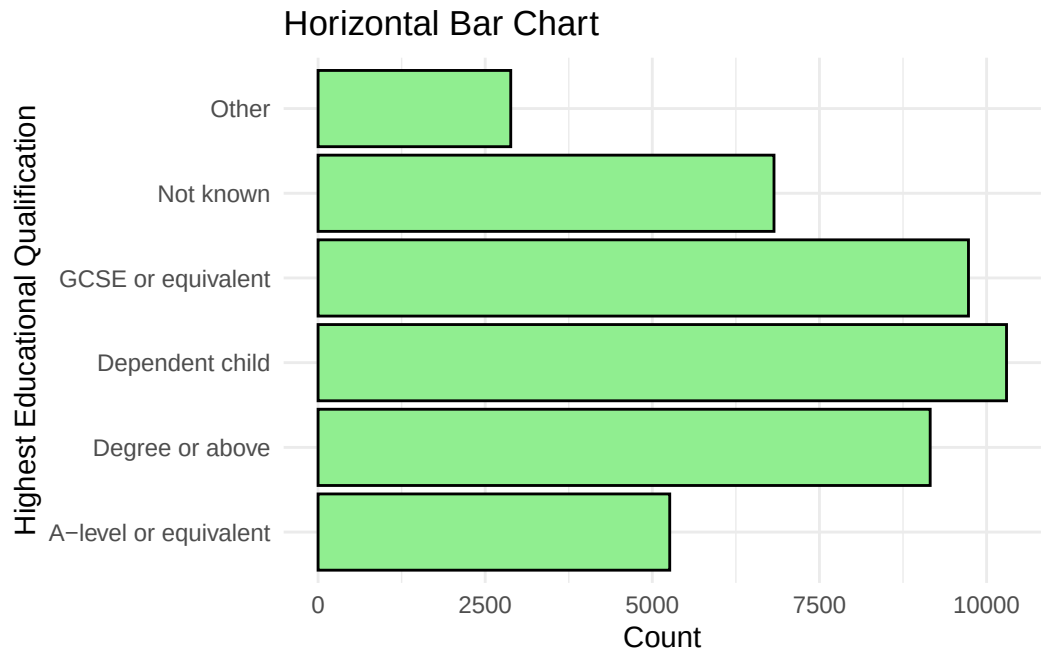
Bar Chart

```
ggplot(frs_data, aes(x = highest_qual)) +  
  geom_bar(fill = "skyblue", color = "black") +  
  labs(title = "Distribution of 1 Categorical Variable", x = "Highest Educational Qualification") +  
  theme_minimal()
```

Horizontal Bar Chart

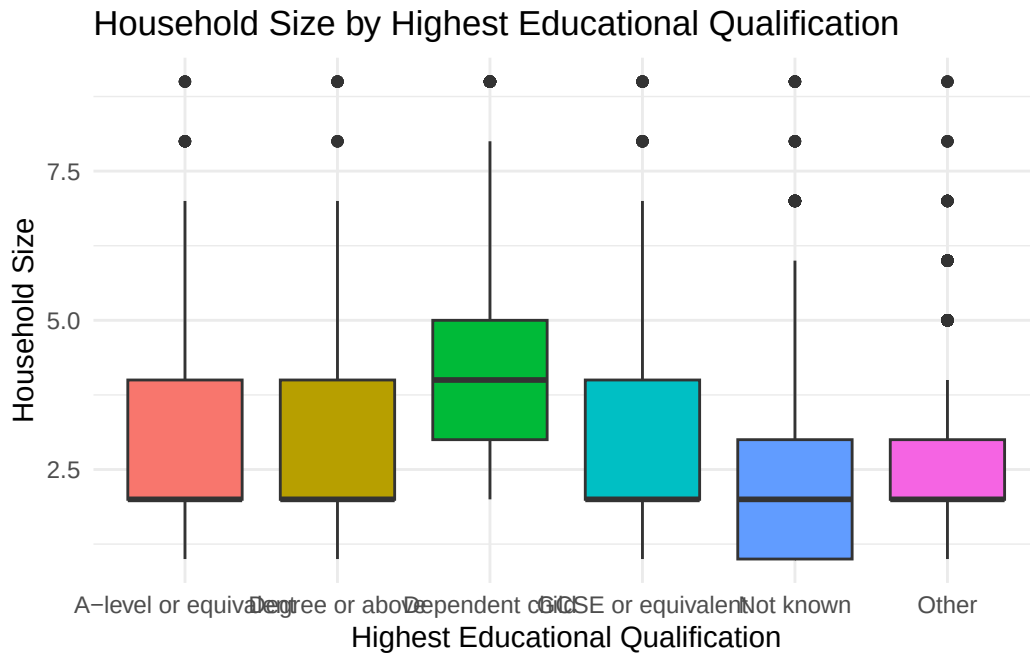
```
ggplot(frs_data, aes(x = highest_qual)) +
  geom_bar(fill = "lightgreen", color = "black") +
  labs(title = "Horizontal Bar Chart", x = "Highest Educational Qualification", y = "Count", )
  coord_flip() + #just flipping
  theme_minimal()
```



6.3.3 Comparing variables

1 numerical, 1 categorical: Boxplot

```
ggplot(frs_data, aes(x = highest_qual, y = hh_size, fill = highest_qual)) +  
  geom_boxplot() +  
  labs(  
    title = "Household Size by Highest Educational Qualification",  
    x = "Highest Educational Qualification",  
    y = "Household Size"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none") # don't need this
```



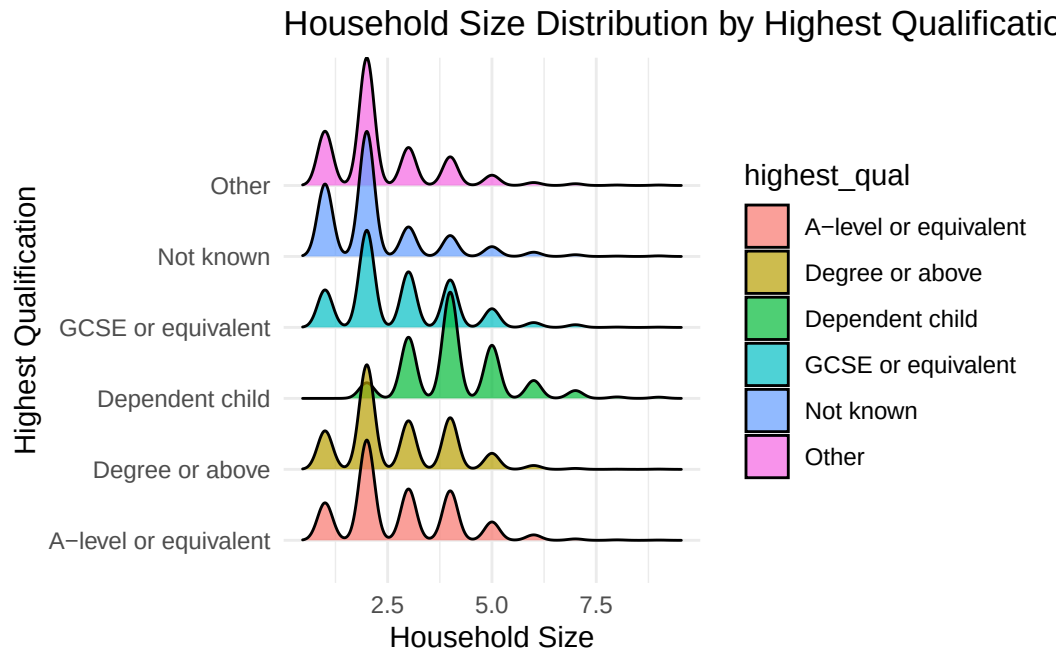
1 numerical, 1 categorical: Density Ridges

```
library(ggribes) # need a library for this
```

Warning: package 'ggribes' was built under R version 4.4.2

```
ggplot(frs_data, aes(x = hh_size, y = highest_qual, fill = highest_qual)) +
  geom_density_ridges(alpha = 0.7) +
  labs(
    title = "Household Size Distribution by Highest Qualification",
    x = "Household Size",
    y = "Highest Qualification"
  ) +
  theme_minimal()
```

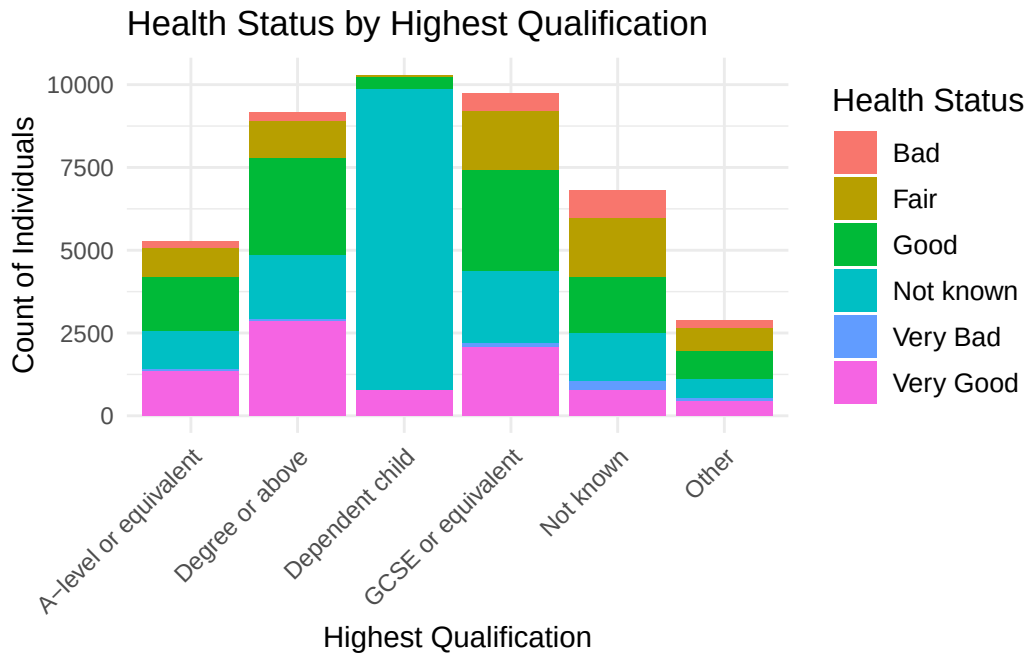
Picking joint bandwidth of 0.18



2 Categorical Variables: Stacked Bar Chart

Note the adjustments made for making the labels look better

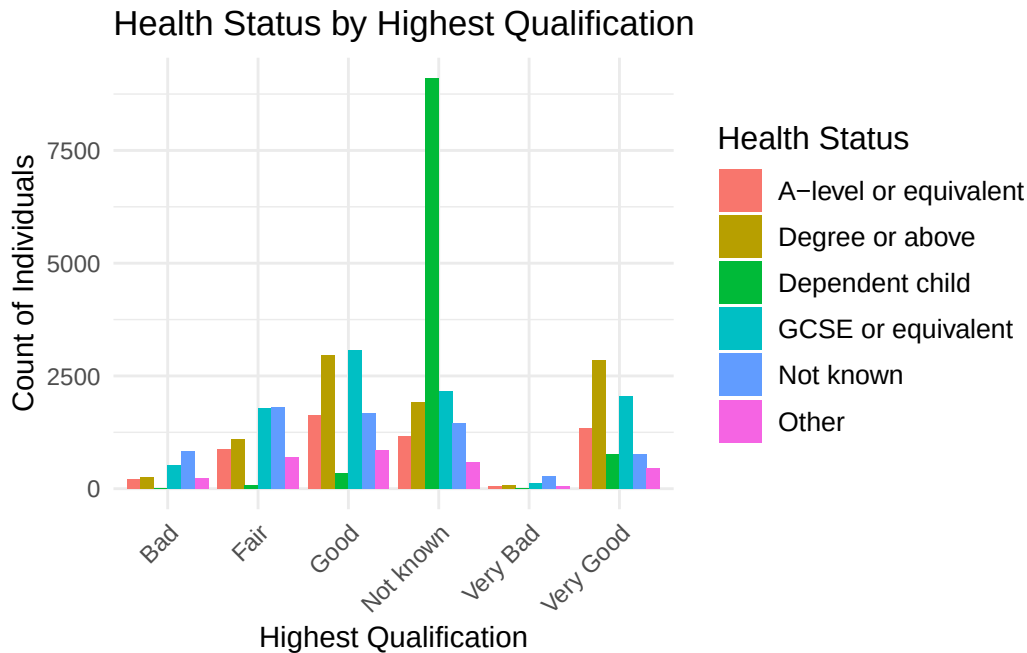
```
ggplot(frs_data, aes(x = highest_qual, fill = health)) +
  geom_bar(position = "stack") +
  labs(
    title = "Health Status by Highest Qualification",
    x = "Highest Qualification",
    y = "Count of Individuals",
    fill = "Health Status"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Rotate X-axis labels for readability
    legend.title = element_text(size = 12),           # Adjust legend title size
    legend.text = element_text(size = 10)             # Adjust legend text size
  )
```



Side-by-Side Bar Chart (Two-Way Frequency Distribution)

This type of chart can get messy.

```
ggplot(frs_data, aes(x = health, fill = highest_qual)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Health Status by Highest Qualification",
    x = "Highest Qualification",
    y = "Count of Individuals",
    fill = "Health Status"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.title = element_text(size = 12),
    legend.text = element_text(size = 10)
  )
```



2 Categorical Variable counts to percentages: Stacked Percentage Graph

One can also convert the counts of two categorical variables to percentages. In this case, we aim to visualise the proportional distribution of one categorical variable (**health**) across levels of another categorical variable (**highest_qual**) using a stacked percentage bar chart. First, we create a cross-tabulation to count the occurrences of each combination of Health and Qualification categories. These counts are then converted into a percentage format relative to the total counts for each Qualification category. This transformation allows us to represent the relative proportions rather than absolute counts, facilitating comparison across categories.

A *cross tabulation* (or contingency table) is a statistical tool used to analyze the relationship between two or more categorical variables. It organizes data into a table format where:

- Rows represent the categories of one variable.
- Columns represent the categories of another variable.
- Cells display the counts or frequencies of data points that fall into each combination of the row and column categories.

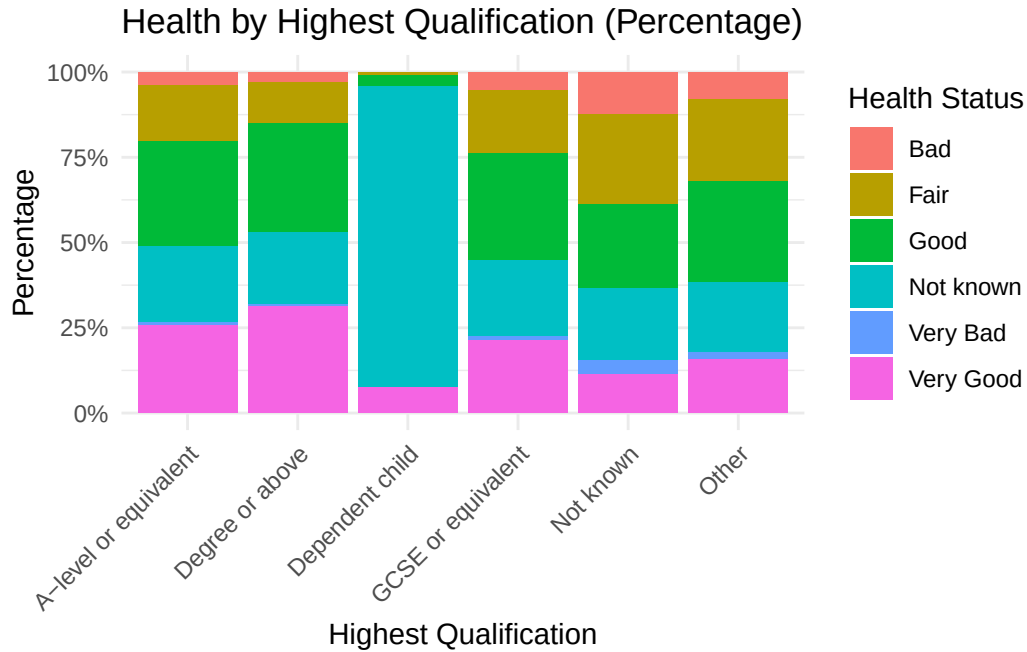
```
# Create a cross-tabulation of observed counts
cross_tab <- table(frs_data$health, frs_data$highest_qual)

# Convert the cross-tabulation to a data frame
cross_tab_df <- as.data.frame(cross_tab)
colnames(cross_tab_df) <- c("Health", "Qualification", "Percentage")
```

The data is reshaped into a data frame for compatibility with **ggplot2**, where a stacked bar

chart with percentage scaling (`position = "fill"`) is generated. The result is a plot that shows how Health statuses are distributed proportionally within each Qualification level.

```
# Create a stacked percentage bar chart
ggplot(cross_tab_df, aes(x = Qualification, y = Percentage, fill = Health)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Health by Highest Qualification (Percentage)",
    x = "Highest Qualification",
    y = "Percentage",
    fill = "Health Status"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



3+ Numerical Variables: Boxplot

To compare three numerical variables using boxplots in `ggplot2`, one needs to reshape the data to a long format so that each numerical variable is treated as a category in a single column.

Here's how you can do it, using `tidyverse`

```
library(tidyverse) # we need tidyverse for this
```

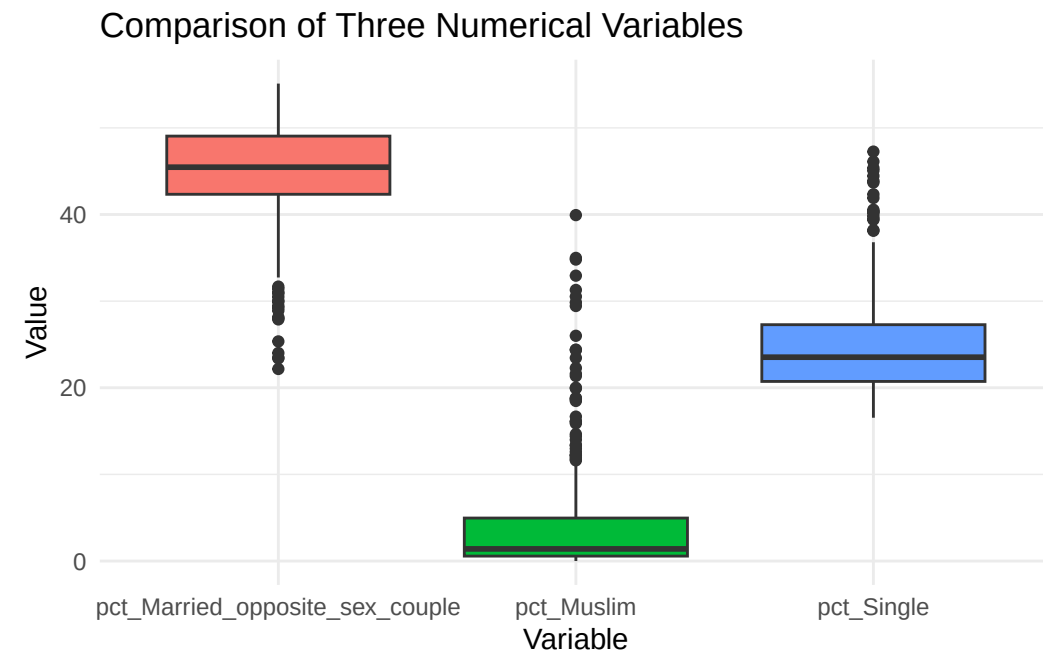
```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.3      v tibble     3.2.1
v purrr      1.0.2      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
```

```
x readr::col_factor() masks scales::col_factor()
x purrr::discard()     masks scales::discard()
x dplyr::filter()      masks stats::filter()
x dplyr::lag()          masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
# Reshape the data into long format
long_data <- census_data %>%
  pivot_longer(
    cols = c(pct_Single, pct_Muslim, pct_Married_opposite_sex_couple),
    names_to = "Variable",
    values_to = "Value"
  )
```

The code above reshapes the data from a wide format to a long format using the `pivot_longer()` function from the `tidyverse` package. In the original dataset, each column represents a separate variable (e.g., `pct_Single`, `pct_Muslim`, ..), and their values are stored in individual columns. The transformation collapses these columns into two new columns: one for the variable names (`Variable`) and another for their corresponding values (`Value`). This format is useful for plotting or analysis where variables are treated uniformly, such as when creating boxplots or facet grids in `ggplot2`.

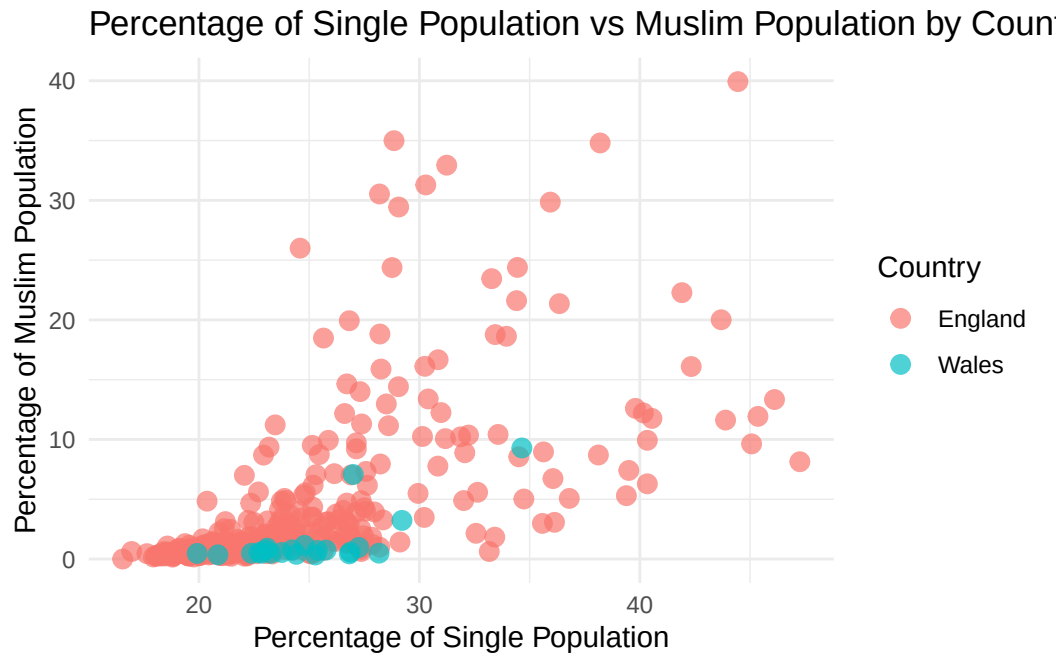
```
# Create the boxplot
ggplot(long_data, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot() +
  labs(
    title = "Comparison of Three Numerical Variables",
    x = "Variable",
    y = "Value"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

6.3.4 Visualising Relationships:

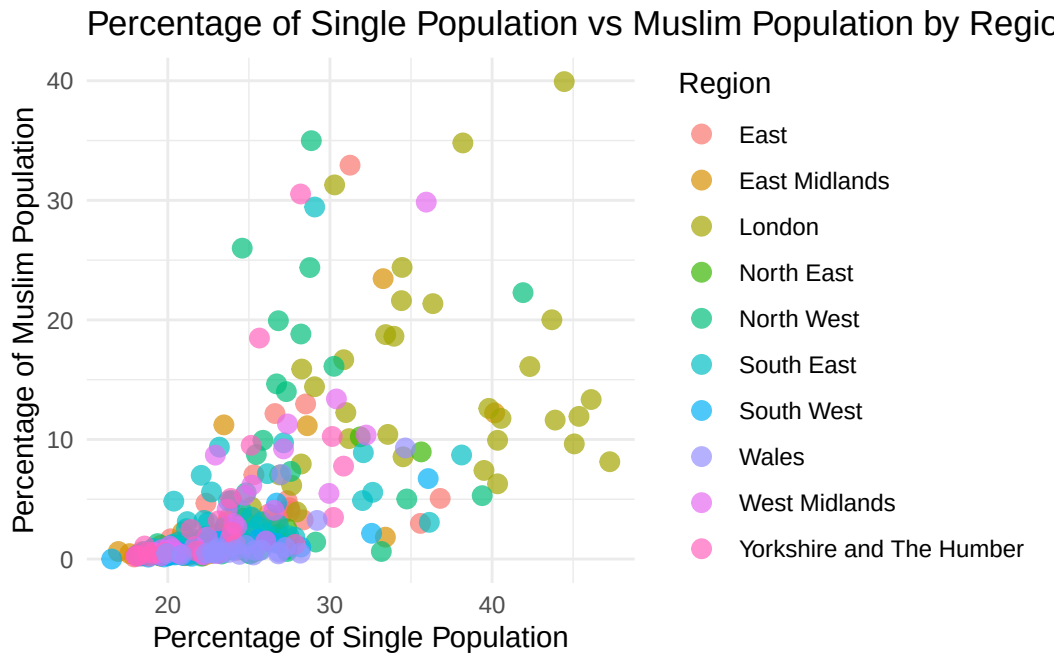
2 Numerical Variables: Scatterplot

```
ggplot(census_data, aes(x = pct_Single, y = pct_Muslim, color = factor(Country))) +
  geom_point(size = 3, alpha = 0.7) +
  labs(
    title = "Percentage of Single Population vs Muslim Population by Country",
    x = "Percentage of Single Population",
    y = "Percentage of Muslim Population",
    color = "Country"
  ) +
  theme_minimal()
```



2 Numerical Variables + 1 Categorical: Scatterplot

```
ggplot(census_data, aes(x = pct_Single, y = pct_Muslim, color = factor(Region))) +  
  geom_point(size = 3, alpha = 0.7) +  
  labs(  
    title = "Percentage of Single Population vs Muslim Population by Region",  
    x = "Percentage of Single Population",  
    y = "Percentage of Muslim Population",  
    color = "Region"  
  ) +  
  theme_minimal()
```



3+ Numerical Variables: Correlogram

A correlogram is a visual representation of a correlation matrix, where the strength and direction of relationships between numerical variables are displayed. The matrix is typically a grid, with variables listed along both the rows and columns. Each cell in the matrix shows the correlation coefficient between two variables.

The library `corrplot` would do it for us. Check: <https://cran.r-project.org/web/packages/corrplot/vignettes/corrplot-intro.html> for deeper customisation.

In the `corrplot` package, a correlogram can use colors, shapes, or numbers to represent the correlation. For example, colors ranging from blue to red often signify negative to positive correlations, while white may indicate no correlation. Similarly, circles or other shapes can vary in size to depict the strength of the relationship. The `corrplot` function simplifies creating correlograms with customizable options. For example, using `method = "number"` places correlation coefficients in each cell, while `method = "circle"` represents correlations graphically.

Correlograms are valuable for quickly identifying patterns or strong relationships between variables, such as spotting which features might influence one another in a dataset.

```
library(corrplot)
```

```
Warning: package 'corrplot' was built under R version 4.4.2
```

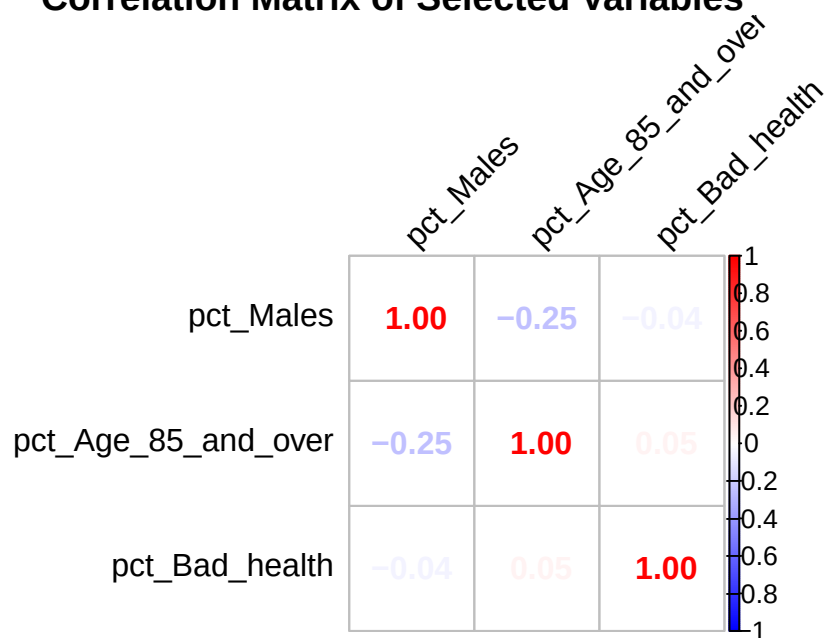
```
corrplot 0.95 loaded
```

```
# Select the necessary variables from the dataset
selected_vars <- census_data[, c("pct_Males", "pct_Age_85_and_over", "pct_Bad_health")]

# Calculate the correlation matrix
cor_matrix <- cor(selected_vars, use = "complete.obs")

# Generate the correlation plot
corrplot(cor_matrix,
  method = "number", # Display numbers
  col = colorRampPalette(c("blue", "white", "red"))(200), # Color gradient
  tl.col = "black", # Text color for labels
  tl.srt = 45, # Rotate text labels
  title = "Correlation Matrix of Selected Variables",
  mar = c(0, 0, 1, 0)) # Adjust margins
```

Correlation Matrix of Selected Variables

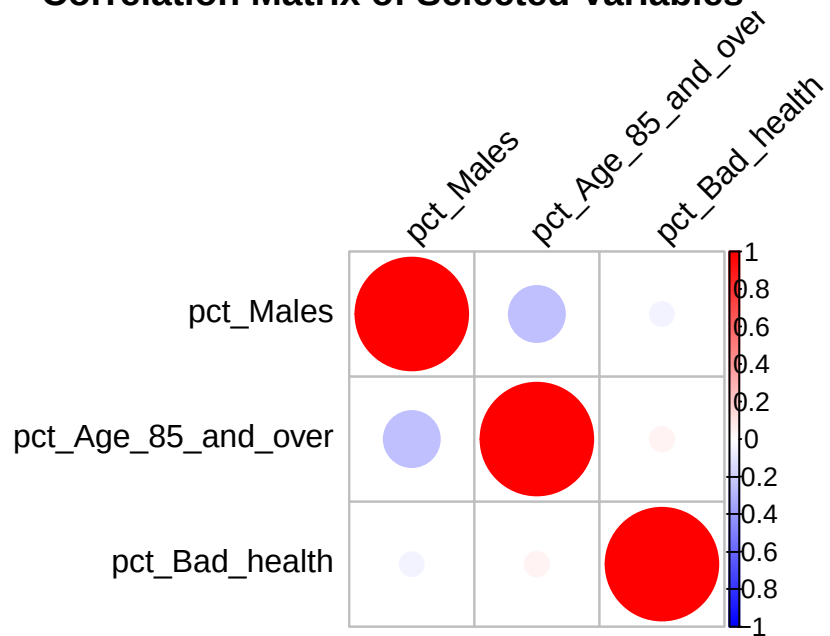


We can also include circles to graphically represent correlations.

```
# Generate the correlation plot with circles
corrplot(cor_matrix,
  method = "circle", # Display circles
  col = colorRampPalette(c("blue", "white", "red"))(200), # Color gradient
  tl.col = "black", # Text color for labels
  tl.srt = 45, # Rotate text labels
```

```
title = "Correlation Matrix of Selected Variables",
mar = c(0, 0, 1, 0)) # Adjust margins
```

Correlation Matrix of Selected Variables



3+ Numerical Variables: Scatterplot matrix

Preparation:

```
library(tidyverse)

# Select the variables
selected_vars <- census_data[, c("pct_Males", "pct_Single", "pct_Bad_health")]

# Create pairwise combinations of variable names
scatter_data <- expand_grid(
  Variable1 = names(selected_vars),
  Variable2 = names(selected_vars)
)

# Add the data values for the pairs
scatter_data <- scatter_data %>%
  rowwise() %>%
  mutate(
    Value1 = list(selected_vars[[Variable1]]), # Extract values for Variable1
```

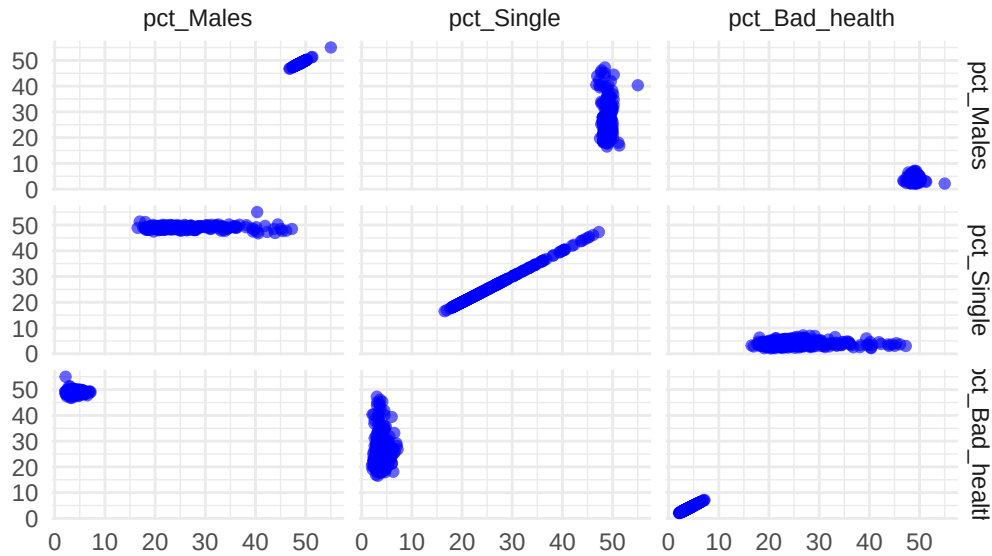
```
Value2 = list(selected_vars[[Variable2]]) # Extract values for Variable2
) %>%
unnest(c(Value1, Value2)) # Unnest the lists into rows
```

This code is used to create a dataset that facilitates generating pairwise scatterplots between selected variables from a dataset. First, it selects the variables of interest (`pct_Males`, `pct_Single`, and `pct_Bad_health`) from the `census_data` dataset and stores them in a new data frame called `selected_vars`. Then, the `expand.grid()` function is used to generate all possible combinations of these variables, resulting in a data frame with two columns, `Variable1` and `Variable2`, where each row represents a pair of variables (e.g., `pct_Males` vs `pct_Single`). Finally, the `mutate()` function is applied to map the actual values of the paired variables (`Variable1` and `Variable2`) to two new columns, `Value1` and `Value2`. These columns contain the corresponding data points for the variable pair, enabling the creation of scatterplots where `Value1` is plotted against `Value2` for each combination.

Visualisation:

```
ggplot(scatter_data, aes(x = Value1, y = Value2)) +
  geom_point(alpha = 0.6, color = "blue") +
  facet_grid(Variable1 ~ Variable2, scales = "free") +
  labs(
    title = "Correlation Matrix",
    x = "",
    y = ""
  ) +
  theme_minimal()
```

Correlation Matrix



You can also include a categorical variable through colouring. Let's try with Country.

```
# Ensure the original data includes the 'Country' variable
selected_vars <- census_data[, c("pct_Males", "pct_Single", "pct_Bad_health", "Country")]

# Create pairwise combinations of variable names (excluding 'Country')
scatter_data <- expand_grid(
  Variable1 = names(selected_vars)[-4], # Exclude 'Country'
  Variable2 = names(selected_vars)[-4]
)

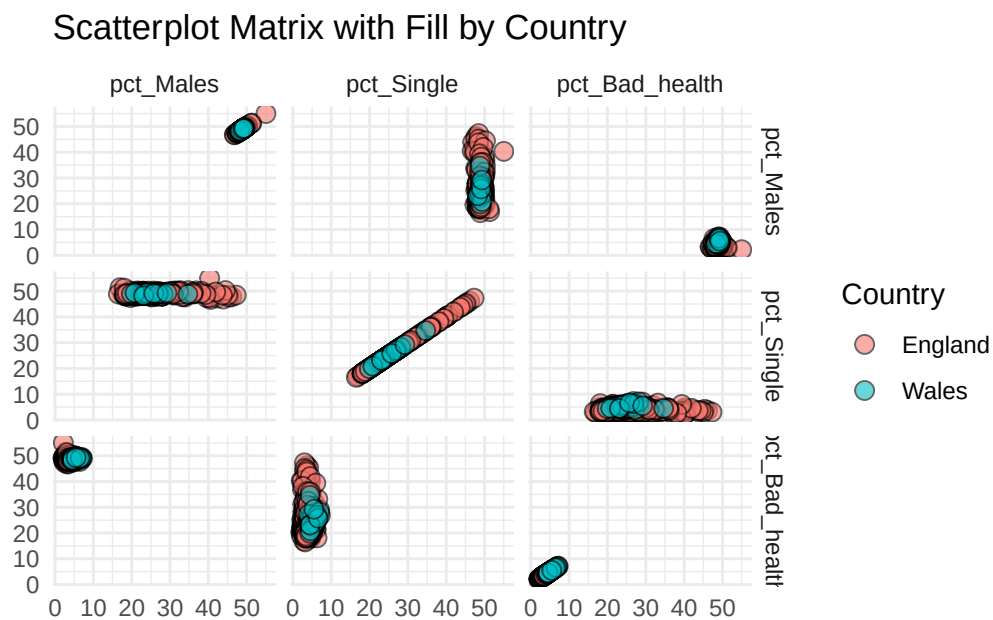
# %>% filter(Variable1 != Variable2) # Remove self-comparisons

# Add the data values for the pairs
scatter_data <- scatter_data %>%
  rowwise() %>%
  mutate(
    Value1 = list(selected_vars[[Variable1]]), # Extract values for Variable1
    Value2 = list(selected_vars[[Variable2]]), # Extract values for Variable2
    Country = list(selected_vars$Country)      # Include 'Country' in the data
  ) %>%
  unnest(c(Value1, Value2, Country)) # Unnest the lists into rows
```

Now, the scatter_data includes the Country column, which can be used to map fill in

ggplot:

```
ggplot(scatter_data, aes(x = Value1, y = Value2, fill = Country)) +
  geom_point(alpha = 0.6, shape = 21, size = 3) + # Use shape with fill (e.g., 21)
  facet_grid(Variable1 ~ Variable2, scales = "free") +
  labs(
    title = "Scatterplot Matrix with Fill by Country",
    x = "",
    y = "",
    fill = "Country"
  ) +
  theme_minimal()
```



1 Numerical Variables, 2+ Categorical Variables: Boxplot faceting

```
ggplot(frs_data, aes(x = marital_status, y = hh_income_gross, fill = marital_status)) +
  geom_boxplot() +
  facet_wrap(~ sex) +
  scale_y_continuous(
    limits = c(0, 100000), # limit the y-axis
    labels = label_comma() # Properly placed within scale_y_continuous()
  ) +
  labs(
```

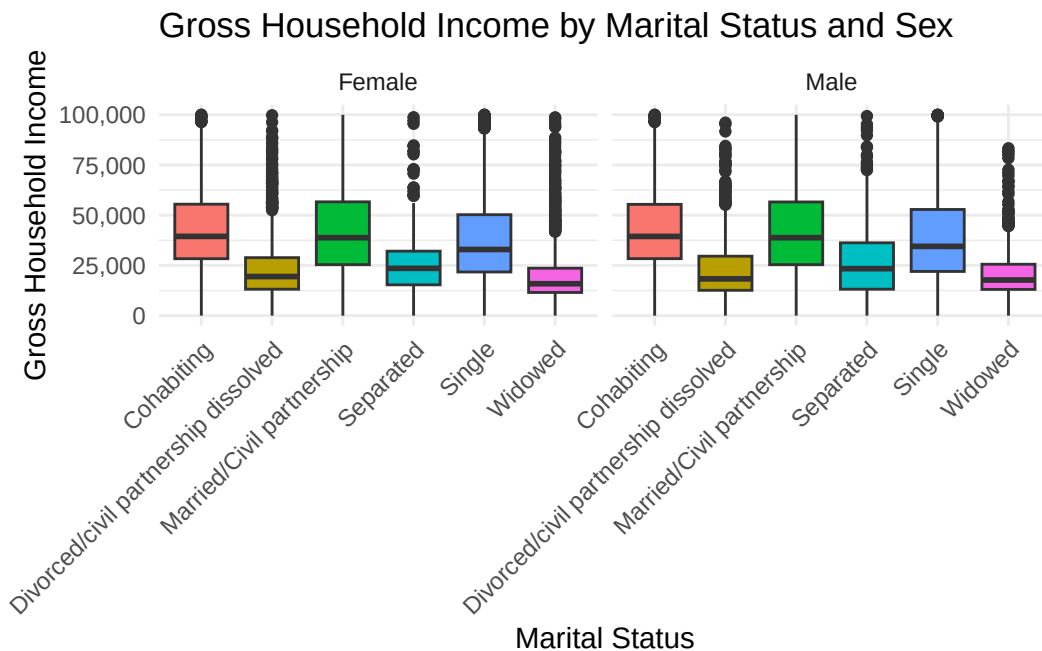


```

title = "Gross Household Income by Marital Status and Sex",
x = "Marital Status",
y = "Gross Household Income"
) + # Removed 'fill' label to avoid legend creation
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1), # Rotate x-axis labels for readability
  legend.position = "none" # Corrected to lowercase 'none'
)

```

Warning: Removed 2769 rows containing non-finite outside the scale range (`stat\_boxplot()`).



1 Numerical Variables, 2+ Cateorical Variables: Violin Plot faceting

A violin plot combines aspects of a box plot and a density plot. It displays the distribution of a continuous variable, showing its probability density across different levels of a categorical variable. The plot consists of a vertical axis representing the variable of interest and horizontal axes that group data by categorical variables. The shape of the “violin” represents the distribution’s density, often mirrored on both sides, making it easier to see the spread and skewness of the data.

```
ggplot(frs_data, aes(x = marital_status, y = hh_income_gross, fill = marital_status)) +
  geom_violin() +
  facet_wrap(~ sex) +
  scale_y_continuous(
    limits = c(0, 100000),
    labels = label_comma() # Properly placed within scale_y_continuous()
  ) +
  labs(
    title = "Gross Household Income by Marital Status and Sex",
    x = "Marital Status",
    y = "Gross Household Income"
  ) + # Removed 'fill' label to avoid legend creation
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Rotate x-axis labels for readability
    legend.position = "none" # Corrected to lowercase 'none'
  )
)
```

Warning: Removed 2769 rows containing non-finite outside the scale range (``stat_ydensity()``).

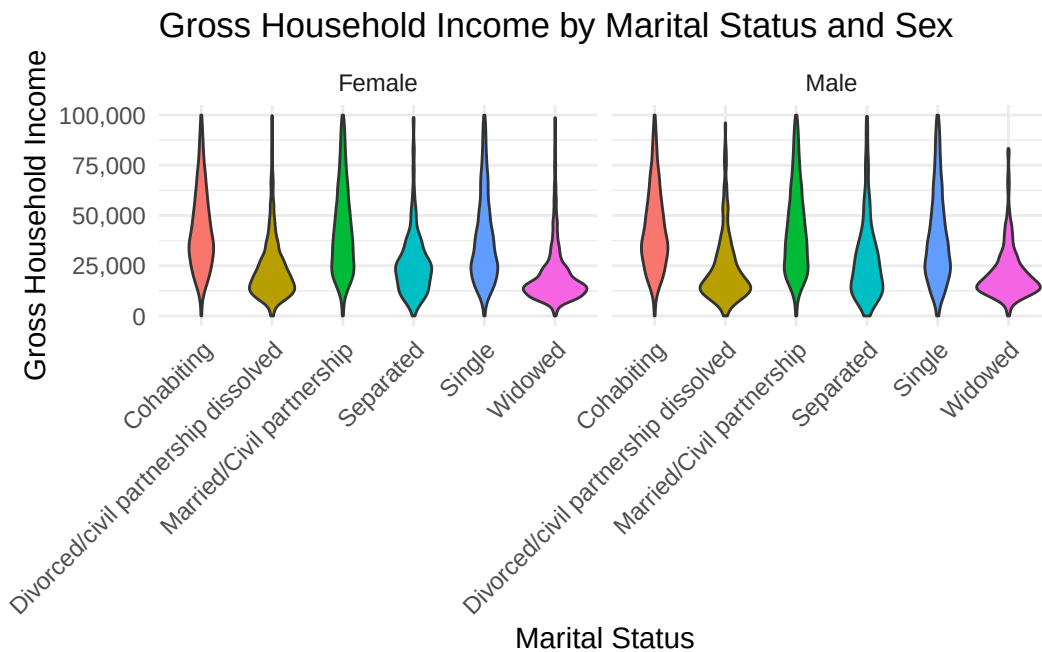


Table 6.6: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
pct_Very_bad_health	331	1.2	0.34	0.55	0.92	1.4	2.4
pct_No_qualifications	331	18	4	6.6	15	20	29
pct_Males	331	49	0.66	47	49	49	55
pct_Higher_manager_prof	331	13	4.7	5.5	9.8	16	40

6.4 Part 4: Publication-Ready Tables

We will be using `kableExtra` to create nicely formatted tables for a series of outputs in RStudio:

```
library(kableExtra)
```

Warning: package 'kableExtra' was built under R version 4.4.2

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

```
group_rows
```

6.4.1 Summarising datasets

Only for summarising statistics we use `vtable` that makes use of `kableExtra` behind the scenes. You can use the function `st()` or `sumtable()`, they do the same thing. See <https://cran.r-project.org/web/packages/vtable/vignettes/sumtable.html>

```
library(vtable)
```

Warning: package 'vtable' was built under R version 4.4.2

```
# Generate the summary table
st(census_data, vars = c("pct_Very_bad_health", "pct_No_qualifications", "pct_Males", "pct_Higher_manager_prof"))
```

6.4.2 Creating a Well-Formatted Table from a Cross Tabulation

We can also convert a basic cross tabulation to a nice table

```
# cross tabulation
cross_tab <- table(frs_data$health, frs_data$highest_qual)

# Convert the cross-tabulation to a data frame
cross_tab_df <- as.data.frame(cross_tab)
```

This will output lots of rows, but it's just an example.

```
colnames(cross_tab_df) <- c("Health", "Qualification", "Percentage")

cross_tab_df %>%
  kbl(caption = "Health and Educational Qualification") %>%
  kable_styling(full_width = FALSE, position = "left") %>%
  add_header_above(c(" " = 1, "Highest Educational Qualification" = 2))
```

- **Rename Columns:** `colnames(cross_tab_df) <- c("Health", "Qualification", "Percentage")`. This assigns new column names to the data frame `cross_tab_df`.
- **Generate a Table with kable:** `cross_tab_df %>% kbl(caption = "Health and Educational Qualification")`. This uses the `kbl` function from the `kableExtra` package to create a basic, well-formatted table. The `caption` argument adds a title to the table: “Health and Educational Qualification”.
- **Style the Table:** `%>% kable_styling(full_width = FALSE, position = "left")`. This enhances the appearance of the table with the `kable_styling` function. `full_width = FALSE` ensures the table is not stretched to the full width of the document or webpage; `position = "left"`: Aligns the table to the left side of the page.
- **Add a Header Row:** `%>% add_header_above(c(" " = 1, "Highest Educational Qualification" = 2))` Adds an additional header row above the table; `" " = 1` the first column (“Health”) has a blank header in this new row; `"Highest Educational Qualification" = 2`: The next two columns (“Qualification” and “Percentage”) are grouped under the label “Highest Educational Qualification”.

6.4.3 Creating a Well-Formatted Table from a Regression Model

Finally, we can derive a table to show the Multiple Linear Regression results.

Table 6.7: Health and Educational Qualification

Health	Highest Educational Qualification	
	Qualification	Percentage
Bad	A-level or equivalent	200
Fair	A-level or equivalent	868
Good	A-level or equivalent	1626
Not known	A-level or equivalent	1163
Very Bad	A-level or equivalent	55
Very Good	A-level or equivalent	1348
Bad	Degree or above	262
Fair	Degree or above	1097
Good	Degree or above	2953
Not known	Degree or above	1923
Very Bad	Degree or above	67
Very Good	Degree or above	2854
Bad	Dependent child	22
Fair	Dependent child	67
Good	Dependent child	342
Not known	Dependent child	9100
Very Bad	Dependent child	7
Very Good	Dependent child	760
Bad	GCSE or equivalent	524
Fair	GCSE or equivalent	1785
Good	GCSE or equivalent	3061
Not known	GCSE or equivalent	2165
Very Bad	GCSE or equivalent	133
Very Good	GCSE or equivalent	2061
Bad	Not known	841
Fair	Not known	1804
Good	Not known	1674
Not known	Not known	1447
Very Bad	Not known	279
Very Good	Not known	775
Bad	Other	231
Fair	Other	696
Good	Other	854
Not known	Other	587
Very Bad	Other	64
Very Good	Other	450

Table 6.8: Regression Results: Predicting Very Bad Health Percentage

Term	Coefficients			
	Estimate	Std. Error	t Value	P Value
(Intercept)	4.003	0.880	4.550	0.000
pct_No_qualifications	0.053	0.006	8.937	0.000
pct_Males	-0.074	0.018	-4.121	0.000
pct_Higher_manager_prof	-0.013	0.005	-2.670	0.008

```
# Load required libraries
library(broom)

# Fit the regression model
model <- lm(pct_Very_bad_health ~ pct_No_qualifications + pct_Males + pct_Higher_manager_prof,
            data = census_data)
```

Let's tidy up the regression output.

```
regression_table <- tidy(model) %>%
  select(term, estimate, std.error, statistic, p.value) %>%
  rename(
    Term = term,
    Estimate = estimate,
    `Std. Error` = std.error,
    `t value` = statistic,
    `P value` = p.value
  )
```

Create and style the regression table

```
regression_table %>%
  kbl(
    caption = "Regression Results: Predicting Very Bad Health Percentage",
    digits = 3,
    col.names = c("Term", "Estimate", "Std. Error", "t Value", "P Value")
  ) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "condensed")) %>%
  column_spec(2:5, width = "3cm") %>% # Adjust column widths
  add_header_above(c(" " = 1, "Coefficients" = 4)) # Add grouped header
```

- `kable_styling()` This function customizes the appearance of a **kable** table. The arguments used are:
 - `full_width = FALSE`: Ensures the table is not stretched to fill the entire width of the page.
 - `bootstrap_options = c("striped", "hover", "condensed")`:
 - * **striped**: Adds alternating row colors for better readability.
 - * **hover**: Highlights rows when hovered over with the mouse (for HTML tables).
 - * **condensed**: Reduces the table’s vertical spacing, making it more compact.
- `column_spec(2:5, width = "3cm")`. This function customizes specific columns of the table.
 - `2:5`: Targets columns 2 through 5 for styling.
 - `width = "3cm"`: Sets the width of these columns to 3 cm, ensuring uniform and readable column sizes.
- `add_header_above(c(" " = 1, "Coefficients" = 4))`. This function adds an additional header row above the table. This is used to customize the header of a table.
 - `" " = 1`: Leaves the first column (e.g., the Term column) without a header, spanning 1 column.
 - `"Coefficients" = 4`: This part means that columns 2, 3, 4, and 5 will be grouped together under one header, which is labeled as “Coefficients” (4 here indicates the other 4 tables). Essentially, these columns will have the same header, which makes the table more readable and organized.

Have a look here for some details on **kableExtra** <https://bookdown.org/yihui/rmarkdown-cookbook/kableextra.html>.

6.5 Part 5: Play with the code

- Experiment with different variables and aesthetics to deepen your understanding of `ggplot2`.
- Test modifying labels, themes, and colour schemes to create tailored visualisations.